Category & Grouping

S#	Category	Group Name
1	MINOR	AI AND COMPETITIVE ADVANTAGE
2	MINOR	ANIMATION AND GAMING
3	MINOR	ANTHROPOLOGY
4	MINOR	ARTS
5	MINOR	BUSINESS ADMINISTRATION
6	MINOR	COMPUTATIONAL MATHEMATICS
7	MINOR	CORPORATE LAW
8	MINOR	CREATIVE ARTS
9	MINOR	DIGITAL MARKETING AND AUTOMATION
10	MINOR	FILM MAKING
11	MINOR	FINTECH AND ALGORITHMIC TRADING
12	MINOR	FOOD TECHNOLOGY
13	MINOR	GRAPHIC DESIGN
14	MINOR	INNOVATION MANAGEMENT AND ENTREPRENEURSHIP
15	MINOR	INTELLECTUAL PROPERTY LAW
16	MINOR	INTERIOR DESIGN
17	MINOR	INTERNATIONAL BUSINESS
18	MINOR	ORGANIZATIONAL BEHAVIOUR
19	MINOR	PHARMACOLOGY
20	MINOR	PRECISION AGRICULTURE
21	MINOR	TECHNOPRENEURSHIP
22	MINOR	WATER SCIENCE AND MANAGEMENT
23	DOUBLE MAJOR	SECOND MAJOR EEE FOR FIRST MAJOR CSE / CSIT / ME
24	DOUBLE MAJOR	SECOND MAJOR ECE FOR FIRST MAJOR CSE / CSIT
25	DOUBLE MAJOR	SECOND MAJOR IOT FOR FIRST MAJOR CSE / CSIT / EEE / ECE / AIDS
26	DOUBLE MAJOR	SECOND MAJOR AIDS FOR FIRST MAJOR ECE / IOT / CSE / CSIT
27	DOUBLE MAJOR	SECOND MAJOR ME FOR FIRST MAJOR ECE / EEE / CSE / CSIT / AIDS / BT / IOT
28	DOUBLE MAJOR	SECOND MAJOR CIVIL FOR FIRST MAJOR ECE / EEE / CSE / CSIT / AIDS / BT / IOT
29	DOUBLE MAJOR	SECOND MAJOR BT FOR FIRST MAJOR B.TECH / BBA / B.COM / BSC / BCA / BA

S#	Category	Group Name
30	DOUBLE MAJOR	SECOND MAJOR ANIMATION AND GAMING FOR FIRST MAJOR B.TECH / BBA / B.COM /
31	DOUBLE MAJOR	SECOND MAJOR BUSINESS ADMINISTRATION FOR FIRST MAJOR B.TECH / B.COM / BSC
32	DOUBLE MAJOR	SECOND MAJOR ECONOMICS AND PUBLIC POLICY FOR FIRST MAJOR B.TECH / B.COM /
33	DOUBLE MAJOR	SECOND MAJOR FOOD TECHNOLOGY FOR FIRST MAJOR B.TECH / B.COM / BBA / BSC
34	DOUBLE MAJOR	SECOND MAJOR COMMERCE FOR FIRST MAJOR B.TECH / BBA / BSC / BCA / BA
35	SPECIALIZATION	AGRI-BIOTECHNOLOGY
36	SPECIALIZATION	AI AND AUTONOMOUS SYSTEMS
37	SPECIALIZATION	AI FOR COMPUTATIONAL INTELLIGENCE
38	SPECIALIZATION	AI SYSTEMS FOR VISUAL INTELLIGENCE
39	SPECIALIZATION	AI-DRIVEN EDGE ARCHITECTURES AND APPLICATIONS
40	SPECIALIZATION	AI-DRIVEN LANGUAGE TECHNOLOGIES
41	SPECIALIZATION	AUTOMOTIVE ELECTRONICS AND AUTOSAR
42	SPECIALIZATION	AUTOMOTIVE ENERGY ENGINEERING
43	SPECIALIZATION	BIOINFORMATICS
44	SPECIALIZATION	BLOCKCHAIN ENGINEERING FOR WEB3
45	SPECIALIZATION	CLOUD AND EDGE COMPUTING
46	SPECIALIZATION	CLOUD INFRASTRUCTURE DESIGN AND ENGINEERING
47	SPECIALIZATION	CLOUD NATIVE SECURITY
48	SPECIALIZATION	CLOUD NATIVE SOFTWARE ENGINEERING
49	SPECIALIZATION	CLOUD-BASED SCIENTIFIC COMPUTING
50	SPECIALIZATION	COMPUTER COMMUNICATION AND 5G TECHNOLOGY
51	SPECIALIZATION	CONSTRUCTION TECHNOLOGY AND MANAGEMENT
52	SPECIALIZATION	CROSS PLATFORM DEVELOPMENT FRAMEWORKS
53	SPECIALIZATION	CYBER PHYSICAL SYSTEMS AND IOT
54	SPECIALIZATION	CYBER SECURITY AND BLOCKCHAIN TECHNOLOGY
55	SPECIALIZATION	DATA COMMUNICATIONS
56	SPECIALIZATION	DATA ENGINEERING FOR AI
57	SPECIALIZATION	DATA SCIENCE AND BIG DATA ANALYTICS
58	SPECIALIZATION	DISTRIBUTED LEDGER ANALYTICS
59	SPECIALIZATION	EMBEDDED SYSTEMS

S#	Category	Group Name
60	SPECIALIZATION	E-MOBILITY ENGINEERING
61	SPECIALIZATION	ENGINEERING DESIGN
62	SPECIALIZATION	GAME DEVELOPMENT AND UX DESIGN
63	SPECIALIZATION	GEOTECHNICAL AND TRANSPORTATION ENGINEERING
64	SPECIALIZATION	GREEN ENERGY TECHNOLOGIES
65	SPECIALIZATION	HARDWARE-SOFTWARE CO-DESIGN FOR SECURITY
66	SPECIALIZATION	HEALTHCARE DATA ANALYTICS
67	SPECIALIZATION	INDUSTRIAL AUTOMATION
68	SPECIALIZATION	INDUSTRIAL BIOTECHNOLOGY
69	SPECIALIZATION	INTELLIGENT MULTIMEDIA PROCESSING
70	SPECIALIZATION	IOT ANALYTICS
71	SPECIALIZATION	MEDICAL BIOTECHNOLOGY
72	SPECIALIZATION	ROBOTICS AND AUTOMATION
73	SPECIALIZATION	SMART GRID TECHNOLOGIES
74	SPECIALIZATION	SMART MANUFACTURING
75	SPECIALIZATION	SOCIAL AND DIGITAL MEDIA ANALYTICS
76	SPECIALIZATION	SOFTWARE MODELLING AND DEVOPS
77	SPECIALIZATION	SPATIAL COMPUTING AND IMMERSIVE TECHNOLOGIES
78	SPECIALIZATION	STRUCTURAL ENGINEERING
79	SPECIALIZATION	VERY LARGE SCALE INTEGRATION
80	SPECIALIZATION	WATER RESOURCE AND ENVIRONMENTAL ENGINEERING
81	PROJECT	RESEARCH PROJECT
82	PROJECT	INNOVATION PROJECT
83	PROJECT	CAPSTONE PROJECT

Category & Grouping Courses

1. MINOR: AI AND COMPETITIVE ADVANTAGE

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	MIN	MIN-CORE	23MRAC01M	AI FOR BUSINESS STRATEGY	M	ABS	4	0	0	0	4	4	
2	MIN	MIN-CORE	23MRAC02M	AI STRATEGY AND GOVERNANCE	M	ASG	4	0	0	0	4	4	
3	MIN	MIN-CORE	23MRAC03M	SUCCESSFUL AI STRATEGIES : A CEO'S PERSPECTIVES	M	SAI	4	0	0	0	4	4	
4	MIN	MIN-CORE	23MRAC04M	CRAFTING A COMPETITIVE ADVANTAGE	M	CCA	4	0	0	0	4	4	
5	MSDC	MIN-SDP	23MRAC05	AI APPLICATIONS FOR BUSINESS SUCCESS	M	AIA	0	0	6	4	4	10	

2. MINOR: ANIMATION AND GAMING

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	MIN	MIN-CORE	23MRAG01	INTRODUCTION TO GAME MATH	R	IGM	0	0	6	4	4	10	
2	MIN	MIN-CORE	23MRAG02	DRAWING BASICS	R	DBS	0	0	6	4	4	10	
3	MIN	MIN-CORE	23MRAG03	GRAPHIC DESIGN	R	GDS	0	0	6	4	4	10	
4	MIN	MIN-CORE	23MRAG04	CONCEPTS OF 3D	R	C3D	0	0	6	4	4	10	
5	MSDC	MIN-SDP	23MRAG05	DIGITAL ART	R	DTA	0	0	6	4	4	10	

3. MINOR: ANTHROPOLOGY

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	MIN	MIN-CORE	23MRAP01M	INTRODUCTION TO ANTHROPOLOGY	M	ITA	4	0	0	0	4	4	
2	MIN	MIN-CORE	23MRAP02M	CULTURAL EVOLUTION	M	CET	4	0	0	0	4	4	
3	MIN	MIN-CORE	23MRAP03M	HUMAN ECOLOGY AND ECOLOGICAL ANTHROPOLOGY	M	HEEA	4	0	0	0	4	4	
4	MIN	MIN-CORE	23MRAP04M	FUNDAMENTALS IN ARCHEOLOGICAL ANTHROPOLOGY CHRONOLOGY	M	FAAC	4	0	0	0	4	4	
5	MSDC	MIN-SDP	23MRAP05	SOCIAL CULTURE ANTHROPOLOGY	M	SCA	0	0	6	4	4	10	

4. MINOR: ARTS

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	MIN	MIN-CORE	23MRAR01	FUNDAMENTALS OF MACRO-ECONOMICS	R	FME	4	0	0	0	4	4	
2	MIN	MIN-CORE	23MRAR02	BASICS OF ECONOMETRICS	R	BOE	4	0	0	0	4	4	
3	MIN	MIN-CORE	23MRAR03	INTERNATIONAL RELATIONS: THEORIES AND CONCEPTS	R	IRTC	4	0	0	0	4	4	
4	MIN	MIN-CORE	23MRAR04	BEHAVIOURAL ECONOMICS AND POLICY	R	BEP	4	0	0	0	4	4	

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
5	MSDC	MIN-SDP	23MRAR05	SOCIAL GEOGRAPHY AND CARTOGRAPHY	R	SGC	0	0	6	4	4	10	

5. MINOR: BUSINESS ADMINISTRATION

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	MIN	MIN-CORE	23MRBA01M	ORGANIZATIONAL BEHAVIOUR	M	OBR	4	0	0	0	4	4	
2	MIN	MIN-CORE	23MRBA02M	MARKETING MANAGEMENT	M	MMG	4	0	0	0	4	4	
3	MIN	MIN-CORE	23MRBA03M	FINANCIAL MANAGEMENT	M	FMG	4	0	0	0	4	4	
4	MIN	MIN-CORE	23MRBA04M	HUMAN RESOURCE MANAGEMENT	M	HRM	4	0	0	0	4	4	
5	MSDC	MIN-SDP	23MRBA05M	LOGISTICS AND SUPPLY CHAIN MANAGEMENT	M	LSCM	0	0	6	4	4	10	

6. MINOR: COMPUTATIONAL MATHEMATICS

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	MIN	MIN-CORE	23MRCM01	LINEAR ALGEBRA	R	LAG	2	2	0	0	4	4	
2	MIN	MIN-CORE	23MRCM02	ADVANCED NUMERICAL ANALYSIS	R	ANA	2	2	0	0	4	4	
3	MIN	MIN-CORE	23MRCM03	MATHEMATICAL MODELLING	R	MML	3	0	2	0	4	5	
4	MIN	MIN-CORE	23MRCM04	STOCHASTIC PROCESSES & OPTIMIZATION	R	SPO	2	2	0	0	4	4	
5	MIN	MIN-CORE	23MRCM05	FUZZY MATHEMATICS AND ITS APPLICATIONS	R	FMAA	2	2	0	0	4	4	
6	MSDC	MIN-SDP	23MRCM06	TRANSFORM TECHNIQUES FOR ENGINEERING	R	TTE	2	0	2	4	4	8	
7	MSDC	MIN-SDP	23MRCM07	STATISTICS WITH R PROGRAMMING	R	SWRP	2	0	2	4	4	8	
8	MSDC	MIN-SDP	23MRCM08	MATRIX COMPUTATION	R	MCT	2	0	2	4	4	8	

7. MINOR: CORPORATE LAW

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	MIN	MIN-CORE	23MRCL01	PRINCIPLES OF COMPANIES ACT AND CORPORATE GOVERNANCE	R	PCACG	3	0	2	0	4	5	
2	MIN	MIN-CORE	23MRCL02	MERGERS, ACQUISITIONS AND PRIVATE EQUITY	R	MAPE	3	0	2	0	4	5	
3	MIN	MIN-CORE	23MRCL03	INSOLVENCY AND BANKRUPTCY LAW	R	IBL	3	0	2	0	4	5	
4	MIN	MIN-CORE	23MRCL04	COMPETITION LAW AND PRACTICE	R	CLP	3	0	2	0	4	5	
5	MSDC	MIN-SDP	23MRCL05	COMMERCIAL CONTRACT DRAFTING	R	CCD	0	0	6	4	4	10	

8. MINOR: CREATIVE ARTS

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	MIN	MIN-CORE	23MRCA01	INTRODUCTION TO CREATIVE ARTS	R	ICA	0	0	6	4	4	10	

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
2	MIN	MIN-CORE	23MRCA02	DRAWING AND SKETCHING	R	DAS	0	0	6	4	4	10	
3	MIN	MIN-CORE	23MRCA03	PAINTING	R	PNT	0	0	6	4	4	10	
4	MIN	MIN-CORE	23MRCA04	CREATIVE WRITING	R	CWT	0	0	6	4	4	10	
5	MSDC	MIN-SDP	23MRCA05	SCULPTURE	R	SCP	0	0	6	4	4	10	

9. MINOR: DIGITAL MARKETING AND AUTOMATION

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	MIN	MIN-CORE	23MRDM01M	FUNDAMENTALS OF DIGITAL MARKETING	M	FDM	4	0	0	0	4	4	
2	MIN	MIN-CORE	23MRDM02M	SOCIAL MEDIA MARKETING	M	SMM	4	0	0	0	4	4	
3	MIN	MIN-CORE	23MRDM03M	SERACH ENGINE OPTIMIZATION	M	SEO	4	0	0	0	4	4	
4	MIN	MIN-CORE	23MRDM04M	ARTIFICIAL INTELLIGENCE IN MARKETING	M	AIM	4	0	0	0	4	4	
5	MSDC	MIN-SDP	23MRDM05	CONTENT MARKETING	M	CM	0	0	6	4	4	10	

10. MINOR: FILM MAKING

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	MIN	MIN-CORE	23MRFM01	FILM AESTHETICS	R	FAS	0	0	6	4	4	10	
2	MIN	MIN-CORE	23MRFM02	SCREEN WRITING	R	SWT	0	0	6	4	4	10	
3	MIN	MIN-CORE	23MRFM03	CINEMATIC LIGHTING	R	CML	0	0	6	4	4	10	
4	MIN	MIN-CORE	23MRFM04	SOUND DESIGN	R	SDS	0	0	6	4	4	10	
5	MSDC	MIN-SDP	23MRFM05	DIGITAL PHOTOGRAPHY	R	DPG	0	0	6	4	4	10	

11. MINOR: FINTECH AND ALGORITHMIC TRADING

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	MIN	MIN-CORE	23MRFA01M	FOUNDATIONS AND APPLICATIONS OF FINANCIAL TECHNOLOGY	M	AFT	4	0	0	0	4	4	
2	MIN	MIN-CORE	23MRFA02M	DECENTRALISED FINANCE: THE FUTURE OF FINANCE	M	DFFF	4	0	0	0	4	4	
3	MIN	MIN-CORE	23MRFA03M	FINTECH INNOVATIONS	M	FIV	4	0	0	0	4	4	
4	MIN	MIN-CORE	23MRFA04M	USING MACHINE LEARNING IN TRADING AND FINANCE	M	ML	4	0	0	0	4	4	
5	MSDC	MIN-SDP	23MRFA05	TRADING ALGORITHMNS	M	TA	0	0	6	4	4	10	

12. MINOR: FOOD TECHNOLOGY

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	MIN	MIN-CORE	23MRFT01	INTRODUCTION TO FOOD SCIENCE AND TECHNOLOGY	R	IFST	3	0	2	0	4	5	

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
2	MIN	MIN-CORE	23MRFT02	FOOD ANALYSIS AND QUALITY ASSURANCE	R	FAQA	3	0	2	0	4	5	
3	MIN	MIN-CORE	23MRFT03	FOOD SAFETY AND REGULATIONS	R	FSR	3	1	0	0	4	4	
4	MIN	MIN-CORE	23MRFT04	FOOD ENGINEERING	R	FEG	3	0	2	0	4	5	
5	MSDC	MIN-SDP	23MRFT05	FOOD PACKAGING TECHNOLOGY	R	FPT	2	0	2	4	4	8	

13. MINOR: GRAPHIC DESIGN

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	MIN	MIN-CORE	23MRGD01	INTRODUCTION TO GRAPHIC DESIGN	R	IGD	0	0	6	4	4	10	
2	MIN	MIN-CORE	23MRGD02	TYPOGRAPHY	R	TPY	0	0	6	4	4	10	
3	MIN	MIN-CORE	23MRGD03	VISUAL IDENTITY AND BRANDING	R	VIB	0	0	6	4	4	10	
4	MIN	MIN-CORE	23MRGD04	PRINT DESIGN	R	PRDG	0	0	6	4	4	10	
5	MSDC	MIN-SDP	23MRGD05	PUBLICATION DESIGN	R	PUDG	0	0	6	4	4	10	

14. MINOR: INNOVATION MANAGEMENT AND ENTREPRENEURSHIP

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	MIN	MIN-CORE	23MRIM01	APPLIED DESIGN THINKING	R	ADT	3	0	2	0	4	5	
2	MIN	MIN-CORE	23MRIM02	BUSINESS MODEL INNOVATION	R	BMI	3	0	2	0	4	5	
3	MIN	MIN-CORE	23MRIM03	VENTURE MANAGEMENT	R	VMT	3	0	2	0	4	5	
4	MIN	MIN-CORE	23MRIM04	BUSINESS AND LEADERSHIP	R	BL	3	0	2	0	4	5	
5	MSDC	MIN-SDP	23MRIM05	PROJECT INNOVATE: STRATEGIES FOR ENTREPRENEURIAL GROWTH	R	PISEG	0	0	6	4	4	10	

15. MINOR: INTELLECTUAL PROPERTY LAW

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	MIN	MIN-CORE	23MRIP01	EMERGENCE AND DEVELOPMENT OF IPR	R	EDIPR	3	0	2	0	4	5	
2	MIN	MIN-CORE	23MRIP02	LAW OF COPYRIGHT	R	LOCR	3	0	2	0	4	5	
3	MIN	MIN-CORE	23MRIP03	LAW OF PATENTS	R	LOPS	3	0	2	0	4	5	
4	MIN	MIN-CORE	23MRIP04	LAW OF TRADEMARK	R	LOTM	3	0	2	0	4	5	
5	MSDC	MIN-SDP	23MRIP05	LAW OF DESIGNS, GEOGRAPHICAL INDICATIONS AND PLANT VARIETIES	R	LODGP	0	0	6	4	4	10	

16. MINOR: INTERIOR DESIGN

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	MIN	MIN-CORE	23MRID01	INTRODUCTION TO INTERIOR DESIGN	R	IID	3	0	2	0	4	5	
2	MIN	MIN-CORE	23MRID02	SPACE PLANNING AND LAYOUT	R	SPL	3	0	2	0	4	5	

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
3	MIN	MIN-CORE	23MRID03	INTERIOR MATERIAL AND FINISHES	R	IMF	3	0	2	0	4	5	
4	MIN	MIN-CORE	23MRID04	FURNITURE AND ACCESSORIES DESIGN	R	FAD	3	0	2	0	4	5	
5	MSDC	MIN-SDP	23MRID05	DESIGN PRESENTATION AND COMMUNICATION	R	DPC	0	0	6	4	4	10	

17. MINOR: INTERNATIONAL BUSINESS

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	MIN	MIN-CORE	23MRIB01M	INTERNATIONAL BUSINESS ESSENTIALS	M	IBET	4	0	0	0	4	4	
2	MIN	MIN-CORE	23MRIB02M	INTERNATIONAL BUSINESS ENVIRONMENT	M	IBEV	4	0	0	0	4	4	
3	MIN	MIN-CORE	23MRIB03M	INTERNATIONAL BUSINESS OPERATIONS	M	IBO	4	0	0	0	4	4	
4	MIN	MIN-CORE	23MRIB04M	INTERNATIONAL BUSINESS MODELS	M	IBM	4	0	0	0	4	4	
5	MSDC	MIN-SDP	23MRIB05	PROFESSIONAL SKILLS FOR INTERNATIONAL BUSINESS	M	PIB	0	0	6	4	4	10	

18. MINOR: ORGANIZATIONAL BEHAVIOUR

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	MIN	MIN-CORE	23MROB01M	ORGANIZATIONAL BEHAVIOUR	M	POM	4	0	0	0	4	4	
2	MIN	MIN-CORE	23MROB02M	ORGANIZATIONAL BEHAVIOUR MODELS	M	OBM	4	0	0	0	4	4	
3	MIN	MIN-CORE	23MROB03M	ORGANIZATIONAL CHANGE AND DEVELOPMENT	M	OCD	4	0	0	0	4	4	
4	MIN	MIN-CORE	23MROB04M	LEADERSHIP IN ORGANIZATION	M	LIO	4	0	0	0	4	4	
5	MSDC	MIN-SDP	23MROB05	TEAM BUIDLING AND COLLABORATION SKILLS FOR MANAGERS	M	TBCM	0	0	6	4	4	10	

19. MINOR: PHARMACOLOGY

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	MIN	MIN-CORE	23MRPY01	BASICS OF PHARMACOLOGY	R	BOP	3	0	2	0	4	5	
2	MIN	MIN-CORE	23MRPY02	ADVANCED PHARMACOLOGY - I	R	AP-I	3	0	2	0	4	5	
3	MIN	MIN-CORE	23MRPY03	EXPERIMENTAL PHARMACOLOGY	R	EPC	3	0	2	0	4	5	
4	MIN	MIN-CORE	23MRPY04	ADVANCED PHARMACOLOGY - II	R	AP-II	3	0	2	0	4	5	
5	MSDC	MIN-SDP	23MRPY05	PRACTICAL PHARMACOLOGY	R	PRI	0	0	6	4	4	10	

20. MINOR: PRECISION AGRICULTURE

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	MIN	MIN-CORE	23MRPA01	FUNDAMENTALS OF AGRONOMY	R	FOA	3	0	2	0	4	5	

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
2	MIN	MIN-CORE	23MRPA02	INTRODUCTORY AGROMETEOROLOGY AND CLIMATE CHANGE	R	IACC	3	0	2	0	4	5	
3	MIN	MIN-CORE	23MRPA03	GEOINFORMATICS AND NANOTECHNOLOGY FOR PRECISION FARMING	R	GNPF	3	0	2	0	4	5	
4	MIN	MIN-CORE	23MRPA04	PROTECTED CULTIVATION AND POST-HARVEST TECHNOLOGIES	R	PCPHT	3	0	2	0	4	5	
5	MSDC	MIN-SDP	23MRPA05	MODERN FARMING TECHNIQUES AND INNOVATIONS	R	AELP	0	0	6	4	4	10	

21. MINOR: TECHNOPRENEURSHIP

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	MIN	MIN-CORE	23MRTP01	INTRODUCTION TO TECHNOLOGICAL ENTREPRENEURSHIP	R	ITEP	3	0	2	0	4	5	
2	MIN	MIN-CORE	23MRTP02	LEAN STARTUP LAUNCHPAD	R	LSLP	3	0	2	0	4	5	
3	MIN	MIN-CORE	23MRTP03	ENTREPRENEURSHIP AND VENTURE CAPITAL	R	EVC	3	0	2	0	4	5	
4	MIN	MIN-CORE	23MRTP04	BUILDING AND SUSTAINING A SUCCESSFUL ENTERPRISE	R	BSSE	3	0	2	0	4	5	
5	MSDC	MIN-SDP	23MRTP05	TECH INNOVATION PROJECT FOR ENTREPRENEURS	R	TIPFE	0	0	6	4	4	10	

22. MINOR: WATER SCIENCE AND MANAGEMENT

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	MIN	MIN-CORE	23MRPA01	FUNDAMENTALS OF AGRONOMY	R	FOA	3	0	2	0	4	5	
2	MIN	MIN-CORE	23MRWS02	SOIL AND WATER CONSERVATION ENGINEERING	R	SWCE	3	0	2	0	4	5	
3	MIN	MIN-CORE	23MRWS04	RAINFED AGRICULTURE AND WATERSHED MANAGEMENT	R	RAWM	3	0	2	0	4	5	
4	MSDC	MIN-SDP	23MRPA05	MODERN FARMING TECHNIQUES AND INNOVATIONS	R	AELP	0	0	6	4	4	10	

23. DOUBLE MAJOR: SECOND MAJOR EEE FOR FIRST MAJOR CSE / CSIT / ME

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	FCC	FC-1	23EE2102F	ELECTRICAL MACHINES	F	ELM	2	0	2	0	3	4	
2	FCC	FC-1	23EE2225F	POWER QUALITY	F	PQ	3	0	0	0	3	3	
3	HFC	HFC-CORE	23EC2106R	PROCESSORS AND CONTROLLERS	R	PRC	3	0	2	0	4	5	
4	OEC	OE-1	23OEEE03	ELECTRICAL POWER ENGINEERING	R	EPE	4	0	0	0	4	4	
5	OEC	OE-2	23EE2207R	CONTROL SYSTEMS	R	CS	3	0	2	0	4	5	
6	OEC	OE-3	23EE2205R	POWER ELECTRONICS	R	PES	3	0	2	0	4	5	

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
7	SMFC	SMFC-	23EE2102R	ELECTRICAL MACHINES	R	ELM	3	0	2	0	4	5	
8	SMFC	SMFC-CORE	23EE2204R	ELECTRICAL POWER GENERATION, TRANSMISSION & DISTRIBUTION	R	EPGTD	2	1	0	0	3	3	
9	SMFC	SMFC-CORE	23EE3106R	POWER SYSTEM ANALYSIS & STABILITY	R	PSAS	2	1	0	0	3	3	
10	SMFC	SMFC-CORE	23EE3208R	POWER SYSTEM PROTECTION & CONTROL	R	PSPC	2	0	2	2	3.5	6	
11	SMFC	SMFC-	23EE2101	ELECTRICAL CIRCUITS	R	ELC	2	0	2	0	3	4	

24. DOUBLE MAJOR: SECOND MAJOR ECE FOR FIRST MAJOR CSE / CSIT

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	FCC	FC-1	23EC2228F	BIOMEDICAL ELECTRONICS & IOT FOR HEALTHCARE	F	BEIH	2	0	2	0	3	4	
2	FCC	FC-1	23EC2226F	WIRELESS COMMUNICATIONS	F	WC	2	0	2	0	3	4	
3	FCC	FC-1	23EC2221F	EMBEDDED SYSTEM DESIGN	F	ESD	2	0	2	0	3	4	
4	FCC	FC-1	23EC2229F	WIRELESS SENSOR NETWORKS	F	WSN	2	0	2	0	3	4	
5	FCC	FC-1	23EC2224F	DEEP NETWORK ARCHITECTURES	F	DNA	2	0	2	0	3	4	
6	FCC	FC-1	23EC2225F	RADIATING SYSTEMS & WAVE PROPAGATION	F	RSWP	2	0	2	0	3	4	
7	FCC	FC-1	23EC2230F	LOW POWER VLSI CIRCUITS	F	LPVC	2	0	2	0	3	4	
8	OEC	OE-1	23EC2105R	SIGNALS & COMMUNICATION SYSTEMS	R	SCS	3	0	2	0	4	5	
9	OEC	OE-2	23EC2208R	DIGITAL COMMUNICATION	R	DC	3	0	2	0	4	5	
10	OEC	OE-3	23EC3112R	DISCRETE TIME SIGNAL PROCESSING	R	DTSP	3	0	2	0	4	5	
11	SMFC	SMFC-	23EC02HF	SENSORS AND TRANSDUCERS	R	SAT	3	0	2	0	4	5	
12	SMFC	SMFC-	23EC03HF	PEER TO PEER NETWORKS	R	PTPN	3	0	2	0	4	5	
13	SMFC	SMFC-CORE	23EC2209R	ELECTROMAGNETIC WAVES & TRANSMISSION LINES	R	EMWTL	3	0	0	0	3	3	
14	SMFC	SMFC-CORE	23EC01HF	VLSI PHYSICAL DESIGN AUTOMATION	R	VPDA	3	0	2	0	4	5	
15	SMFC	SMFC-CORE	23EC2104R	ANALOG ELECTRONIC CIRCUIT DESIGN	R	AECD	3	0	2	2	4.5	7	
16	SMFC	SMFC-	23EC2211R	VLSI DESIGN	R	VLSID	3	0	2	2	4.5	7	

25. DOUBLE MAJOR: SECOND MAJOR IOT FOR FIRST MAJOR CSE / CSIT / EEE / ECE / AIDS

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	FCC	FC-1	23IN2222F	CLOUD COMPUTING FOR IOT	F	CCIOT	2	0	2	0	3	4	
2	FCC	FC-1	23IN3104F	REAL TIME OPERATING SYSTEMS	F	RTOS	2	0	2	0	3	4	

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
3	OEC	OE-1	23IN2101R	IOT SYSTEM DESIGN	R	IOTPA	3	0	2	0	4	5	
4	OEC	OE-2	23IN2102R	ELECTRONIC DEVICES AND INTEGRATED CIRCUITS	R	EDAIC	3	0	2	0	4	5	
5	OEC	OE-3	23IN2203R	WIRELESS TECHNOLOGIES FOR IOT	R	WTIOT	3	0	2	0	4	5	
6	SMFC	SMFC-	23IN01HF	5G TECHNOLOGIES FOR IOT	R	5TAI	3	0	2	0	4	5	
7	SMFC	SMFC-CORE	23IN02HF	DEEP LEARNING FOR COMPUTER VISION	R	DPNV	3	0	2	0	4	5	
8	SMFC	SMFC-CORE	23IN03HF	AR & VR APPLICATION DEVELOPMENT	R	ARVRAD	3	0	2	0	4	5	
9	SMFC	SMFC-	23IN04HF	DEVOPS FOR IOT	R	DFIOT	3	0	2	0	4	5	
10	SMFC	SMFC-	23IN3104R	REAL TIME OPERATING SYSTEMS	R	RTOS	3	0	2	0	4	5	
11	SMFC	SMFC-	23IN3105R	EMBEDDED SYSTEMS DESIGN	R	ESD	2	0	2	0	3	4	

26. DOUBLE MAJOR: SECOND MAJOR AIDS FOR FIRST MAJOR ECE / IOT / CSE / CSIT

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	FCC	FC-1	23AD2224F	CLOUD COMPUTING	F	CC	2	1	0	0	3	3	
2	OEC	OE-1	23AD2204R	DATA MANAGEMENT AND WAREHOUSING	R	DMW	3	0	2	0	4	5	
3	OEC	OE-2	23AD3106R	BIG DATA ENGINEERING	R	BDE	3	0	2	0	4	5	
4	OEC	OE-3	23AD3207R	NATURAL LANGUAGE PROCESSING	R	NLP	3	0	2	0	4	5	
5	SMFC	SMFC-CORE	23AD01HF	MACHINE LEARNING FOR HEALTHCARE	R	MLH	3	0	2	0	4	5	
6	SMFC	SMFC-	23AD03HF	REINFORCEMENT LEARNING	R	RIL	3	0	2	0	4	5	
7	SMFC	SMFC-	23AD04HF	EXPLAINABLE AI	R	EAI	3	0	2	0	4	5	
8	SMFC	SMFC-	23AD2205R	DEEP LEARNING	R	DL	3	0	2	4	5	9	
9	SMFC	SMFC-CORE	23AD4108	ETHICS FOR ARTIFICIAL INTELLIGENCE	R	EAI	1	0	0	0	1	1	
10	SMFC	SMFC-CORE	23AD2103R	SYSTEM DESIGN & INTRODUCTION TO CLOUD	R	SDC	3	0	2	0	4	5	
11	SMFC	SMFC-CORE	23AD02HF	DATA ENGINEERING AND ARCHITECTURE	R	DEA	3	0	2	0	4	5	

27. DOUBLE MAJOR: SECOND MAJOR ME FOR FIRST MAJOR ECE / EEE / CSE / CSIT / AIDS / BT / IOT

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	FCC	FC-1	23ME2107F	THERMODYNAMICS	F	TD	2	1	0	0	3	3	
2	FCC	FC-1	23ME1001F	ENGINEERING MECHANICS	F	EM	2	1	0	0	3	3	
3	OEC	OE-1	23ME2106R	MECHANICS OF SOLIDS	R	MOS	3	0	2	0	4	5	

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
4	OEC	OE-2	23ME2116	FLUID MECHANICS & HYDRAULIC MACHINES	R	FMHM	3	0	2	0	4	5	
5	OEC	OE-3	23ME2208	MANUFACTURING PROCESSES	R	MP	3	0	2	0	4	5	
6	SMFC	SMFC-	23ME2107R	THERMODYNAMICS	R	TD	3	0	0	0	3	3	
7	SMFC	SMFC-	23ME3110R	HEAT TRANSFER	R	HT	3	0	2	0	4	5	
8	SMFC	SMFC-	23ME1002	ENGINEERING GRAPHICS	R	EG	0	0	4	0	2	4	
9	SMFC	SMFC-CORE	23ME2209R	KINEMATICS & DYNAMICS OF MACHINES	R	KDOM	3	1	2	0	5	6	
10	SMFC	SMFC-	23ME3111R	MECHANICAL ENGINEERING DESIGN	R	MED	3	0	0	0	3	3	
11	SMFC	SMFC-CORE	23ME1004	WORKSHOP PRACTICES FOR ENGINEERS	R	WPE	0	0	4	0	2	4	
12	SMFC	SMFC-	23ME3113R	MANUFACTURING TECHNOLOGY	R	MT	2	0	2	0	3	4	

28. DOUBLE MAJOR: SECOND MAJOR CIVIL FOR FIRST MAJOR ECE / EEE / CSE / CSIT / AIDS / BT / IOT

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	FCC	FC-1	23ME1001F	ENGINEERING MECHANICS	F	EM	2	1	0	0	3	3	
2	FCC	FC-1	23CE2102F	FLUID MECHANICS & HYDRAULICS	F	FMH	2	0	2	0	3	4	
3	OEC	OE-1	23CE2208R	GEOTECHNICAL ENGINEERING	R	GTE	3	0	2	0	4	5	
4	OEC	OE-2	23CE2209R	TRANSPORTATION ENGINEERING	R	TPE	3	0	2	0	4	5	
5	OEC	OE-3	23CE3109	ENVIRONMENTAL ENGINEERING	R	EVE	3	0	2	0	4	5	
6	SMFC	SMFC-	23CE2103R	SURVEYING	R	SVY	3	0	2	4	5	9	
7	SMFC	SMFC-	23CE2206	STRUCTURAL ANALYSIS	R	STA	3	1	0	0	4	4	
8	SMFC	SMFC-	23CE2207	CONCRETE TECHNOLOGY	R	CT	3	0	2	4	5	9	
9	SMFC	SMFC-	22CE1201	ENGINEERING GEOLOGY	R	EGY	3	0	2	0	4	5	
10	SMFC	SMFC-	23CE2102R	FLUID MECHANICS & HYDRAULICS	R	FMH	3	0	2	0	4	5	

29. DOUBLE MAJOR: SECOND MAJOR BT FOR FIRST MAJOR B.TECH / BBA / B.COM / BSC / BCA / BA

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	BSC	BSC-SE-3	23BT2103	BIOCHEMISTRY	R	BCM	3	0	2	0	4	5	
2	FCC	FC-1	23BT2224F	BIOREACTOR OPERATIONS	F	BO	2	0	2	0	3	4	
3	SDC	SDP-1	23SDBT01A	MEDICAL LAB TECHNOLOGY	A	MLT	0	0	6	4	4	10	
4	SDC	SDP-1	23SDBT01E	MEDICAL LAB TECHNOLOGY	E	MLT	0	0	6	4	4	10	
5	SDC	SDP-1	23SDBT01R	MEDICAL LAB TECHNOLOGY	R	MLT	0	0	2	4	2	6	
6	OEC	OE-1	23BT2209R	BIOCHEMICAL REACTION ENGINEERING	R	BCRE	3	0	2	0	4	5	

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
7	OEC	OE-2	23BT3212	PLANT AND ANIMAL BIOTECHNOLOGY	R	PABT	3	0	2	0	4	5	
8	OEC	OE-3	23BT3213R	DOWNSTREAM PROCESSING	R	DSP	2	0	4	0	4	6	
9	SMFC	SMFC-	23BT2104	MICROBIOLOGY	R	MBG	3	0	2	0	4	5	
10	SMFC	SMFC-	23BT2215	PROCESS ENGINEERING PRINCIPLES	R	PEP	2	0	2	0	3	4	
11	SMFC	SMFC-	23BT3214	BIOINFORMATICS	R	BI	2	0	2	0	3	4	
12	SMFC	SMFC-	23BT1101	CELL BIOLOGY	R	CB	3	0	0	0	3	3	
13	SMFC	SMFC-	23BT2105R	IMMUNOLOGY	R	IMM	3	0	2	0	4	5	
14	SMFC	SMFC-	23BT2206	MOLECULAR BIOLOGY	R	MBY	3	0	0	0	3	3	

33. DOUBLE MAJOR: SECOND MAJOR FOOD TECHNOLOGY FOR FIRST MAJOR B.TECH / B.COM / BBA / BSC / BCA / BA

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	OEC	OE-1	22FT2212R	PROCESSING OF AQUATIC FOODS	R	PAF	3	0	2	0	4	5	
2	OEC	OE-2	22FT22C1R	FOOD ANALYSIS AND QUALITY ASSURANCE	R	FAQA	3	0	2	0	4	5	
3	OEC	OE-3	22FT3215R	TECHNOLOGY OF MEAT AND POULTRY	R	TMP	3	0	2	0	4	5	
4	SMFC	SMFC-	22FT1102	FOOD CHEMISTRY	R	FC	3	0	2	0	4	5	
5	SMFC	SMFC-	22FT1105	FOOD BIOCHEMISTRY	R	FBC	3	0	2	0	4	5	
6	SMFC	SMFC-CORE	22FT2109R	PROCESSING OF HORTICULTURAL PRODUCE	R	PHP	3	0	2	0	4	5	
7	SMFC	SMFC-CORE	22FT12C1	PRINCIPLES OF FOOD PRESERVATION	R	PFP	1	0	0	4	2	5	
8	SMFC	SMFC-	22FT32C2R	FOOD PACKAGING TECHNOLOGY	R	FPT	1	0	0	4	2	5	
9	SMFC	SMFC-	22FT1207R	FOOD MICROBIOLOGY	R	FMB	3	0	2	0	4	5	
10	SMFC	SMFC-CORE	22FT2213R	BAKERY, CONFECTIONERY AND SNACKS TECHNOLOGY	R	BCST	2	0	4	0	4	6	
11	SMFC	SMFC-CORE	22FT1104	INTRODUCTION TO FOOD SCIENCE AND TECHNOLOGY	R	IFST	3	0	2	0	4	5	

34. DOUBLE MAJOR: SECOND MAJOR COMMERCE FOR FIRST MAJOR B.TECH / BBA / BSC / BCA / BA

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	OEC	OE-1	23ACCABT	BUSINESS TECHNOLOGY	R	BT	3	0	0	4	4	7	
2	OEC	OE-2	23ACCAFA	FINANCIAL ACCOUNTING	R	FA	2	2	0	4	5	8	
3	OEC	OE-3	23ACCAMA	MANAGEMENT ACCOUNTING	R	MA	2	2	0	4	5	8	
4	SMFC	SMFC-	23ACCAPM	PERFORMANCE MANAGEMENT	R	PM	2	2	0	4	5	8	
5	SMFC	SMFC-	22CM1206	ADVANCED COST ACCOUNTING	R	ACA	3	2	0	0	5	5	

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
6	SMFC	SMFC-CORE	22CM31B5	ACCOUNTING & REPORTING STANDARDS	R	ARS	3	2	0	0	5	5	
7	SMFC	SMFC-	23ACCATX	TAXATION	R	TXT	2	2	0	4	5	8	
8	SMFC	SMFC-	22CM21B2	INCOME TAX LAW & PRACTICE	R	ITLP	3	2	0	0	5	5	
9	SMFC	SMFC-	22CM1203	DATA ANALYSIS WITH EXCEL	R	DAWE	0	0	0	4	1	4	

35. SPECIALIZATION: AGRI-BIOTECHNOLOGY

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDBT10A	MASS SPECTROMETRY BASED PROTEOMICS	A	MSBP	0	0	6	4	4	10	
2	SDC	SDP-4	23SDBT10E	MASS SPECTROMETRY BASED PROTEOMICS	E	MSBP	0	0	6	4	4	10	
3	SDC	SDP-4	23SDBT10R	MASS SPECTROMETRY BASED PROTEOMICS	R	MSBP	0	0	2	4	2	6	
4	PEC	PE-1	23ABT3101A	BIOSENSORS AND BIOELECTRONICS	A	BSBE	4	0	4	4	7	12	GN(1)
5	PEC	PE-1	23ABT3101E	BIOSENSORS AND BIOELECTRONICS	E	BSBE	4	0	4	4	7	12	GN(1)
6	PEC	PE-1	23ABT3101R	BIOSENSORS AND BIOELECTRONICS	R	BSBE	3	0	2	4	5	9	GN(1)
7	PEC	PE-2	23ABT3202A	IOT APPLICATIONS IN PRECISION AGRICULTURE	A	IOTAPA	5	0	0	0	5	5	GN(1)
8	PEC	PE-2	23ABT3202E	IOT APPLICATIONS IN PRECISION AGRICULTURE	E	IOTAPA	5	0	0	0	5	5	GN(1)
9	PEC	PE-2	23ABT3202R	IOT APPLICATIONS IN PRECISION AGRICULTURE	R	IOTAPA	3	0	0	0	3	3	GN(1)
10	PEC	PE-2	23ABT3203A	MACHINE LEARNING FOR PRECISION AGRICULTURE	A	MLPA	5	0	0	0	5	5	GN(1)
11	PEC	PE-2	23ABT3203E	MACHINE LEARNING FOR PRECISION AGRICULTURE	E	MLPA	5	0	0	0	5	5	GN(1)
12	PEC	PE-2	23ABT3203R	MACHINE LEARNING FOR PRECISION AGRICULTURE	R	MLPA	3	0	0	0	3	3	GN(1)
13	PEC	PE-3	23ABT3304A	IMAGE PROCESSING FOR AGRICULTURAL APPLICATIONS	A	IPAA	4	0	4	4	7	12	GN(1)
14	PEC	PE-3	23ABT3304E	IMAGE PROCESSING FOR AGRICULTURAL APPLICATIONS	E	IPAA	4	0	4	4	7	12	GN(1)
15	PEC	PE-3	23ABT3304R	IMAGE PROCESSING FOR AGRICULTURAL APPLICATIONS	R	IPAA	3	0	2	4	5	9	GN(1)
16	PEC	PE-4	23ABT3405M	BIOINSTRUMENTATION	M	BIM	4	0	0	0	4	4	GN(1)
17	PEC	PE-4	23ABT3406M	DATA ACQUISITION AND COMMUNICATION SYSTEMS FOR AGRICULTURE	M	DACSA	3	0	0	0	3	3	GN(1)
18	PEC	PE-5	23ABT3507	MICROFLUIDICS FOR AGRICULTURAL APPLICATIONS	R	MAP	3	0	0	0	3	3	GN(1)

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
19	PEC	PE-5	23ABT3508	BIOINFORMATICS FOR AGRICULTURE	R	BIA	3	0	0	0	3	3	GN(1)

36. SPECIALIZATION: AI AND AUTONOMOUS SYSTEMS

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDEE06A	EVT HARDWARE PROTOTYPING	A	EVTHP	0	0	6	4	4	10	
2	SDC	SDP-4	23SDEE06E	EVT HARDWARE PROTOTYPING	E	EVTHP	0	0	6	4	4	10	
3	SDC	SDP-4	23SDEE06R	EVT HARDWARE PROTOTYPING	R	EVTHP	0	0	2	4	2	6	
4	SDC	SDP-4	23SDAD09A	AUTONOMOUS VEHICLE SYSTEMS	A	AVS	0	0	6	4	4	10	AIML(1)
5	SDC	SDP-4	23SDAD09E	AUTONOMOUS VEHICLE SYSTEMS	E	AVS	0	0	6	4	4	10	AIML(1)
6	SDC	SDP-4	23SDAD09R	AUTONOMOUS VEHICLE SYSTEMS	R	AVS	0	0	2	4	2	6	AIML(1)
7	PEC	PE-1	23ASS3101A	AUTONOMOUS DRIVER ASSISTIVE SYSTEMS	A	ADAS	4	0	4	4	7	12	AIML(1) Rule:1
8	PEC	PE-1	23ASS3101E	AUTONOMOUS DRIVER ASSISTIVE SYSTEMS	E	ADAS	4	0	4	4	7	12	AIML(1) Rule:1
9	PEC	PE-1	23ASS3101R	AUTONOMOUS DRIVER ASSISTIVE SYSTEMS	R	ADAS	3	0	2	4	5	9	AIML(1) Rule:1
10	PEC	PE-1	23ASS3102A	AI AND CONTROLS FOR AUTONOMOUS FUTURE VEHICLE	A	AICAFT	4	0	4	4	7	12	
11	PEC	PE-1	23ASS3102E	AI AND CONTROLS FOR AUTONOMOUS FUTURE VEHICLE	E	AICAFT	4	0	4	4	7	12	
12	PEC	PE-1	23ASS3102R	AI AND CONTROLS FOR AUTONOMOUS FUTURE VEHICLE	R	AICAFT	3	0	2	4	5	9	
13	PEC	PE-2	23ASS3202A	INTRODUCTION TO INTELLIGENT DRONES	A	IID	5	0	0	0	5	5	AIML(1) Rule:1
14	PEC	PE-2	23ASS3202E	INTRODUCTION TO INTELLIGENT DRONES	E	IID	5	0	0	0	5	5	AIML(1) Rule:1
15	PEC	PE-2	23ASS3202R	INTRODUCTION TO INTELLIGENT DRONES	R	IID	3	0	0	0	3	3	AIML(1) Rule:1
16	PEC	PE-2	23ASS3203A	AI AND IOT FOR AUTONOMOUS VEHICLE	A	AIAV	5	0	0	0	5	5	AIML(1) Rule:1
17	PEC	PE-2	23ASS3203E	AI AND IOT FOR AUTONOMOUS VEHICLE	E	AIAV	5	0	0	0	5	5	AIML(1) Rule:1
18	PEC	PE-2	23ASS3203R	AI AND IOT FOR AUTONOMOUS VEHICLE	R	AIAV	3	0	0	0	3	3	AIML(1) Rule:1
19	PEC	PE-2	23ASS3205A	AUTONOMOUS VEHICLE ECU DESIGN WITH AUTOSAR	A	AVECUDA	3	0	4	0	5	7	
20	PEC	PE-2	23ASS3205E	AUTONOMOUS VEHICLE ECU DESIGN WITH AUTOSAR	E	AVECUDA	3	0	4	0	5	7	
21	PEC	PE-2	23ASS3205R	AUTONOMOUS VEHICLE ECU DESIGN WITH AUTOSAR	R	AVECUDA	2	0	2	0	3	4	
22	PEC	PE-3	23ASS3304A	APPLIED DEEP LEARNING FOR AUTONOMOUS SYSTEMS	A	ADLAS	4	0	4	4	7	12	AIML(1) Rule:1

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
23	PEC	PE-3	23ASS3304E	APPLIED DEEP LEARNING FOR AUTONOMOUS SYSTEMS	E	ADLAS	4	0	4	4	7	12	AIML(1) Rule:1
24	PEC	PE-3	23ASS3304R	APPLIED DEEP LEARNING FOR AUTONOMOUS SYSTEMS	R	ADLAS	3	0	2	4	5	9	AIML(1) Rule:1
25	PEC	PE-3	23ASS3309A	AUTONOMOUS ELECTRIC MOBILITY - SYSTEMS AND APPLICATIONS	A	AAAV	4	0	4	4	7	12	
26	PEC	PE-3	23ASS3309E	AUTONOMOUS ELECTRIC MOBILITY - SYSTEMS AND APPLICATIONS	E	AAAV	4	0	4	4	7	12	
27	PEC	PE-3	23ASS3309R	AUTONOMOUS ELECTRIC MOBILITY - SYSTEMS AND APPLICATIONS	R	AAAV	3	0	2	4	5	9	
28	PEC	PE-4	23ASS3405M	EXPERT SYSTEMS	M	ES	3	0	0	0	3	3	AIML(1) Rule:1
29	PEC	PE-4	23ASS3406M	SELF-DRIVING CARS	M	SDC	3	0	0	0	3	3	AIML(1) Rule:1
30	PEC	PE-4	23ASS3410M	CYBERSECURITY IN AUTONOMOUS VEHICLE	M	CAV	3	0	0	0	3	3	
31	PEC	PE-5	23ASS3507	LOCALIZATION & PROGRAMMING REAL-TIME AUTONOMOUS SYSTEMS	R	LPRAS	3	0	0	0	3	3	
32	PEC	PE-5	23ASS3508	REINFORCEMENT LEARNING FOR AUTONOMOUS SYSTEM	R	RLA	3	0	0	0	3	3	AIML(1) Rule:1
33	PEC	PE-5	23ASS3511	AI DRIVEN ECU DESIGN FOR AUTONOMOUS VEHICLES	R	AEDATT	2	0	2	0	3	4	

37. SPECIALIZATION: AI FOR COMPUTATIONAL INTELLIGENCE

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDCS17A	GENERATIVE ADVERSARIAL NETWORKS FOR IMAGES	A	GANI	0	0	6	4	4	10	AIML(1) Rule:1
2	SDC	SDP-4	23SDCS17E	GENERATIVE ADVERSARIAL NETWORKS FOR IMAGES	E	GANI	0	0	6	4	4	10	AIML(1) Rule:1
3	SDC	SDP-4	23SDCS17R	GENERATIVE ADVERSARIAL NETWORKS FOR IMAGES	R	GANI	0	0	2	4	2	6	AIML(1) Rule:1
4	PEC	PE-1	23ACI3101A	GENERATIVE ADVERSARIAL NETWORKS	A	GANS	4	0	4	4	7	12	AIML(1) Rule:1
5	PEC	PE-1	23ACI3101E	GENERATIVE ADVERSARIAL NETWORKS	E	GANS	4	0	4	4	7	12	AIML(1) Rule:1
6	PEC	PE-1	23ACI3101R	GENERATIVE ADVERSARIAL NETWORKS	R	GANS	3	0	2	4	5	9	AIML(1) Rule:1
7	PEC	PE-2	23ACI3202A	EXPLAINABLE AI AND COMMUNICATIONS	A	EAIC	5	0	0	0	5	5	AIML(1) Rule:1
8	PEC	PE-2	23ACI3202E	EXPLAINABLE AI AND COMMUNICATIONS	E	EAIC	5	0	0	0	5	5	AIML(1) Rule:1
9	PEC	PE-2	23ACI3202R	EXPLAINABLE AI AND COMMUNICATIONS	R	EAIC	3	0	0	0	3	3	AIML(1) Rule:1
10	PEC	PE-3	23ACI3303A	FUZZY LOGIC WITH SET THEORY	A	FLST	4	0	4	4	7	12	AIML(1) Rule:1

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
11	PEC	PE-3	23ACI3303E	FUZZY LOGIC WITH SET THEORY	E	FLST	4	0	4	4	7	12	AIML(1) Rule:1
12	PEC	PE-3	23ACI3303R	FUZZY LOGIC WITH SET THEORY	R	FLST	3	0	2	4	5	9	AIML(1) Rule:1
13	PEC	PE-4	23ACI3404M	HEURISTIC ALGORITHMS	M	HAG	3	0	0	0	3	3	AIML(1) Rule:1
14	PEC	PE-4	23ACI3405M	SWARM INTELLIGENCE APPROCAHES	M	SIA	3	0	0	0	3	3	AIML(1) Rule:1
15	PEC	PE-5	23ACI3506	DATA MANIPULATION	R	DMP	3	0	0	0	3	3	AIML(1) Rule:1
16	PEC	PE-5	23ACI3507	REINFORCEMENT LEARNING	R	RLG	3	0	0	0	3	3	AIML(1) Rule:1
17	PEC	PE-5	23ACI3508	EVOLUTIONARY COMPUTATION	R	ECP	3	0	0	0	3	3	AIML(1)

38. SPECIALIZATION: AI SYSTEMS FOR VISUAL INTELLIGENCE

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDCS19A	VISUAL RECOGNITION AND SCENE UNDERSTANDING	A	VRSU	0	0	6	4	4	10	AIML(1) Rule:1
2	SDC	SDP-4	23SDCS19E	VISUAL RECOGNITION AND SCENE UNDERSTANDING	E	VRSU	0	0	6	4	4	10	AIML(1) Rule:1
3	SDC	SDP-4	23SDCS19R	VISUAL RECOGNITION AND SCENE UNDERSTANDING	R	VRSU	0	0	2	4	2	6	AIML(1) Rule:1
4	PEC	PE-1	23AVI3101A	DEEP LEARNING	A	DL	4	0	4	4	7	12	AIML(1) Rule:1
5	PEC	PE-1	23AVI3101E	DEEP LEARNING	E	DL	4	0	4	4	7	12	AIML(1) Rule:1
6	PEC	PE-1	23AVI3101R	DEEP LEARNING	R	DL	3	0	2	4	5	9	AIML(1) Rule:1
7	PEC	PE-2	23AVI3202A	COMPUTER VISION	A	CV	5	0	0	0	5	5	AIML(1) Rule:1
8	PEC	PE-2	23AVI3202E	COMPUTER VISION	E	CV	5	0	0	0	5	5	AIML(1) Rule:1
9	PEC	PE-2	23AVI3202R	COMPUTER VISION	R	CV	3	0	0	0	3	3	AIML(1) Rule:1
10	PEC	PE-3	23AVI3303A	OBJECT DETECTION	A	OD	4	0	4	4	7	12	AIML(1) Rule:1
11	PEC	PE-3	23AVI3303E	OBJECT DETECTION	E	OD	4	0	4	4	7	12	AIML(1) Rule:1
12	PEC	PE-3	23AVI3303R	OBJECT DETECTION	R	OD	3	0	2	4	5	9	AIML(1) Rule:1
13	PEC	PE-4	23AVI3404M	VISUAL RECOGNITION AND SCENE UNDERSTANDING	M	VRASU	3	0	0	0	3	3	AIML(1)
14	PEC	PE-4	23AVI3405M	DEEP REINFORCEMENT LEARNING FOR VISION	M	DRLV	3	0	0	0	3	3	AIML(1) Rule:1

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
15	PEC	PE-5	23AVI3506	HUMAN COMPUTER INTERACTION	R	HCI	3	0	0	0	3	3	AIML(1) Rule:1
16	PEC	PE-5	23AVI3507	OBJECT LOCALIZATION	R	OL	3	0	0	0	3	3	AIML(1) Rule:1
17	PEC	PE-5	23AVI3508	ADVANCED IMAGE PROCESSING AND ANALYSIS	R	AIPA	3	0	0	0	3	3	AIML(1) Rule:1

39. SPECIALIZATION: AI-DRIVEN EDGE ARCHITECTURES AND APPLICATIONS

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDCS20A	SMART IOT AND SENSOR NETWORKS	A	SISN	0	0	6	4	4	10	OS(1) Rule:1
2	SDC	SDP-4	23SDCS20E	SMART IOT AND SENSOR NETWORKS	E	SISN	0	0	6	4	4	10	OS(1) Rule:1
3	SDC	SDP-4	23SDCS20R	SMART IOT AND SENSOR NETWORKS	R	SISN	0	0	2	4	2	6	OS(1) Rule:1
4	PEC	PE-1	23ADE3101A	EDGE COMPUTING FUNDAMENTALS	A	ECF	4	0	4	4	7	12	OS(1) Rule:1
5	PEC	PE-1	23ADE3101E	EDGE COMPUTING FUNDAMENTALS	E	ECF	4	0	4	4	7	12	OS(1) Rule:1
6	PEC	PE-1	23ADE3101R	EDGE COMPUTING FUNDAMENTALS	R	ECF	3	0	2	4	5	9	OS(1) Rule:1
7	PEC	PE-2	23ADE3202A	AI AT THE EDGE	A	AIAE	5	0	0	0	5	5	OS(1) Rule:1
8	PEC	PE-2	23ADE3202E	AI AT THE EDGE	E	AIAE	5	0	0	0	5	5	OS(1) Rule:1
9	PEC	PE-2	23ADE3202R	AI AT THE EDGE	R	AIAE	3	0	0	0	3	3	OS(1) Rule:1
10	PEC	PE-3	23ADE3303A	EDGE AI ARCHITECTURES AND APPLICATIONS	A	EAIAA	4	0	4	4	7	12	OS(1)
11	PEC	PE-3	23ADE3303E	EDGE AI ARCHITECTURES AND APPLICATIONS	E	EAIAA	4	0	4	4	7	12	OS(1)
12	PEC	PE-3	23ADE3303R	EDGE AI ARCHITECTURES AND APPLICATIONS	R	EAIAA	3	0	2	4	5	9	OS(1)
13	PEC	PE-4	23ADE3404M	EDGE AI AND 5G	M	EAI5G	3	0	0	0	3	3	OS(1) Rule:1
14	PEC	PE-4	23ADE3405M	EDGE AI SECURITY AND PROTECTION	M	EAISP	3	0	0	0	3	3	OS(1) Rule:1
15	PEC	PE-5	23ADE3506	EDGE AI DEVELOPMENT FRAMEWORKS AND TOOLS	R	EAIDFT	3	0	0	0	3	3	OS(1) Rule:1
16	PEC	PE-5	23ADE3507	EDGE AI ANALYTICS AND DATA MANAGEMENT	R	EAIADM	3	0	0	0	3	3	OS(1) Rule:1
17	PEC	PE-5	23ADE3508	EDGE AI AND SUSTAINABILITY	R	EAIAS	3	0	0	0	3	3	OS(1) Rule:1

40. SPECIALIZATION: AI-DRIVEN LANGUAGE TECHNOLOGIES

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDCS18A	TEXT ANALYTICS	A	TAT	0	0	6	4	4	10	AIML(1) Rule:1
2	SDC	SDP-4	23SDCS18E	TEXT ANALYTICS	E	TAT	0	0	6	4	4	10	AIML(1) Rule:1
3	SDC	SDP-4	23SDCS18R	TEXT ANALYTICS	R	TAT	0	0	2	4	2	6	AIML(1) Rule:1
4	PEC	PE-1	23ALT3101A	APPLIED MACHINE LEARNING FOR TEXT ANALYSIS	A	AMLFTA	4	0	4	4	7	12	AIML(1) Rule:1
5	PEC	PE-1	23ALT3101E	APPLIED MACHINE LEARNING FOR TEXT ANALYSIS	E	AMLFTA	4	0	4	4	7	12	AIML(1) Rule:1
6	PEC	PE-1	23ALT3101R	APPLIED MACHINE LEARNING FOR TEXT ANALYSIS	R	AMLFTA	3	0	2	4	5	9	AIML(1) Rule:1
7	PEC	PE-2	23ALT3202A	DEEP LEARNING FOR NATURAL LANGUAGE PROCESSING	A	DLFNLP	5	0	0	0	5	5	AIML(1) Rule:1
8	PEC	PE-2	23ALT3202E	DEEP LEARNING FOR NATURAL LANGUAGE PROCESSING	E	DLFNLP	5	0	0	0	5	5	AIML(1) Rule:1
9	PEC	PE-2	23ALT3202R	DEEP LEARNING FOR NATURAL LANGUAGE PROCESSING	R	DLFNLP	3	0	0	0	3	3	AIML(1) Rule:1
10	PEC	PE-3	23ALT3303A	CONVERSATIONAL AI AND CHATBOT DEVELOPMENT	A	CAICBD	4	0	4	4	7	12	AIML(1) Rule:1
11	PEC	PE-3	23ALT3303E	CONVERSATIONAL AI AND CHATBOT DEVELOPMENT	E	CAICBD	4	0	4	4	7	12	AIML(1) Rule:1
12	PEC	PE-3	23ALT3303R	CONVERSATIONAL AI AND CHATBOT DEVELOPMENT	R	CAICBD	3	0	2	4	5	9	AIML(1) Rule:1
13	PEC	PE-4	23ALT3404M	TEXT UNDERSTANDING	M	TUS	3	0	0	0	3	3	AIML(1) Rule:1
14	PEC	PE-4	23ALT3405M	NATURAL LANGUAGE PROCESSING	M	NLP	3	0	0	0	3	3	AIML(1) Rule:1
15	PEC	PE-5	23ALT3506	LANGUAGE UNDERSTANDING WITH PRE-TRAINED MODELS	R	LUWPTM	3	0	0	0	3	3	AIML(1) Rule:1
16	PEC	PE-5	23ALT3507	LARGE LANGUAGE MODEL FOR SPEECH PROCESSING	R	LLMFSP	3	0	0	0	3	3	AIML(1) Rule:1

41. SPECIALIZATION: AUTOMOTIVE ELECTRONICS AND AUTOSAR

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	PEC	PE-1	23AEA3101A	AUTOMOTIVE SENSOR AND APPLICATIONS	A	ASA	4	0	4	4	7	12	
2	PEC	PE-1	23AEA3101E	AUTOMOTIVE SENSOR AND APPLICATIONS	E	ASA	4	0	4	4	7	12	
3	PEC	PE-1	23AEA3101R	AUTOMOTIVE SENSOR AND APPLICATIONS	R	ASA	3	0	2	4	5	9	
4	PEC	PE-2	23AEA3202A	AUTOTRONICS	A	AT	5	0	0	0	5	5	
5	PEC	PE-2	23AEA3202E	AUTOTRONICS	E	AT	5	0	0	0	5	5	

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
6	PEC	PE-2	23AEA3202R	AUTOTRONICS	R	AT	3	0	0	0	3	3	
7	PEC	PE-2	23AEA3203A	AUTOMOTIVE POLLUTION AND ITS CONTROL	A	APC	5	0	0	0	5	5	
8	PEC	PE-2	23AEA3203E	AUTOMOTIVE POLLUTION AND ITS CONTROL	E	APC	5	0	0	0	5	5	
9	PEC	PE-2	23AEA3203R	AUTOMOTIVE POLLUTION AND ITS CONTROL	R	APC	3	0	0	0	3	3	
10	PEC	PE-3	23AEA3304A	ALTERNATE DRIVES, TRACTION AND CONTROLS	A	ADTC	4	0	4	4	7	12	AIML(1)
11	PEC	PE-3	23AEA3304E	ALTERNATE DRIVES, TRACTION AND CONTROLS	E	ADTC	4	0	4	4	7	12	AIML(1)
12	PEC	PE-3	23AEA3304R	ALTERNATE DRIVES, TRACTION AND CONTROLS	R	ADTC	3	0	2	4	5	9	AIML(1)
13	PEC	PE-4	23AEA3405M	VEHICLE CONTROL SYSTEMS	M	VCS	3	0	0	0	3	3	
14	PEC	PE-4	23AEA3406M	AUTOMOTIVE ELECTRICAL AND ELECTRONIC SYSTEMS	M	AEES	3	0	0	0	3	3	
15	PEC	PE-5	23AEA3507	SOFT COMPUTING TECHNIQUES FOR AUTOMOTIVE APPLICATIONS	R	SCTAA	3	0	0	0	3	3	
16	PEC	PE-5	23AEA3508	AUTOMOTIVE NETWORKING AND PROTOCOLS	R	ANP	3	0	0	0	3	3	

42. SPECIALIZATION: AUTOMOTIVE ENERGY ENGINEERING

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDME04A	ANALYSIS OF ENERGY SYSTEMS	A	AES	0	0	6	4	4	10	
2	SDC	SDP-4	23SDME04E	ANALYSIS OF ENERGY SYSTEMS	E	AES	0	0	6	4	4	10	
3	SDC	SDP-4	23SDME04R	ANALYSIS OF ENERGY SYSTEMS	R	AES	0	0	2	4	2	6	
4	PEC	PE-1	23ECF3101A	AUTOMOBILE ENGINEERING	A	AE	4	0	4	4	7	12	
5	PEC	PE-1	23ECF3101E	AUTOMOBILE ENGINEERING	E	AE	4	0	4	4	7	12	
6	PEC	PE-1	23ECF3101R	AUTOMOBILE ENGINEERING	R	AE	3	0	2	4	5	9	
7	PEC	PE-2	23ECF3202A	AUTOTRONICS & SAFETY	A	ATS	5	0	0	0	5	5	
8	PEC	PE-2	23ECF3202E	AUTOTRONICS & SAFETY	E	ATS	5	0	0	0	5	5	
9	PEC	PE-2	23ECF3202R	AUTOTRONICS & SAFETY	R	ATS	3	0	0	0	3	3	
10	PEC	PE-2	23ECF3203A	AUTONOMOUS VEHICLES	A	AV	5	0	0	0	5	5	
11	PEC	PE-2	23ECF3203E	AUTONOMOUS VEHICLES	E	AV	5	0	0	0	5	5	
12	PEC	PE-2	23ECF3203R	AUTONOMOUS VEHICLES	R	AV	3	0	0	0	3	3	
13	PEC	PE-3	23ECF3304A	HYBRID AND ELECTRIC VEHICLE DESIGN	A	HEVD	4	0	4	4	7	12	
14	PEC	PE-3	23ECF3304E	HYBRID AND ELECTRIC VEHICLE DESIGN	E	HEVD	4	0	4	4	7	12	

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
15	PEC	PE-3	23ECF3304R	HYBRID AND ELECTRIC VEHICLE DESIGN	R	HEVD	3	0	2	4	5	9	
16	PEC	PE-4	23ECF3405M	SPECIAL PURPOSE VEHICLES	M	SPV	3	0	0	0	3	3	
17	PEC	PE-4	23ECF3406M	VEHICLE DYNAMICS	M	VD	3	0	0	0	3	3	
18	PEC	PE-5	23ECF3507	THERMAL MANAGEMENT OF ELECTRIC AND ELECTRONIC SYSTEMS	R	TMEES	3	0	0	0	3	3	
19	PEC	PE-5	23ECF3508	ALTERNATE ENERGY SOURCES FOR AUTOMOBILES	R	AESFA	3	0	0	0	3	3	

43. SPECIALIZATION: BIOINFORMATICS

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDBT08A	DRUG DESIGN: PRINCIPLES AND ENGINEERING	A	DDPE	0	0	6	4	4	10	CDS(1) Rule:1
2	SDC	SDP-4	23SDBT08E	DRUG DESIGN: PRINCIPLES AND ENGINEERING	E	DDPE	0	0	6	4	4	10	CDS(1) Rule:1
3	SDC	SDP-4	23SDBT08R	DRUG DESIGN: PRINCIPLES AND ENGINEERING	R	DDPE	0	0	2	4	2	6	CDS(1) Rule:1
4	PEC	PE-1	23BIS3101A	MOLECULAR MODELLING AND DRUG DESIGN	A	MMDD	4	0	4	4	7	12	CB(1)
5	PEC	PE-1	23BIS3101E	MOLECULAR MODELLING AND DRUG DESIGN	E	MMDD	4	0	4	4	7	12	CB(1)
6	PEC	PE-1	23BIS3101R	MOLECULAR MODELLING AND DRUG DESIGN	R	MMDD	3	0	2	4	5	9	CB(1)
7	PEC	PE-2	23BIS3202A	BIOMEDICAL INFORMATICS	A	BMI	5	0	0	0	5	5	CB(1)
8	PEC	PE-2	23BIS3202E	BIOMEDICAL INFORMATICS	E	BMI	5	0	0	0	5	5	CB(1)
9	PEC	PE-2	23BIS3202R	BIOMEDICAL INFORMATICS	R	BMI	3	0	0	0	3	3	CB(1)
10	PEC	PE-2	23BIS3203A	PYTHON AND R PROGRAMMING	A	P&RP	5	0	0	0	5	5	CB(1)
11	PEC	PE-2	23BIS3203E	PYTHON AND R PROGRAMMING	E	P&RP	5	0	0	0	5	5	CB(1)
12	PEC	PE-2	23BIS3203R	PYTHON AND R PROGRAMMING	R	P&RP	3	0	0	0	3	3	CB(1)
13	PEC	PE-3	23BIS3304A	STRUCTURAL BIOLOGY	A	SB	4	0	4	4	7	12	CB(1)
14	PEC	PE-3	23BIS3304E	STRUCTURAL BIOLOGY	E	SB	4	0	4	4	7	12	CB(1)
15	PEC	PE-3	23BIS3304R	STRUCTURAL BIOLOGY	R	SB	3	0	2	4	5	9	CB(1)
16	PEC	PE-4	23BIS3405M	APPLIED BIOINFORMATICS	M	ABI	3	0	0	0	3	3	CB(1)
17	PEC	PE-4	23BIS3406M	NGS SEQUENCING AND DATA ANALYSIS	M	NGSDA	3	0	0	0	3	3	CB(1)
18	PEC	PE-5	23BIS3507	SYSTEMS BIOLOGY	R	SSB	3	0	0	0	3	3	CB(1)
19	PEC	PE-5	23BIS3508	DATABASE MANAGEMENT SYSTEM FOR BIOLOGIST	R	DBMS	3	0	0	0	3	3	CB(1)

44. SPECIALIZATION: BLOCKCHAIN ENGINEERING FOR WEB3

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDAD08A	EMERGING BLOCKCHAIN MODELS FOR DIGITAL CURRENCIES	A	EBCMDC	0	0	6	4	4	10	DBMS(1) Rule:1
2	SDC	SDP-4	23SDAD08E	EMERGING BLOCKCHAIN MODELS FOR DIGITAL CURRENCIES	E	EBCMDC	0	0	6	4	4	10	DBMS(1) Rule:1
3	SDC	SDP-4	23SDAD08R	EMERGING BLOCKCHAIN MODELS FOR DIGITAL CURRENCIES	R	EBCMDC	0	0	2	4	2	6	DBMS(1) Rule:1
4	PEC	PE-1	23BEW3101A	INTRODUCTION TO BLOCKCHAIN TECHNOLOGY, CRYPTOCURRENCIES AND TOKENS	A	ITBTCT	4	0	4	4	7	12	
5	PEC	PE-1	23BEW3101E	INTRODUCTION TO BLOCKCHAIN TECHNOLOGY, CRYPTOCURRENCIES AND TOKENS	E	ITBTCT	4	0	4	4	7	12	
6	PEC	PE-1	23BEW3101R	INTRODUCTION TO BLOCKCHAIN TECHNOLOGY, CRYPTOCURRENCIES AND TOKENS	R	ITBTCT	3	0	2	4	5	9	
7	PEC	PE-2	23BEW3202A	SMART CONTRACTS	A	SCR	5	0	0	0	5	5	
8	PEC	PE-2	23BEW3202E	SMART CONTRACTS	E	SCR	5	0	0	0	5	5	
9	PEC	PE-2	23BEW3202R	SMART CONTRACTS	R	SCR	3	0	0	0	3	3	
10	PEC	PE-3	23BEW3303A	DECENTRALIZED APPLICATIONS AND DECENTRALIZED WEB	A	DADW	4	0	4	4	7	12	
11	PEC	PE-3	23BEW3303E	DECENTRALIZED APPLICATIONS AND DECENTRALIZED WEB	E	DADW	4	0	4	4	7	12	
12	PEC	PE-3	23BEW3303R	DECENTRALIZED APPLICATIONS AND DECENTRALIZED WEB	R	DADW	3	0	2	4	5	9	
13	PEC	PE-4	23BEW3404M	BLOCKCHAIN PLATFORMS AND PROTOCOLS	M	BCPAP	3	0	0	0	3	3	
14	PEC	PE-4	23BEW3405M	BLOCKCHAIN SECURITY	M	BCS	3	0	0	0	3	3	
15	PEC	PE-4	23BEW3406M	ORACLES AND DATA FEEDS	M	ODF	3	0	0	0	3	3	
16	PEC	PE-5	23BEW3507	BLOCKCHAIN SCALABILITY AND LAYER 2 SOLUTIONS	R	BCSALS	3	0	0	0	3	3	
17	PEC	PE-5	23BEW3508	REGULATORY LANDSCAPE AND COMPLIANCE	R	RLAC	3	0	0	0	3	3	
18	PEC	PE-5	23BEW3509	FUTURE TRENDS IN BLOCKCHAIN AND WEB3	R	FTBCAW	3	0	0	0	3	3	

45. SPECIALIZATION: CLOUD AND EDGE COMPUTING

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	FCC	FC-1	23CS2223F	CLOUD INFRASTRUCTURE AND SERVICES	F	CIS	2	0	2	0	3	4	OS(1) Rule:1
2	FCC	FC-1	23CS2236F	FUNCTIONAL & CONCURRENT PROGRAMMING	F	FCP	2	0	2	0	3	4	OS(1) Rule:1

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
3	SDC	SDP-4	23SDCS04A	CLOUD BASED SOLUTIONS ARCHITECT	A	CBSA	0	0	6	4	4	10	OS(1) Rule:1
4	SDC	SDP-4	23SDCS04E	CLOUD BASED SOLUTIONS ARCHITECT	E	CBSA	0	0	6	4	4	10	OS(1) Rule:1
5	SDC	SDP-4	23SDCS04R	CLOUD BASED SOLUTIONS ARCHITECT	R	CBSA	0	0	2	4	2	6	OS(1) Rule:1
6	PEC	PE-1	23CEC3101A	CLOUD INFRASTRUCTURE AND SERVICES	A	CIS	4	0	4	4	7	12	OS(1) Rule:1
7	PEC	PE-1	23CEC3101E	CLOUD INFRASTRUCTURE AND SERVICES	E	CIS	4	0	4	4	7	12	OS(1) Rule:1
8	PEC	PE-1	23CEC3101R	CLOUD INFRASTRUCTURE AND SERVICES	R	CIS	3	0	2	4	5	9	OS(1) Rule:1
9	PEC	PE-2	23CEC3202A	ADVANCED OPERATING SYSTEMS	A	AOS	5	0	0	0	5	5	OS(1)
10	PEC	PE-2	23CEC3202E	ADVANCED OPERATING SYSTEMS	E	AOS	5	0	0	0	5	5	OS(1)
11	PEC	PE-2	23CEC3202R	ADVANCED OPERATING SYSTEMS	R	AOS	3	0	0	0	3	3	OS(1)
12	PEC	PE-2	23CEC3203A	FUNCTIONAL & CONCURRENT PROGRAMMING	A	FCP	5	0	0	0	5	5	OS(1) Rule:1
13	PEC	PE-2	23CEC3203E	FUNCTIONAL & CONCURRENT PROGRAMMING	E	FCP	5	0	0	0	5	5	OS(1) Rule:1
14	PEC	PE-2	23CEC3203R	FUNCTIONAL & CONCURRENT PROGRAMMING	R	FCP	3	0	0	0	3	3	OS(1) Rule:1
15	PEC	PE-2	23CEC3204A	CLOUD DEVOPS	A	CD	5	0	0	0	5	5	OS(1)
16	PEC	PE-2	23CEC3204E	CLOUD DEVOPS	E	CD	5	0	0	0	5	5	OS(1)
17	PEC	PE-2	23CEC3204R	CLOUD DEVOPS	R	CD	3	0	0	0	3	3	OS(1)
18	PEC	PE-3	23CEC3305A	CLOUD & SERVERLESS COMPUTING	A	CSC	4	0	4	4	7	12	OS(1)
19	PEC	PE-3	23CEC3305E	CLOUD & SERVERLESS COMPUTING	E	CSC	4	0	4	4	7	12	OS(1)
20	PEC	PE-3	23CEC3305R	CLOUD & SERVERLESS COMPUTING	R	CSC	3	0	2	4	5	9	OS(1)
21	PEC	PE-4	23CEC3406M	ADVANCED COMPUTER ARCHITECTURE	M	ACA	3	0	0	0	3	3	OS(1)
22	PEC	PE-4	23CEC3407M	PARALLEL ALGORITHMS	M	PA	3	0	0	0	3	3	OS(1)
23	PEC	PE-4	23CEC3408M	CLOUD SECURITY	M	CS	3	0	0	0	3	3	OS(1)
24	PEC	PE-4	23CEC3409M	ARCHITECTING CLOUD SOLUTIONS	M	ACS	3	0	0	0	3	3	OS(1)
25	PEC	PE-5	23CEC3510	EDGE COMPUTING	R	EC	3	0	0	0	3	3	OS(1)
26	PEC	PE-5	23CEC3511	HIGH PERFORMANCE COMPUTING	R	HPC	3	0	0	0	3	3	OS(1)
27	PEC	PE-5	23CEC3512	DESIGN OF DISTRIBUTED APPLICATIONS ON CLOUD	R	DDAC	3	0	0	0	3	3	OS(1)
28	PEC	PE-5	23CEC3513	CLOUD NETWORKING	R	CN	3	0	0	0	3	3	OS(1)

46. SPECIALIZATION: CLOUD INFRASTRUCTURE DESIGN AND ENGINEERING

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDCS16A	CLOUD INFRASTRUCTURE DESIGN	A	CISD	0	0	6	4	4	10	OS(1) Rule:1
2	SDC	SDP-4	23SDCS16E	CLOUD INFRASTRUCTURE DESIGN	E	CISD	0	0	6	4	4	10	OS(1) Rule:1
3	SDC	SDP-4	23SDCS16R	CLOUD INFRASTRUCTURE DESIGN	R	CISD	0	0	2	4	2	6	OS(1) Rule:1
4	PEC	PE-1	23CID3101A	CLOUD INFRASTRUCTURE ENGINEERING	A	CIE	4	0	4	4	7	12	OS(1)
5	PEC	PE-1	23CID3101E	CLOUD INFRASTRUCTURE ENGINEERING	E	CIE	4	0	4	4	7	12	OS(1)
6	PEC	PE-1	23CID3101R	CLOUD INFRASTRUCTURE ENGINEERING	R	CIE	3	0	2	4	5	9	OS(1)
7	PEC	PE-2	23CID3202A	ENTERPRISE CLOUD COMPUTING: TECHNOLOGY, ARCHITECTURE, APPLICATIONS	A	ECT	5	0	0	0	5	5	OS(1)
8	PEC	PE-2	23CID3202E	ENTERPRISE CLOUD COMPUTING: TECHNOLOGY, ARCHITECTURE, APPLICATIONS	E	ECT	5	0	0	0	5	5	OS(1)
9	PEC	PE-2	23CID3202R	ENTERPRISE CLOUD COMPUTING: TECHNOLOGY, ARCHITECTURE, APPLICATIONS	R	ECT	3	0	0	0	3	3	OS(1)
10	PEC	PE-2	23CID3203A	ARCHITECTING HIGHLY AVAILABLE AND SCALABLE CLOUD ENVIRONMENTS	A	AHASCE	5	0	0	0	5	5	OS(1)
11	PEC	PE-2	23CID3203E	ARCHITECTING HIGHLY AVAILABLE AND SCALABLE CLOUD ENVIRONMENTS	E	AHASCE	5	0	0	0	5	5	OS(1)
12	PEC	PE-2	23CID3203R	ARCHITECTING HIGHLY AVAILABLE AND SCALABLE CLOUD ENVIRONMENTS	R	AHASCE	3	0	0	0	3	3	OS(1)
13	PEC	PE-3	23CID3304A	CLOUD INFRASTRUCTURE ARCHITECTURE AND ENGINEERING	A	CIA	4	0	4	4	7	12	OS(1)
14	PEC	PE-3	23CID3304E	CLOUD INFRASTRUCTURE ARCHITECTURE AND ENGINEERING	E	CIA	4	0	4	4	7	12	OS(1)
15	PEC	PE-3	23CID3304R	CLOUD INFRASTRUCTURE ARCHITECTURE AND ENGINEERING	R	CIA	3	0	2	4	5	9	OS(1)
16	PEC	PE-4	23CID3405M	RELIABLE CLOUD INFRASTRUCTURE: DESIGN & PROCESS	M	RID	3	0	0	0	3	3	OS(1)
17	PEC	PE-4	23CID3406M	HYBRID CLOUD INTEGRATION AND MANAGEMENT	M	HCIM	3	0	0	0	3	3	OS(1)
18	PEC	PE-5	23CID3507	CLOUD ENGINEERING	R	CE	3	0	0	0	3	3	OS(1)
19	PEC	PE-5	23CID3508	CLOUD DATA MANAGEMENT AND STORAGE SOLUTIONS	R	CDMSS	3	0	0	0	3	3	OS(1)

47. SPECIALIZATION: CLOUD NATIVE SECURITY

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDCS05A	CLOUD BASED SECURITY SPECIALITY	A	CBSS	0	0	6	4	4	10	OS(1) Rule:1
2	SDC	SDP-4	23SDCS05E	CLOUD BASED SECURITY SPECIALITY	E	CBSS	0	0	6	4	4	10	OS(1) Rule:1
3	SDC	SDP-4	23SDCS05R	CLOUD BASED SECURITY SPECIALITY	R	CBSS	0	0	2	4	2	6	OS(1) Rule:1
4	PEC	PE-1	23CNS3101A	INTRODUCTION TO CLOUD NATIVE COMPUTING	A	ITCNC	4	0	4	4	7	12	OS(1)
5	PEC	PE-1	23CNS3101E	INTRODUCTION TO CLOUD NATIVE COMPUTING	E	ITCNC	4	0	4	4	7	12	OS(1)
6	PEC	PE-1	23CNS3101R	INTRODUCTION TO CLOUD NATIVE COMPUTING	R	ITCNC	3	0	2	4	5	9	OS(1)
7	PEC	PE-2	23CNS3202A	FUNDAMENTALS OF CLOUD SECURITY	A	FOCS	5	0	0	0	5	5	OS(1)
8	PEC	PE-2	23CNS3202E	FUNDAMENTALS OF CLOUD SECURITY	E	FOCS	5	0	0	0	5	5	OS(1)
9	PEC	PE-2	23CNS3202R	FUNDAMENTALS OF CLOUD SECURITY	R	FOCS	3	0	0	0	3	3	OS(1)
10	PEC	PE-3	23CNS3303A	CONTAINER SECURITY	A	CTSY	4	0	4	4	7	12	OS(1)
11	PEC	PE-3	23CNS3303E	CONTAINER SECURITY	E	CTSY	4	0	4	4	7	12	OS(1)
12	PEC	PE-3	23CNS3303R	CONTAINER SECURITY	R	CTSY	3	0	2	4	5	9	OS(1)
13	PEC	PE-4	23CNS3404M	MICROSERVICES SECURITY	M	MSS	3	0	0	0	3	3	OS(1)
14	PEC	PE-4	23CNS3405M	INFRASTRUCTURE AS CODE SECURITY	M	ISACS	3	0	0	0	3	3	OS(1)
15	PEC	PE-4	23CNS3406M	IDENTITY AND ACCESS MANAGEMENT IN CLOUD NATIVE ENVIRONMENTS	M	IAMCNE	3	0	0	0	3	3	OS(1)
16	PEC	PE-4	23CNS3407M	NETWORK SECURITY IN CLOUD NATIVE ARCHITECTURES	M	NSCNA	3	0	0	0	3	3	OS(1)
17	PEC	PE-5	23CNS3508	MONITORING AND LOGGING FOR CLOUD NATIVE SECURITY	R	MLCNS	3	0	0	0	3	3	OS(1)
18	PEC	PE-5	23CNS3509	COMPLIANCE AND GOVERNANCE IN CLOUD NATIVE ENVIRONMENTS	R	CGCNE	3	0	0	0	3	3	OS(1)
19	PEC	PE-5	23CNS3510	SERVERLESS SECURITY	R	SLSY	3	0	0	0	3	3	OS(1)
20	PEC	PE-5	23CNS3511	CLOUD NATIVE SECURITY TOOLS AND SOLUTIONS	R	CNSTAS	3	0	0	0	3	3	OS(1)
21	PEC	PE-5	23CNS3512	FUTURE TRENDS IN CLOUD NATIVE SECURITY	R	FTCNS	3	0	0	0	3	3	OS(1)

48. SPECIALIZATION: CLOUD NATIVE SOFTWARE ENGINEERING

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDCS15A	SOFTWARE DESIGN WITH CLOUD NATIVE COMPUTING	A	SDCNC	0	0	6	4	4	10	OS(1) Rule:1
2	SDC	SDP-4	23SDCS15E	SOFTWARE DESIGN WITH CLOUD NATIVE COMPUTING	E	SDCNC	0	0	6	4	4	10	OS(1) Rule:1
3	SDC	SDP-4	23SDCS15R	SOFTWARE DESIGN WITH CLOUD NATIVE COMPUTING	R	SDCNC	0	0	2	4	2	6	OS(1) Rule:1
4	PEC	PE-1	23CNE3101A	FOUNDATIONS OF CLOUD NATIVE COMPUTING	A	FCNC	4	0	4	4	7	12	OS(1)
5	PEC	PE-1	23CNE3101E	FOUNDATIONS OF CLOUD NATIVE COMPUTING	E	FCNC	4	0	4	4	7	12	OS(1)
6	PEC	PE-1	23CNE3101R	FOUNDATIONS OF CLOUD NATIVE COMPUTING	R	FCNC	3	0	2	4	5	9	OS(1)
7	PEC	PE-2	23CNE3202A	CLOUD NATIVE APPLICATION DEVELOPMENT	A	CNAD	5	0	0	0	5	5	OS(1)
8	PEC	PE-2	23CNE3202E	CLOUD NATIVE APPLICATION DEVELOPMENT	E	CNAD	5	0	0	0	5	5	OS(1)
9	PEC	PE-2	23CNE3202R	CLOUD NATIVE APPLICATION DEVELOPMENT	R	CNAD	3	0	0	0	3	3	OS(1)
10	PEC	PE-2	23CNE3203A	CLOUD NATIVE DESIGN PROCESS	A	CNDP	5	0	0	0	5	5	OS(1)
11	PEC	PE-2	23CNE3203E	CLOUD NATIVE DESIGN PROCESS	E	CNDP	5	0	0	0	5	5	OS(1)
12	PEC	PE-2	23CNE3203R	CLOUD NATIVE DESIGN PROCESS	R	CNDP	3	0	0	0	3	3	OS(1)
13	PEC	PE-3	23CNE3304A	SOFTWARE ENGINEERING FOR CLOUD COMPUTING	A	SEC	4	0	4	4	7	12	OS(1)
14	PEC	PE-3	23CNE3304E	SOFTWARE ENGINEERING FOR CLOUD COMPUTING	E	SEC	4	0	4	4	7	12	OS(1)
15	PEC	PE-3	23CNE3304R	SOFTWARE ENGINEERING FOR CLOUD COMPUTING	R	SEC	3	0	2	4	5	9	OS(1)
16	PEC	PE-4	23CNE3405M	SERVERLESS COMPUTING AND EVENT-DRIVEN ARCHITECTURES	M	SCEDA	3	0	0	0	3	3	OS(1)
17	PEC	PE-4	23CNE3406M	CLOUD NATIVE SOFTWARE ARCHITECTURES	M	CNSA	3	0	0	0	3	3	OS(1)
18	PEC	PE-5	23CNE3507	CLOUD NATIVE NETWORKING AND STORAGE SYSTEMS	R	CNNSS	3	0	0	0	3	3	OS(1)
19	PEC	PE-5	23CNE3508	CLOUD INFRASTRUCTURE AND VIRTUALIZATION	R	CIAV	3	0	0	0	3	3	OS(1)

49. SPECIALIZATION: CLOUD-BASED SCIENTIFIC COMPUTING

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDCS14A	CLOUD ARCHITECTURES FOR SCIENTIFIC WORKLOADS	A	CASW	0	0	6	4	4	10	OS(1) Rule:1
2	SDC	SDP-4	23SDCS14E	CLOUD ARCHITECTURES FOR SCIENTIFIC WORKLOADS	E	CASW	0	0	6	4	4	10	OS(1) Rule:1

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
3	SDC	SDP-4	23SDCS14R	CLOUD ARCHITECTURES FOR SCIENTIFIC WORKLOADS	R	CASW	0	0	2	4	2	6	OS(1) Rule:1
4	PEC	PE-1	23CSC3101A	FUNDAMENTALS OF CLOUD-BASED SCIENTIFIC COMPUTING	A	FCS	4	0	4	4	7	12	OS(1)
5	PEC	PE-1	23CSC3101E	FUNDAMENTALS OF CLOUD-BASED SCIENTIFIC COMPUTING	E	FCS	4	0	4	4	7	12	OS(1)
6	PEC	PE-1	23CSC3101R	FUNDAMENTALS OF CLOUD-BASED SCIENTIFIC COMPUTING	R	FCS	3	0	2	4	5	9	OS(1)
7	PEC	PE-2	23CSC3202A	DEVELOPMENT OF SCIENTIFIC APPLICATIONS USING CLOUD-BASED TECHNOLOGY	A	DSA	5	0	0	0	5	5	OS(1)
8	PEC	PE-2	23CSC3202E	DEVELOPMENT OF SCIENTIFIC APPLICATIONS USING CLOUD-BASED TECHNOLOGY	E	DSA	5	0	0	0	5	5	OS(1)
9	PEC	PE-2	23CSC3202R	DEVELOPMENT OF SCIENTIFIC APPLICATIONS USING CLOUD-BASED TECHNOLOGY	R	DSA	3	0	0	0	3	3	OS(1)
10	PEC	PE-2	23CSC3203A	DISTRIBUTED SCIENTIFIC COMPUTING IN THE CLOUD	A	DSCC	5	0	0	0	5	5	OS(1)
11	PEC	PE-2	23CSC3203E	DISTRIBUTED SCIENTIFIC COMPUTING IN THE CLOUD	E	DSCC	5	0	0	0	5	5	OS(1)
12	PEC	PE-2	23CSC3203R	DISTRIBUTED SCIENTIFIC COMPUTING IN THE CLOUD	R	DSCC	3	0	0	0	3	3	OS(1)
13	PEC	PE-3	23CSC3304A	CLOUD BASED HIGH PERFORMANCE COMPUTING	A	CHC	4	0	4	4	7	12	OS(1)
14	PEC	PE-3	23CSC3304E	CLOUD BASED HIGH PERFORMANCE COMPUTING	E	CHC	4	0	4	4	7	12	OS(1)
15	PEC	PE-3	23CSC3304R	CLOUD BASED HIGH PERFORMANCE COMPUTING	R	CHC	3	0	2	4	5	9	OS(1)
16	PEC	PE-4	23CSC3405M	CLOUD-BASED GRID COMPUTING	M	CGC	3	0	0	0	3	3	OS(1)
17	PEC	PE-4	23CSC3406M	CLOUD COMPUTING FOR SCIENTIFIC DISCOVERY	M	CCFSD	3	0	0	0	3	3	OS(1)
18	PEC	PE-5	23CSC3508	PARALLEL COMPUTING FOR SCIENTISTS USING CLOUD PLATFORMS	R	PCSCP	3	0	0	0	3	3	OS(1)
19	PEC	PE-5	23CSC3507	CLOUD-BASED SCALABLE COMPUTING AND BIG DATA ANALYTICS	R	CSCBD	3	0	0	0	3	3	OS(1)

50. SPECIALIZATION: COMPUTER COMMUNICATION AND 5G TECHNOLOGY

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDEC12A	NETWORK PROGRAMMABILITY AND AUTOMATION	A	NPA	0	0	6	4	4	10	
2	SDC	SDP-4	23SDEC12E	NETWORK PROGRAMMABILITY AND AUTOMATION	E	NPA	0	0	6	4	4	10	

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
3	SDC	SDP-4	23SDEC12R	NETWORK PROGRAMMABILITY AND AUTOMATION	R	NPA	0	0	2	4	2	6	
4	PEC	PE-1	23CCF3101A	TCP/IP & OTHER PROTOCOL SUITE	A	TCP	4	0	4	4	7	12	
5	PEC	PE-1	23CCF3101E	TCP/IP & OTHER PROTOCOL SUITE	E	TCP	4	0	4	4	7	12	
6	PEC	PE-1	23CCF3101R	TCP/IP & OTHER PROTOCOL SUITE	R	TCP	3	0	2	4	5	9	
7	PEC	PE-2	23CCF3202A	CLOUD COMPUTING AND NETWORKS SECURITY	A	CCNS	5	0	0	0	5	5	
8	PEC	PE-2	23CCF3202E	CLOUD COMPUTING AND NETWORKS SECURITY	E	CCNS	5	0	0	0	5	5	
9	PEC	PE-2	23CCF3202R	CLOUD COMPUTING AND NETWORKS SECURITY	R	CCNS	3	0	0	0	3	3	
10	PEC	PE-3	23CCF3303A	VOIP AND BROADBAND NETWORKS	A	VBN	4	0	4	4	7	12	
11	PEC	PE-3	23CCF3303E	VOIP AND BROADBAND NETWORKS	E	VBN	4	0	4	4	7	12	
12	PEC	PE-3	23CCF3303R	VOIP AND BROADBAND NETWORKS	R	VBN	3	0	2	4	5	9	
13	PEC	PE-4	23CCF3404M	5G MOBILE AND IEEE STANDARDS	M	5GMI	3	0	0	0	3	3	
14	PEC	PE-5	23CCF3505	IP MULTIMEDIA SUB-SYSTEM & EMERGING TECHNOLOGIES	R	IMSET	3	0	0	0	3	3	
15	PEC	PE-5	23CCF3506	IT SECURITY: DEFENCE AGAINST THE DIGITAL DARK ARTS	R	IMSET	3	0	0	0	3	3	

51. SPECIALIZATION: CONSTRUCTION TECHNOLOGY AND MANAGEMENT

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDCE04A	PLANNING AND SCHEDULING OF RESIDENTIAL BUILDING USING PRIMAVERA SOFTWARE	A	PSRBPS	0	0	6	4	4	10	
2	SDC	SDP-4	23SDCE04E	PLANNING AND SCHEDULING OF RESIDENTIAL BUILDING USING PRIMAVERA SOFTWARE	E	PSRBPS	0	0	6	4	4	10	
3	SDC	SDP-4	23SDCE04R	PLANNING AND SCHEDULING OF RESIDENTIAL BUILDING USING PRIMAVERA SOFTWARE	R	PSRBPS	0	0	2	4	2	6	
4	PEC	PE-1	23CTM3101A	BUILDING INFORMATION MODELLING	A	BIM	4	0	4	4	7	12	
5	PEC	PE-1	23CTM3101E	BUILDING INFORMATION MODELLING	E	BIM	4	0	4	4	7	12	
6	PEC	PE-1	23CTM3101R	BUILDING INFORMATION MODELLING	R	BIM	3	0	2	4	5	9	
7	PEC	PE-2	23CTM3202A	ADVANCED CONSTRUCTION TECHNOLOGY	A	ACT	5	0	0	0	5	5	
8	PEC	PE-2	23CTM3202E	ADVANCED CONSTRUCTION TECHNOLOGY	E	ACT	5	0	0	0	5	5	
9	PEC	PE-2	23CTM3202R	ADVANCED CONSTRUCTION TECHNOLOGY	R	ACT	3	0	0	0	3	3	

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
10	PEC	PE-2	23CTM3203A	SUSTAINABLE CONSTRUCTION TECHNOLOGY	A	SCT	5	0	0	0	5	5	
11	PEC	PE-2	23CTM3203E	SUSTAINABLE CONSTRUCTION TECHNOLOGY	E	SCT	5	0	0	0	5	5	
12	PEC	PE-2	23CTM3203R	SUSTAINABLE CONSTRUCTION TECHNOLOGY	R	SCT	3	0	0	0	3	3	
13	PEC	PE-3	23CTM3304A	CONSTRUCTION PLANNING & SCHEDULING	A	CPS	4	0	4	4	7	12	
14	PEC	PE-3	23CTM3304E	CONSTRUCTION PLANNING & SCHEDULING	E	CPS	4	0	4	4	7	12	
15	PEC	PE-3	23CTM3304R	CONSTRUCTION PLANNING & SCHEDULING	R	CPS	3	0	2	4	5	9	
16	PEC	PE-4	23CTM3405M	CONSTRUCTION CONTRACTS	M	CC	3	0	0	0	3	3	
17	PEC	PE-4	23CTM3406M	CONSTRUCTION FORMULATION APPRAISAL	M	CFA	3	0	0	0	3	3	
18	PEC	PE-5	23CTM3507	QUALITY AND SAFETY IN CONSTRUCTION	R	QSC	3	0	0	0	3	3	
19	PEC	PE-5	23CTM3508	GREEN BUILDING	R	GB	3	0	0	0	3	3	

52. SPECIALIZATION: CROSS PLATFORM DEVELOPMENT FRAMEWORKS

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDCI04A	ADVANCED ANDROID MOBILE APPLICATION WITH CLOUD-BASED WEB SERVICES	A	AAMA	0	0	6	4	4	10	
2	SDC	SDP-4	23SDCI04E	ADVANCED ANDROID MOBILE APPLICATION WITH CLOUD-BASED WEB SERVICES	E	AAMA	0	0	6	4	4	10	
3	SDC	SDP-4	23SDCI04R	ADVANCED ANDROID MOBILE APPLICATION WITH CLOUD-BASED WEB SERVICES	R	AAMA	0	0	2	4	2	6	
4	PEC	PE-1	23CPD3101A	FUNDAMENTALS OF MOBILE APPLICATION DEVELOPMENT	A	FMAD	4	0	4	4	7	12	
5	PEC	PE-1	23CPD3101E	FUNDAMENTALS OF MOBILE APPLICATION DEVELOPMENT	E	FMAD	4	0	4	4	7	12	
6	PEC	PE-1	23CPD3101R	FUNDAMENTALS OF MOBILE APPLICATION DEVELOPMENT	R	FMAD	3	0	2	4	5	9	
7	PEC	PE-2	23CPD3202A	REACT NATIVE FOR ANDROID AND IOS DEVELOPMENT	A	RNA	5	0	0	0	5	5	
8	PEC	PE-2	23CPD3202E	REACT NATIVE FOR ANDROID AND IOS DEVELOPMENT	E	RNA	5	0	0	0	5	5	
9	PEC	PE-2	23CPD3202R	REACT NATIVE FOR ANDROID AND IOS DEVELOPMENT	R	RNA	3	0	0	0	3	3	
10	PEC	PE-2	23CPD3203A	FRAMEWORK BASED CROSS PLATFORM APP DEVELOPMENT	A	FBCPD	5	0	0	0	5	5	

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
11	PEC	PE-2	23CPD3203E	FRAMEWORK BASED CROSS PLATFORM APP DEVELOPMENT	E	FBCPD	5	0	0	0	5	5	
12	PEC	PE-2	23CPD3203R	FRAMEWORK BASED CROSS PLATFORM APP DEVELOPMENT	R	FBCPD	3	0	0	0	3	3	
13	PEC	PE-3	23CPD3304A	SECURE MOBILE APPLICATION DEVELOPMENT	A	SMAD	4	0	4	4	7	12	
14	PEC	PE-3	23CPD3304E	SECURE MOBILE APPLICATION DEVELOPMENT	E	SMAD	4	0	4	4	7	12	
15	PEC	PE-3	23CPD3304R	SECURE MOBILE APPLICATION DEVELOPMENT	R	SMAD	3	0	2	4	5	9	
16	PEC	PE-4	23CPD3406M	META REACT NATIVE SPECIALIZATION	M	MRNS	3	0	0	0	3	3	
17	PEC	PE-4	23CPD3405M	UX DESIGN	M	UXD	3	0	0	0	3	3	
18	PEC	PE-5	23CPD3507	ADVANCED MOBILE APPLICATION DEVELOPMENT	R	AMAD	3	0	0	0	3	3	
19	PEC	PE-5	23CPD3508	CROSS PLATFORM USER INTERFACE DESIGN	R	CPED	3	0	0	0	3	3	

53. SPECIALIZATION: CYBER PHYSICAL SYSTEMS AND IOT

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDIN06A	EMBEDDED AND IOT ROGRAMMING	A	IOTHP	0	0	6	4	4	10	
2	SDC	SDP-4	23SDIN06E	EMBEDDED AND IOT ROGRAMMING	E	IOTHP	0	0	6	4	4	10	
3	SDC	SDP-4	23SDIN06R	EMBEDDED AND IOT ROGRAMMING	R	IOTHP	0	0	2	4	2	6	
4	PEC	PE-1	23CPS3101A	IOT SENSING AND ACTUATING DEVICES	A	ISAD	4	0	4	4	7	12	
5	PEC	PE-1	23CPS3101E	IOT SENSING AND ACTUATING DEVICES	E	ISAD	4	0	4	4	7	12	
6	PEC	PE-1	23CPS3101R	IOT SENSING AND ACTUATING DEVICES	R	ISAD	3	0	2	4	5	9	
7	PEC	PE-2	23CPS3202A	INTERNET OF THINGS ARCHITECTURES AND PROTOCOLS	A	IOTAP	5	0	0	0	5	5	
8	PEC	PE-2	23CPS3202E	INTERNET OF THINGS ARCHITECTURES AND PROTOCOLS	E	IOTAP	5	0	0	0	5	5	
9	PEC	PE-2	23CPS3202R	INTERNET OF THINGS ARCHITECTURES AND PROTOCOLS	R	IOTAP	3	0	0	0	3	3	
10	PEC	PE-3	23CPS3303A	CYBER PHYSICAL SYSTEMS	A	CPS	4	0	4	4	7	12	
11	PEC	PE-3	23CPS3303E	CYBER PHYSICAL SYSTEMS	E	CPS	4	0	4	4	7	12	
12	PEC	PE-3	23CPS3303R	CYBER PHYSICAL SYSTEMS	R	CPS	3	0	2	4	5	9	
13	PEC	PE-4	23CPS3404M	FOUNDATIONS OF HYBRID AND EMBEDED SYSTEMS	M	DVP	3	0	0	0	3	3	
14	PEC	PE-5	23CPS3505	CLOUD COMPUTING FOR IOT ENGINEERS	R	CCIOTE	3	0	0	0	3	3	

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
15	PEC	PE-5	23CPS3506	WIRELESS SENSOR NETWORKS	R	WSN	3	0	0	0	3	3	
16	PEC	PE-5	23CPS3507	EDGE COMPUTING	R	EC	3	0	0	0	3	3	

54. SPECIALIZATION: CYBER SECURITY AND BLOCKCHAIN TECHNOLOGY

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	FCC	FC-1	23CS2224F	CRYPT ANALYSIS & CYBER DEFENSE	F	CACD	2	0	2	0	3	4	
2	FCC	FC-1	23CS2234F	NETWORK AND INFRASTRUCTURE SECURITY	F	NIS	2	0	2	0	3	4	
3	FCC	FC-1	23CS2233F	INTRODUCTION TO BLOCKCHAIN AND CRYPTO CURRENCIES	F	IBCC	2	0	2	0	3	4	
4	SDC	SDP-4	23SDCS05A	CLOUD BASED SECURITY SPECIALITY	A	CBSS	0	0	6	4	4	10	OS(1) Rule:1
5	SDC	SDP-4	23SDCS05E	CLOUD BASED SECURITY SPECIALITY	E	CBSS	0	0	6	4	4	10	OS(1) Rule:1
6	SDC	SDP-4	23SDCS05R	CLOUD BASED SECURITY SPECIALITY	R	CBSS	0	0	2	4	2	6	OS(1) Rule:1
7	PEC	PE-1	23CSB3101A	CRYPT ANALYSIS & CYBER DEFENSE	A	CACD	4	0	4	4	7	12	
8	PEC	PE-1	23CSB3101E	CRYPT ANALYSIS & CYBER DEFENSE	E	CACD	4	0	4	4	7	12	
9	PEC	PE-1	23CSB3101R	CRYPT ANALYSIS & CYBER DEFENSE	R	CACD	3	0	2	4	5	9	
10	PEC	PE-2	23CSB3202A	NETWORK AND INFRASTRUCTURE SECURITY	A	NIS	5	0	0	0	5	5	
11	PEC	PE-2	23CSB3202E	NETWORK AND INFRASTRUCTURE SECURITY	E	NIS	5	0	0	0	5	5	
12	PEC	PE-2	23CSB3202R	NETWORK AND INFRASTRUCTURE SECURITY	R	NIS	3	0	0	0	3	3	
13	PEC	PE-2	23CSB3203A	INTRODUCTION TO BLOCKCHAIN AND CRYPTO CURRENCIES	A	IBCC	3	0	4	0	5	7	
14	PEC	PE-2	23CSB3203E	INTRODUCTION TO BLOCKCHAIN AND CRYPTO CURRENCIES	E	IBCC	3	0	4	0	5	7	
15	PEC	PE-2	23CSB3203R	INTRODUCTION TO BLOCKCHAIN AND CRYPTO CURRENCIES	R	IBCC	2	0	2	0	3	4	
16	PEC	PE-3	23CSB3304A	DIGITAL FORENSICS	A	DF	4	0	4	4	7	12	
17	PEC	PE-3	23CSB3304E	DIGITAL FORENSICS	E	DF	4	0	4	4	7	12	
18	PEC	PE-3	23CSB3304R	DIGITAL FORENSICS	R	DF	3	0	2	4	5	9	
19	PEC	PE-4	23CSB3405M	DATABASE SYSTEM AND SECURITY	M	DSS	3	0	0	0	3	3	
20	PEC	PE-4	23CSB3406M	PROGRAMMING FOR SMART CONTRACTS	M	PSC	3	0	0	0	3	3	
21	PEC	PE-4	23CSB3407M	CLOUD SECURITY	M	CS	3	0	0	0	3	3	
22	PEC	PE-5	23CSB3508	SECURE SOFTWARE ENGINEERING	R	SSE	3	0	0	0	3	3	

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
23	PEC	PE-5	23CSB3509	WEB SECURITY	R	WS	3	0	0	0	3	3	
24	PEC	PE-5	23CSB3510	SECURITY GOVERNANCE & MANAGEMENT	R	SGM	3	0	0	0	3	3	

55. SPECIALIZATION: DATA COMMUNICATIONS

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDEC13A	5G PRIVATE AND INDUSTRIAL AUTOMATION NETWORKS	A	5PIA	0	0	6	4	4	10	
2	SDC	SDP-4	23SDEC13E	5G PRIVATE AND INDUSTRIAL AUTOMATION NETWORKS	E	5PIA	0	0	6	4	4	10	
3	SDC	SDP-4	23SDEC13R	5G PRIVATE AND INDUSTRIAL AUTOMATION NETWORKS	R	5PIA	0	0	2	4	2	6	
4	PEC	PE-1	23DCS3101A	4G WIRELESS TECHNOLOGIES AND CELLULAR COMMUNICATION	A	4GW	4	0	4	4	7	12	
5	PEC	PE-1	23DCS3101E	4G WIRELESS TECHNOLOGIES AND CELLULAR COMMUNICATION	E	4GW	4	0	4	4	7	12	
6	PEC	PE-1	23DCS3101R	4G WIRELESS TECHNOLOGIES AND CELLULAR COMMUNICATION	R	4GW	3	0	2	4	5	9	
7	PEC	PE-2	23DCS3202A	MODERN SATELLITE COMMUNICATION SYSTEMS	A	MSCS	5	0	0	0	5	5	
8	PEC	PE-2	23DCS3202E	MODERN SATELLITE COMMUNICATION SYSTEMS	E	MSCS	5	0	0	0	5	5	
9	PEC	PE-2	23DCS3202R	MODERN SATELLITE COMMUNICATION SYSTEMS	R	MSCS	3	0	0	0	3	3	
10	PEC	PE-3	23DCS3303A	5G WIRELESS TECHNOLOGIES	A	5GWT	4	0	4	4	7	12	
11	PEC	PE-3	23DCS3303E	5G WIRELESS TECHNOLOGIES	E	5GWT	4	0	4	4	7	12	
12	PEC	PE-3	23DCS3303R	5G WIRELESS TECHNOLOGIES	R	5GWT	3	0	2	4	5	9	
13	PEC	PE-4	23DCS3404M	OPTICAL WIRELESS COMMUNICATIONS	M	PWC	3	0	0	0	3	3	
14	PEC	PE-5	23DCS3505	MACHINE LEARNING FOR WIRELESS COMMUNICATION	R	MLWC	3	0	0	0	3	3	

56. SPECIALIZATION: DATA ENGINEERING FOR AI

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDAD11A	AI-DRIVEN DATA ENGINEERING	A	ADDE	0	0	6	4	4	10	DBMS(1)
2	SDC	SDP-4	23SDAD11E	AI-DRIVEN DATA ENGINEERING	E	ADDE	0	0	6	4	4	10	DBMS(1)
3	SDC	SDP-4	23SDAD11R	AI-DRIVEN DATA ENGINEERING	R	ADDE	0	0	2	4	2	6	DBMS(1)
4	PEC	PE-1	23DEA3101A	FUNDAMENTALS OF DATA ENGINEERING	A	FDE	4	0	4	4	7	12	DBMS(1)
5	PEC	PE-1	23DEA3101E	FUNDAMENTALS OF DATA ENGINEERING	E	FDE	4	0	4	4	7	12	DBMS(1)

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
6	PEC	PE-1	23DEA3101R	FUNDAMENTALS OF DATA ENGINEERING	R	FDE	3	0	2	4	5	9	DBMS(1)
7	PEC	PE-2	23DEA3202A	DATA EXPLORATION AND PREPARATION	A	DEP	5	0	0	0	5	5	DBMS(1)
8	PEC	PE-2	23DEA3202E	DATA EXPLORATION AND PREPARATION	E	DEP	5	0	0	0	5	5	DBMS(1)
9	PEC	PE-2	23DEA3202R	DATA EXPLORATION AND PREPARATION	R	DEP	3	0	0	0	3	3	DBMS(1)
10	PEC	PE-3	23DEA3303A	BIG DATA TECHNOLOGIES	A	BDT	4	0	4	4	7	12	DBMS(1)
11	PEC	PE-3	23DEA3303E	BIG DATA TECHNOLOGIES	E	BDT	4	0	4	4	7	12	DBMS(1)
12	PEC	PE-3	23DEA3303R	BIG DATA TECHNOLOGIES	R	BDT	3	0	2	4	5	9	DBMS(1)
13	PEC	PE-4	23DEA3404M	DATA WAREHOUSING AND BUSINESS INTELLIGENCE	M	DWBI	3	0	0	0	3	3	DBMS(1)
14	PEC	PE-5	23DEA3505	MACHINE LEARNING ENGINEERING FOR BIG DATA	R	MLEBD	3	0	0	0	3	3	DBMS(1)

57. SPECIALIZATION: DATA SCIENCE AND BIG DATA ANALYTICS

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	FCC	FC-1	23CS2227F	DATA ANALYTICS AND VISUALIZATION	F	DAV	2	0	2	0	3	4	DBMS(1) Rule:1
2	FCC	FC-1	23CS2232F	ADVANCED DATABASES	F	AD	2	0	2	0	3	4	DBMS(1) Rule:1
3	SDC	SDP-4	23SDCS08A	CLOUD BASED DATA ANALYTICS SPECIALITY	A	CBDAS	0	0	6	4	4	10	DBMS(1)
4	SDC	SDP-4	23SDCS08E	CLOUD BASED DATA ANALYTICS SPECIALITY	E	CBDAS	0	0	6	4	4	10	DBMS(1)
5	SDC	SDP-4	23SDCS08R	CLOUD BASED DATA ANALYTICS SPECIALITY	R	CBDAS	0	0	2	4	2	6	DBMS(1)
6	PEC	PE-1	23DSB3101A	DATA ANALYTICS AND VISUALIZATION	A	DAV	4	0	4	4	7	12	DBMS(1) Rule:1
7	PEC	PE-1	23DSB3101E	DATA ANALYTICS AND VISUALIZATION	E	DAV	4	0	4	4	7	12	DBMS(1) Rule:1
8	PEC	PE-1	23DSB3101R	DATA ANALYTICS AND VISUALIZATION	R	DAV	3	0	2	4	5	9	DBMS(1) Rule:1
9	PEC	PE-2	23DSB3202A	DATA WAREHOUSING AND MINING	A	DWHM	5	0	0	0	5	5	DBMS(1)
10	PEC	PE-2	23DSB3202E	DATA WAREHOUSING AND MINING	E	DWHM	5	0	0	0	5	5	DBMS(1)
11	PEC	PE-2	23DSB3202R	DATA WAREHOUSING AND MINING	R	DWHM	3	0	0	0	3	3	DBMS(1)
12	PEC	PE-3	23DSB3303A	BIG DATA ANALYTICS	A	BDA	4	0	4	4	7	12	DBMS(1)
13	PEC	PE-3	23DSB3303E	BIG DATA ANALYTICS	E	BDA	4	0	4	4	7	12	DBMS(1)
14	PEC	PE-3	23DSB3303R	BIG DATA ANALYTICS	R	BDA	3	0	2	4	5	9	DBMS(1)
15	PEC	PE-4	23DSB3404M	BIG DATA OPTIMIZATION	M	BDO	3	0	0	0	3	3	DBMS(1)

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
16	PEC	PE-4	23DSB3407M	DATA ANALYTICS ON CLOUD	M	DAC	3	0	0	0	3	3	DBMS(1)
17	PEC	PE-4	23DSB3408M	DIGITAL MEDIA ANALYTICS	M	DMA	3	0	0	0	3	3	DBMS(1)
18	PEC	PE-4	23DSB3406M	DIGITAL VIDEO PROCESSING	M	DVP	3	0	0	0	3	3	DBMS(1)
19	PEC	PE-5	23DSB3509	ADVANCED DATABASES	R	AD	3	0	0	0	3	3	DBMS(1) Rule:1
20	PEC	PE-5	23DSB3510	BUSINESS ANALYTICS	R	BA	3	0	0	0	3	3	DBMS(1)
21	PEC	PE-5	23DSB3511	GRAPH & WEB ANALYTICS	R	GWA	3	0	0	0	3	3	DBMS(1)
22	PEC	PE-5	23DSB3512	DATA GOVERNANCE ON CLOUD	R	DGC	3	0	0	0	3	3	DBMS(1)

58. SPECIALIZATION: DISTRIBUTED LEDGER ANALYTICS

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDAD08A	EMERGING BLOCKCHAIN MODELS FOR DIGITAL CURRENCIES	A	EBCMDC	0	0	6	4	4	10	DBMS(1) Rule:1
2	SDC	SDP-4	23SDAD08E	EMERGING BLOCKCHAIN MODELS FOR DIGITAL CURRENCIES	E	EBCMDC	0	0	6	4	4	10	DBMS(1) Rule:1
3	SDC	SDP-4	23SDAD08R	EMERGING BLOCKCHAIN MODELS FOR DIGITAL CURRENCIES	R	EBCMDC	0	0	2	4	2	6	DBMS(1) Rule:1
4	PEC	PE-1	23DLA3101A	SYSTEM & NETWORK TRAFFIC SECURITY ANALYTICS	A	SNTSA	4	0	4	4	7	12	DBMS(1)
5	PEC	PE-1	23DLA3101E	SYSTEM & NETWORK TRAFFIC SECURITY ANALYTICS	E	SNTSA	4	0	4	4	7	12	DBMS(1)
6	PEC	PE-1	23DLA3101R	SYSTEM & NETWORK TRAFFIC SECURITY ANALYTICS	R	SNTSA	3	0	2	4	5	9	DBMS(1)
7	PEC	PE-2	23DLA3202A	AUTOMATED NETWORK ANALYSIS	A	ANA	5	0	0	0	5	5	DBMS(1)
8	PEC	PE-2	23DLA3202E	AUTOMATED NETWORK ANALYSIS	E	ANA	5	0	0	0	5	5	DBMS(1)
9	PEC	PE-2	23DLA3202R	AUTOMATED NETWORK ANALYSIS	R	ANA	3	0	0	0	3	3	DBMS(1)
10	PEC	PE-2	23DLA3203A	BLOCKCHAIN TECHNOLOGY FOR DIGITAL TRANSFORMATION	A	BTDT	5	0	0	0	5	5	DBMS(1)
11	PEC	PE-2	23DLA3203E	BLOCKCHAIN TECHNOLOGY FOR DIGITAL TRANSFORMATION	E	BTDT	5	0	0	0	5	5	DBMS(1)
12	PEC	PE-2	23DLA3203R	BLOCKCHAIN TECHNOLOGY FOR DIGITAL TRANSFORMATION	R	BTDT	3	0	0	0	3	3	DBMS(1)
13	PEC	PE-3	23DLA3304A	MULTI AGENT SYSTEMS	A	MAS	4	0	4	4	7	12	DBMS(1)
14	PEC	PE-3	23DLA3304E	MULTI AGENT SYSTEMS	E	MAS	4	0	4	4	7	12	DBMS(1)
15	PEC	PE-3	23DLA3304R	MULTI AGENT SYSTEMS	R	MAS	3	0	2	4	5	9	DBMS(1)
16	PEC	PE-4	23DLA3405M	BLOCKCHAIN ANALYTICS	M	BCA	3	0	0	0	3	3	DBMS(1)
17	PEC	PE-4	23DLA3406M	CRYPTOCURRENCY AND BLOCKCHAIN TECHNOLOGY	M	CBT	3	0	0	0	3	3	DBMS(1)

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
18	PEC	PE-5	23DLA3507	DISTRIBUTED LEDGER ARCHITECTURE FOR AUTOMATION	R	DLAA	3	0	0	0	3	3	DBMS(1)
19	PEC	PE-5	23DLA3508	PERMISSIONED DISTRIBUTED LEDGER	R	PDL	3	0	0	0	3	3	DBMS(1)

59. SPECIALIZATION: EMBEDDED SYSTEMS

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDEC14A	EMBEDDED PROTOTYPE	A	EPT	0	0	6	4	4	10	
2	SDC	SDP-4	23SDEC14E	EMBEDDED PROTOTYPE	E	EPT	0	0	6	4	4	10	
3	SDC	SDP-4	23SDEC14R	EMBEDDED PROTOTYPE	R	EPT	0	0	2	4	2	6	
4	PEC	PE-1	23EDS3101A	ADVANCED EMBEDDED SYSTEMS	A	AES	4	0	4	4	7	12	
5	PEC	PE-1	23EDS3101E	ADVANCED EMBEDDED SYSTEMS	E	AES	4	0	4	4	7	12	
6	PEC	PE-1	23EDS3101R	ADVANCED EMBEDDED SYSTEMS	R	AES	3	0	2	4	5	9	
7	PEC	PE-2	23EDS3202A	EMBEDDED SYSTEMS FOR IOT	A	ESIOT	5	0	0	0	5	5	
8	PEC	PE-2	23EDS3202E	EMBEDDED SYSTEMS FOR IOT	E	ESIOT	5	0	0	0	5	5	
9	PEC	PE-2	23EDS3202R	EMBEDDED SYSTEMS FOR IOT	R	ESIOT	3	0	0	0	3	3	
10	PEC	PE-3	23EDS3303A	EMBEDDED AND REAL-TIME SYSTEMS	A	ERTS	4	0	4	4	7	12	
11	PEC	PE-3	23EDS3303E	EMBEDDED AND REAL-TIME SYSTEMS	E	ERTS	4	0	4	4	7	12	
12	PEC	PE-3	23EDS3303R	EMBEDDED AND REAL-TIME SYSTEMS	R	ERTS	3	0	2	4	5	9	
13	PEC	PE-4	23EDS3404M	CLOUD ARCHITECTURE IN IOT	M	CAIOT	3	0	0	0	3	3	
14	PEC	PE-5	23EDS3505	EDGE COMPUTING & DATA ANALYTICS IN IOT	R	ECDAI	3	0	0	0	3	3	

60. SPECIALIZATION: E-MOBILITY ENGINEERING

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDEE06A	EVT HARDWARE PROTOTYPING	A	EVTHP	0	0	6	4	4	10	
2	SDC	SDP-4	23SDEE06E	EVT HARDWARE PROTOTYPING	E	EVTHP	0	0	6	4	4	10	
3	SDC	SDP-4	23SDEE06R	EVT HARDWARE PROTOTYPING	R	EVTHP	0	0	2	4	2	6	
4	PEC	PE-1	23EME3101A	POWER TRAIN DESIGN FOR ELECTRIC VEHICLE	A	PTDEV	4	0	4	4	7	12	
5	PEC	PE-1	23EME3101E	POWER TRAIN DESIGN FOR ELECTRIC VEHICLE	E	PTDEV	4	0	4	4	7	12	
6	PEC	PE-1	23EME3101R	POWER TRAIN DESIGN FOR ELECTRIC VEHICLE	R	PTDEV	3	0	2	4	5	9	
7	PEC	PE-2	23EME3202A	COMMUNICATION PROTOCOLS & TESTING OF ELECTRIC VEHICLE	A	CPTEV	5	0	0	0	5	5	

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
8	PEC	PE-2	23EME3202E	COMMUNICATION PROTOCOLS & TESTING OF ELECTRIC VEHICLE	E	CPTEV	5	0	0	0	5	5	
9	PEC	PE-2	23EME3202R	COMMUNICATION PROTOCOLS & TESTING OF ELECTRIC VEHICLE	R	CPTEV	3	0	0	0	3	3	
10	PEC	PE-2	23EME3203A	AUTONOMOUS VEHICLES & AUTOMOTIVE ELECTRONICS	A	AVAE	5	0	0	0	5	5	
11	PEC	PE-2	23EME3203E	AUTONOMOUS VEHICLES & AUTOMOTIVE ELECTRONICS	E	AVAE	5	0	0	0	5	5	
12	PEC	PE-2	23EME3203R	AUTONOMOUS VEHICLES & AUTOMOTIVE ELECTRONICS	R	AVAE	3	0	0	0	3	3	
13	PEC	PE-3	23EME3304A	CHARGING STATIONS FOR ELECTRIC VEHICLES	A	CSEV	4	0	4	4	7	12	
14	PEC	PE-3	23EME3304E	CHARGING STATIONS FOR ELECTRIC VEHICLES	E	CSEV	4	0	4	4	7	12	
15	PEC	PE-3	23EME3304R	CHARGING STATIONS FOR ELECTRIC VEHICLES	R	CSEV	3	0	2	4	5	9	
16	PEC	PE-4	23EME3405M	INTRODUCTION TO BATTERY-MANAGEMENT SYSTEMS	M	IBMS	3	0	0	0	3	3	
17	PEC	PE-4	23EME3406M	BATTERY STATE ESTIMATION ALGORITHMS FOR ELECTRIC VEHICLE	M	BSAEV	3	0	0	0	3	3	
18	PEC	PE-5	23EME3507	AI AND IOT FOR ELECTRIC VEHICLE	R	AIEV	3	0	0	0	3	3	
19	PEC	PE-5	23EME3508	EV SYSTEM AND WIRING DESIGN	R	ESWS	3	0	0	0	3	3	

61. SPECIALIZATION: ENGINEERING DESIGN

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDME05A	3D MODELLING AND DIGITAL PROTOTYPING	A	3DMDP	0	0	6	4	4	10	
2	SDC	SDP-4	23SDME05E	3D MODELLING AND DIGITAL PROTOTYPING	E	3DMDP	0	0	6	4	4	10	
3	SDC	SDP-4	23SDME05R	3D MODELLING AND DIGITAL PROTOTYPING	R	3DMDP	0	0	2	4	2	6	
4	PEC	PE-1	23EGD3101A	MODELING ANALYSIS & DESIGN OF ROBOTIC SYSTEMS	A	MADRS	4	0	4	4	7	12	EM(1) Rule:1
5	PEC	PE-1	23EGD3101E	MODELING ANALYSIS & DESIGN OF ROBOTIC SYSTEMS	E	MADRS	4	0	4	4	7	12	EM(1) Rule:1
6	PEC	PE-1	23EGD3101R	MODELING ANALYSIS & DESIGN OF ROBOTIC SYSTEMS	R	MADRS	3	0	2	4	5	9	EM(1) Rule:1
7	PEC	PE-2	23EGD3202A	CREEP, FATIGUE AND FRACTURE MECHANICS	A	CFFM	5	0	0	0	5	5	EM(1) Rule:1
8	PEC	PE-2	23EGD3202E	CREEP, FATIGUE AND FRACTURE MECHANICS	E	CFFM	5	0	0	0	5	5	EM(1) Rule:1
9	PEC	PE-2	23EGD3202R	CREEP, FATIGUE AND FRACTURE MECHANICS	R	CFFM	3	0	0	0	3	3	EM(1) Rule:1

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
10	PEC	PE-2	23EGD3203A	THEORY OF ELASTICITY AND PLASTICITY	A	TEP	5	0	0	0	5	5	EM(1) Rule:1
11	PEC	PE-2	23EGD3203E	THEORY OF ELASTICITY AND PLASTICITY	E	TEP	5	0	0	0	5	5	EM(1) Rule:1
12	PEC	PE-2	23EGD3203R	THEORY OF ELASTICITY AND PLASTICITY	R	TEP	3	0	0	0	3	3	EM(1) Rule:1
13	PEC	PE-3	23EGD3304A	SUSTAINABLE DESIGN & SOCIAL INNOVATION IN ENGINEERING DESIGN	A	SDSIED	4	0	4	4	7	12	EM(1) Rule:1
14	PEC	PE-3	23EGD3304E	SUSTAINABLE DESIGN & SOCIAL INNOVATION IN ENGINEERING DESIGN	E	SDSIED	4	0	4	4	7	12	EM(1) Rule:1
15	PEC	PE-3	23EGD3304R	SUSTAINABLE DESIGN & SOCIAL INNOVATION IN ENGINEERING DESIGN	R	SDSIED	3	0	2	4	5	9	EM(1) Rule:1
16	PEC	PE-4	23EGD3405M	ADVANCED VIBRATIONS	M	AV	3	0	0	0	3	3	EM(1) Rule:1
17	PEC	PE-4	23EGD3406M	MECHANICS OF COMPOSITE MATERIALS	M	MOCM	3	0	0	0	3	3	EM(1) Rule:1
18	PEC	PE-5	23EGD3507	ADVANCED STRENGTH OF MATERIALS	R	ASM	3	0	0	0	3	3	EM(1) Rule:1
19	PEC	PE-5	23EGD3508	HYBRID AND ELECTRIC VEHICLE DESIGN	R	HEVD	3	0	0	0	3	3	EM(1) Rule:1

62. SPECIALIZATION: GAME DEVELOPMENT AND UX DESIGN

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	FCC	FC-1	23CS2221F	UX DESIGN	F	UD	2	0	2	0	3	4	CTSD(1) Rule:1
2	SDC	SDP-4	23SDCS06A	CERTIFIED GAME DEVELOPER	A	CGD	0	0	6	4	4	10	CTSD(1)
3	SDC	SDP-4	23SDCS06E	CERTIFIED GAME DEVELOPER	E	CGD	0	0	6	4	4	10	CTSD(1)
4	SDC	SDP-4	23SDCS06R	CERTIFIED GAME DEVELOPER	R	CGD	0	0	2	4	2	6	CTSD(1)
5	PEC	PE-1	23GDU3101A	PROGRAMMING FOR GAME DEVELOPMENT	A	PGD	4	0	4	4	7	12	CTSD(1)
6	PEC	PE-1	23GDU3101E	PROGRAMMING FOR GAME DEVELOPMENT	E	PGD	4	0	4	4	7	12	CTSD(1)
7	PEC	PE-1	23GDU3101R	PROGRAMMING FOR GAME DEVELOPMENT	R	PGD	3	0	2	4	5	9	CTSD(1)
8	PEC	PE-2	23GDU3202A	UX DESIGN	A	UD	5	0	0	0	5	5	CTSD(1) Rule:1
9	PEC	PE-2	23GDU3202E	UX DESIGN	E	UD	5	0	0	0	5	5	CTSD(1) Rule:1
10	PEC	PE-2	23GDU3202R	UX DESIGN	R	UD	3	0	0	0	3	3	CTSD(1) Rule:1
11	PEC	PE-3	23GDU3303A	AR & VR APPLICATION DEVELOPMENT	A	AR&VR	4	0	4	4	7	12	CTSD(1)

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
12	PEC	PE-3	23GDU3303E	AR & VR APPLICATION DEVELOPMENT	E	AR&VR	4	0	4	4	7	12	CTSD(1)
13	PEC	PE-3	23GDU3303R	AR & VR APPLICATION DEVELOPMENT	R	AR&VR	3	0	2	4	5	9	CTSD(1)
14	PEC	PE-4	23GDU3404M	COMPUTER GRAPHICS	M	CG	3	0	0	0	3	3	CTSD(1)
15	PEC	PE-4	23GDU3405M	3D MODELLING & ANIMATION	M	3DMA	3	0	0	0	3	3	CTSD(1)
16	PEC	PE-5	23GDU3506	PRINCIPLES OF GAME DESIGN	R	PRGD	3	0	0	0	3	3	CTSD(1)
17	PEC	PE-5	23GDU3507	BUSINESS OF GAMES & ENTREPRENEURSHIP	R	BGE	3	0	0	0	3	3	CTSD(1)

63. SPECIALIZATION: GEOTECHNICAL AND TRANSPORTATION ENGINEERING

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDCE05A	DESIGN OF THE PAVEMENT BY USING MX ROADS	A	DPMX	0	0	6	4	4	10	
2	SDC	SDP-4	23SDCE05E	DESIGN OF THE PAVEMENT BY USING MX ROADS	E	DPMX	0	0	6	4	4	10	
3	SDC	SDP-4	23SDCE05R	DESIGN OF THE PAVEMENT BY USING MX ROADS	R	DPMX	0	0	2	4	2	6	
4	PEC	PE-1	23GTE3101A	ADVANCED FOUNDATION ENGINEERING	A	AFE	4	0	4	4	7	12	
5	PEC	PE-1	23GTE3101E	ADVANCED FOUNDATION ENGINEERING	E	AFE	4	0	4	4	7	12	
6	PEC	PE-1	23GTE3101R	ADVANCED FOUNDATION ENGINEERING	R	AFE	3	0	2	4	5	9	
7	PEC	PE-2	23GTE3202A	INTELLIGENT TRANSPORTATION SYSTEMS	A	ITS	5	0	0	0	5	5	
8	PEC	PE-2	23GTE3202E	INTELLIGENT TRANSPORTATION SYSTEMS	E	ITS	5	0	0	0	5	5	
9	PEC	PE-2	23GTE3202R	INTELLIGENT TRANSPORTATION SYSTEMS	R	ITS	3	0	0	0	3	3	
10	PEC	PE-2	23GTE3203A	GROUND IMPROVEMENT TECHNIQUES	A	GIT	5	0	0	0	5	5	
11	PEC	PE-2	23GTE3203E	GROUND IMPROVEMENT TECHNIQUES	E	GIT	5	0	0	0	5	5	
12	PEC	PE-2	23GTE3203R	GROUND IMPROVEMENT TECHNIQUES	R	GIT	3	0	0	0	3	3	
13	PEC	PE-3	23GTE3304A	PAVEMENT MATERIALS &DESIGN	A	PMD	4	0	4	4	7	12	
14	PEC	PE-3	23GTE3304E	PAVEMENT MATERIALS &DESIGN	E	PMD	4	0	4	4	7	12	
15	PEC	PE-3	23GTE3304R	PAVEMENT MATERIALS &DESIGN	R	PMD	3	0	2	4	5	9	
16	PEC	PE-4	23GTE3405M	URBAN TRANSPORTATION SYSTEMS PLANNING	M	UTSP	3	0	0	0	3	3	
17	PEC	PE-4	23GTE3406M	GEOTECHNICAL EARTHQUAKE ENGINEERING	M	BIE	3	0	0	0	3	3	

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
18	PEC	PE-5	23GTE3507	TRAFFIC ENGINEERING AND MANAGEMENT	R	TEM	3	0	0	0	3	3	
19	PEC	PE-5	23GTE3508	DESIGN OF EARTH RETAINING STRUCTURES	R	DERS	3	0	0	0	3	3	

64. SPECIALIZATION: GREEN ENERGY TECHNOLOGIES

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDEE05A	MODELING AND SIMULATION OF GREEN ENERGY SYSTEMS	A	GML	0	0	6	4	4	10	
2	SDC	SDP-4	23SDEE05E	MODELING AND SIMULATION OF GREEN ENERGY SYSTEMS	E	GML	0	0	6	4	4	10	
3	SDC	SDP-4	23SDEE05R	MODELING AND SIMULATION OF GREEN ENERGY SYSTEMS	R	GML	0	0	2	4	2	6	
4	PEC	PE-1	23GET3101A	SOLAR PV AND MICRO ENERGY TECHNOLOGIES	A	SPMET	4	0	4	4	7	12	
5	PEC	PE-1	23GET3101E	SOLAR PV AND MICRO ENERGY TECHNOLOGIES	E	SPMET	4	0	4	4	7	12	
6	PEC	PE-1	23GET3101R	SOLAR PV AND MICRO ENERGY TECHNOLOGIES	R	SPMET	3	0	2	4	5	9	
7	PEC	PE-2	23GET3202A	WIND AND ENERGY STORAGE TECHNOLOGIES	A	WEST	5	0	0	0	5	5	
8	PEC	PE-2	23GET3202E	WIND AND ENERGY STORAGE TECHNOLOGIES	E	WEST	5	0	0	0	5	5	
9	PEC	PE-2	23GET3202R	WIND AND ENERGY STORAGE TECHNOLOGIES	R	WEST	3	0	0	0	3	3	
10	PEC	PE-2	23GET3203A	ENERGY ECONOMICS AND POLICY	A	EEP	5	0	0	0	5	5	
11	PEC	PE-2	23GET3203E	ENERGY ECONOMICS AND POLICY	E	EEP	5	0	0	0	5	5	
12	PEC	PE-2	23GET3203R	ENERGY ECONOMICS AND POLICY	R	EEP	3	0	0	0	3	3	
13	PEC	PE-3	23GET3304A	GRID INTEGRATION OF RENEWABLE ENERGY SOURCES	A	GIRES	4	0	4	4	7	12	
14	PEC	PE-3	23GET3304E	GRID INTEGRATION OF RENEWABLE ENERGY SOURCES	E	GIRES	4	0	4	4	7	12	
15	PEC	PE-3	23GET3304R	GRID INTEGRATION OF RENEWABLE ENERGY SOURCES	R	GIRES	3	0	2	4	5	9	
16	PEC	PE-4	23GET3405M	ENERGY MANAGEMENT AND GREEN BUILDING	M	EMGB	3	0	0	0	3	3	
17	PEC	PE-4	23GET3406M	HYDROGEN FUEL CELL TECHNOLOGY	M	HFCT	3	0	0	0	3	3	
18	PEC	PE-5	23GET3507	AI AND IOT FOR GREEN ENERGY INTEGRATION	R	AIGEI	3	0	0	0	3	3	
19	PEC	PE-5	23GET3508	POWER ELECTRONICS FOR RENEWABLE ENERGY SYSTEMS	R	PERES	3	0	0	0	3	3	

65. SPECIALIZATION: HARDWARE-SOFTWARE CO-DESIGN FOR SECURITY

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDCS21A	HARDWARE SOFTWARE CODESIGN IMPLEMENTATION FOR SECURITY SPECILAITY	A	HSCISS	0	0	6	4	4	10	
2	SDC	SDP-4	23SDCS21E	HARDWARE SOFTWARE CODESIGN IMPLEMENTATION FOR SECURITY SPECILAITY	E	HSCISS	0	0	6	4	4	10	
3	SDC	SDP-4	23SDCS21R	HARDWARE SOFTWARE CODESIGN IMPLEMENTATION FOR SECURITY SPECILAITY	R	HSCISS	0	0	2	4	2	6	
4	PEC	PE-1	23HSS3101A	INTRODUCTION TO HARDWARE-SOFTWARE CO-DESIGN	A	ITHSCD	4	0	4	4	7	12	
5	PEC	PE-1	23HSS3101E	INTRODUCTION TO HARDWARE-SOFTWARE CO-DESIGN	E	ITHSCD	4	0	4	4	7	12	
6	PEC	PE-1	23HSS3101R	INTRODUCTION TO HARDWARE-SOFTWARE CO-DESIGN	R	ITHSCD	3	0	2	4	5	9	
7	PEC	PE-2	23HSS3202A	FUNDAMENTALS OF COMPUTER SECURITY	A	FOCS	5	0	0	0	5	5	
8	PEC	PE-2	23HSS3202E	FUNDAMENTALS OF COMPUTER SECURITY	E	FOCS	5	0	0	0	5	5	
9	PEC	PE-2	23HSS3202R	FUNDAMENTALS OF COMPUTER SECURITY	R	FOCS	3	0	0	0	3	3	
10	PEC	PE-3	23HSS3303A	HARDWARE SECURITY MECHANISMS	A	HSM	4	0	4	4	7	12	
11	PEC	PE-3	23HSS3303E	HARDWARE SECURITY MECHANISMS	E	HSM	4	0	4	4	7	12	
12	PEC	PE-3	23HSS3303R	HARDWARE SECURITY MECHANISMS	R	HSM	3	0	2	4	5	9	
13	PEC	PE-4	23HSS3404M	SOFTWARE SECURITY PRINCIPLES	M	SSP	3	0	0	0	3	3	
14	PEC	PE-4	23HSS3405M	SECURE CO-DESIGN METHODOLOGIES	M	SCDM	3	0	0	0	3	3	
15	PEC	PE-4	23HSS3406M	HARDWARE TROJAN DETECTION AND PREVENTION	M	HTDAP	3	0	0	0	3	3	
16	PEC	PE-5	23HSS3507	SIDE-CHANNEL ATTACKS AND COUNTERMEASURES	R	SCACM	3	0	0	0	3	3	
17	PEC	PE-5	23HSS3508	FORMAL METHODS FOR HARDWARE-SOFTWARE SECURITY	R	FMHSS	3	0	0	0	3	3	
18	PEC	PE-5	23HSS3509	TRUSTED PLATFORM MODULES AND SECURE ELEMENTS	R	TPMASE	3	0	0	0	3	3	
19	PEC	PE-5	23HSS3510	SECURE COMMUNICATION PROTOCOLS AND APIS	R	SCPA	3	0	0	0	3	3	
20	PEC	PE-5	23HSS3511	HARDWARE-SOFTWARE CO-DESIGN	R	HSCCS	3	0	0	0	3	3	
21	PEC	PE-5	23HSS3512	FUTURE TRENDS IN HARDWARE-SOFTWARE CO-DESIGN FOR SECURITY	R	FTHSCS	3	0	0	0	3	3	

66. SPECIALIZATION: HEALTHCARE DATA ANALYTICS

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDBT08A	DRUG DESIGN: PRINCIPLES AND ENGINEERING	A	DDPE	0	0	6	4	4	10	CDS(1) Rule:1
2	SDC	SDP-4	23SDBT08E	DRUG DESIGN: PRINCIPLES AND ENGINEERING	E	DDPE	0	0	6	4	4	10	CDS(1) Rule:1
3	SDC	SDP-4	23SDBT08R	DRUG DESIGN: PRINCIPLES AND ENGINEERING	R	DDPE	0	0	2	4	2	6	CDS(1) Rule:1
4	PEC	PE-1	23HDA3101A	INTELLIGENT SYSTEMS FOR DISEASE PREDICTION & DRUG DISCOVERY	A	ISDPDD	4	0	4	4	7	12	CDS(1) Rule:1
5	PEC	PE-1	23HDA3101E	INTELLIGENT SYSTEMS FOR DISEASE PREDICTION & DRUG DISCOVERY	E	ISDPDD	4	0	4	4	7	12	CDS(1) Rule:1
6	PEC	PE-1	23HDA3101R	INTELLIGENT SYSTEMS FOR DISEASE PREDICTION & DRUG DISCOVERY	R	ISDPDD	3	0	2	4	5	9	CDS(1) Rule:1
7	PEC	PE-2	23HDA3202A	BIO MEDICAL INFORMATICS	A	BMI	5	0	0	0	5	5	CDS(1)
8	PEC	PE-2	23HDA3202E	BIO MEDICAL INFORMATICS	E	BMI	5	0	0	0	5	5	CDS(1)
9	PEC	PE-2	23HDA3202R	BIO MEDICAL INFORMATICS	R	BMI	3	0	0	0	3	3	CDS(1)
10	PEC	PE-2	23HDA3203A	GENETIC PROGRAMMING	A	GP	5	0	0	0	5	5	CDS(1)
11	PEC	PE-2	23HDA3203E	GENETIC PROGRAMMING	E	GP	5	0	0	0	5	5	CDS(1)
12	PEC	PE-2	23HDA3203R	GENETIC PROGRAMMING	R	GP	3	0	0	0	3	3	CDS(1)
13	PEC	PE-2	23HDA3204A	PYTHON FOR GENOMIC DATA SCIENCE	A	PFGDS	5	0	0	0	5	5	CDS(1)
14	PEC	PE-2	23HDA3204E	PYTHON FOR GENOMIC DATA SCIENCE	E	PFGDS	5	0	0	0	5	5	CDS(1)
15	PEC	PE-2	23HDA3204R	PYTHON FOR GENOMIC DATA SCIENCE	R	PFGDS	3	0	0	0	3	3	CDS(1)
16	PEC	PE-3	23HDA3305A	COMPUTATIONAL NEUROSCIENCE	A	CNSE	4	0	4	4	7	12	CDS(1) Rule:1
17	PEC	PE-3	23HDA3305E	COMPUTATIONAL NEUROSCIENCE	E	CNSE	4	0	4	4	7	12	CDS(1) Rule:1
18	PEC	PE-3	23HDA3305R	COMPUTATIONAL NEUROSCIENCE	R	CNSE	3	0	2	4	5	9	CDS(1) Rule:1
19	PEC	PE-4	23HDA3407M	INTRODUCTION TO GENOMIC TECHNOLOGIES	M	IGT	3	0	0	0	3	3	CDS(1) Rule:1
20	PEC	PE-4	23HDA3406M	NGS SEQUENCING AND DATA ANALYSIS	M	NGSDA	3	0	0	0	3	3	CDS(1)
21	PEC	PE-5	23HDA3511	EPI GENETIC CONTROL OF GENE EXPRESSION	R	ECBI	3	0	0	0	3	3	CDS(1) Rule:1
22	PEC	PE-5	23HDA3509	MOLECULAR MODELING AND DRUG DESIGN	R	MMDD	3	0	0	0	3	3	CDS(1)
23	PEC	PE-5	23HDA3510	GENOMIC DATA SCIENCE & CLUSTERING	R	GDSC	3	0	0	0	3	3	CDS(1)

67. SPECIALIZATION: INDUSTRIAL AUTOMATION

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDEE07A	IA HARDWARE PROTOTYPING	A	IAHP	0	0	6	4	4	10	ELM(1)
2	SDC	SDP-4	23SDEE07E	IA HARDWARE PROTOTYPING	E	IAHP	0	0	6	4	4	10	ELM(1)
3	SDC	SDP-4	23SDEE07R	IA HARDWARE PROTOTYPING	R	IAHP	0	0	2	4	2	6	ELM(1)
4	PEC	PE-1	23ILA3101A	INTRODUCTION TO INDUSTRIAL INTERNET OF THINGS	A	IIIoT	4	0	4	4	7	12	ELM(1)
5	PEC	PE-1	23ILA3101E	INTRODUCTION TO INDUSTRIAL INTERNET OF THINGS	E	IIIoT	4	0	4	4	7	12	ELM(1)
6	PEC	PE-1	23ILA3101R	INTRODUCTION TO INDUSTRIAL INTERNET OF THINGS	R	IIIoT	3	0	2	4	5	9	ELM(1)
7	PEC	PE-2	23ILA3202A	INDUSTRIAL AUTOMATION AND ROBOTICS	A	IAR	5	0	0	0	5	5	ELM(1)
8	PEC	PE-2	23ILA3202E	INDUSTRIAL AUTOMATION AND ROBOTICS	E	IAR	5	0	0	0	5	5	ELM(1)
9	PEC	PE-2	23ILA3202R	INDUSTRIAL AUTOMATION AND ROBOTICS	R	IAR	3	0	0	0	3	3	ELM(1)
10	PEC	PE-2	23ILA3203A	EDGE COMPUTING FOR INDUSTRY 4.0	A	ECI	5	0	0	0	5	5	ELM(1)
11	PEC	PE-2	23ILA3203E	EDGE COMPUTING FOR INDUSTRY 4.0	E	ECI	5	0	0	0	5	5	ELM(1)
12	PEC	PE-2	23ILA3203R	EDGE COMPUTING FOR INDUSTRY 4.0	R	ECI	3	0	0	0	3	3	ELM(1)
13	PEC	PE-3	23ILA3304A	INDUSTRIAL DRIVES AND CONTROL	A	ID&C	4	0	4	4	7	12	ELM(1)
14	PEC	PE-3	23ILA3304E	INDUSTRIAL DRIVES AND CONTROL	E	ID&C	4	0	4	4	7	12	ELM(1)
15	PEC	PE-3	23ILA3304R	INDUSTRIAL DRIVES AND CONTROL	R	ID&C	3	0	2	4	5	9	ELM(1)
16	PEC	PE-4	23ILA3405M	INDUSTRIAL COMMUNICATION PROTOCOLS AND CYBER SECURITY	M	ICPCS	3	0	0	0	3	3	ELM(1)
17	PEC	PE-4	23ILA3406M	DIGITAL MANUFACTURING AND DESIGN	M	DMD	3	0	0	0	3	3	ELM(1)
18	PEC	PE-5	23ILA3508	PLC PROGRAMMING & CONTROL	R	PLCPC	3	0	0	0	3	3	ELM(1)
19	PEC	PE-5	23ILA3507	SMART SENSORS AND SENSOR NETWORKING	R	SSSN	3	0	0	0	3	3	ELM(1)

68. SPECIALIZATION: INDUSTRIAL BIOTECHNOLOGY

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDBT07A	ASPECTS OF BIOCHEMICAL ENGINEERING	A	ABCE	0	0	6	4	4	10	
2	SDC	SDP-4	23SDBT07E	ASPECTS OF BIOCHEMICAL ENGINEERING	E	ABCE	0	0	6	4	4	10	
3	SDC	SDP-4	23SDBT07R	ASPECTS OF BIOCHEMICAL ENGINEERING	R	ABCE	0	0	2	4	2	6	

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
4	PEC	PE-1	23IBT3101A	PHARMACEUTICAL BIOTECHNOLOGY	A	PBT	4	0	4	4	7	12	
5	PEC	PE-1	23IBT3101E	PHARMACEUTICAL BIOTECHNOLOGY	E	PBT	4	0	4	4	7	12	
6	PEC	PE-1	23IBT3101R	PHARMACEUTICAL BIOTECHNOLOGY	R	PBT	3	0	2	4	5	9	
7	PEC	PE-2	23IBT3202A	PHARMACOVIGILANCE AND SAFETY	A	PCS	5	0	0	0	5	5	
8	PEC	PE-2	23IBT3202E	PHARMACOVIGILANCE AND SAFETY	E	PCS	5	0	0	0	5	5	
9	PEC	PE-2	23IBT3202R	PHARMACOVIGILANCE AND SAFETY	R	PCS	3	0	0	0	3	3	
10	PEC	PE-2	23IBT3203A	BIOPROCESS ECONOMICS AND PLANT DESIGN	A	BEPD	5	0	0	0	5	5	
11	PEC	PE-2	23IBT3203E	BIOPROCESS ECONOMICS AND PLANT DESIGN	E	BEPD	5	0	0	0	5	5	
12	PEC	PE-2	23IBT3203R	BIOPROCESS ECONOMICS AND PLANT DESIGN	R	BEPD	3	0	0	0	3	3	
13	PEC	PE-3	23IBT3304A	ENZYME ENGINEERING	A	EE	4	0	4	4	7	12	
14	PEC	PE-3	23IBT3304E	ENZYME ENGINEERING	E	EE	4	0	4	4	7	12	
15	PEC	PE-3	23IBT3304R	ENZYME ENGINEERING	R	EE	3	0	2	4	5	9	
16	PEC	PE-4	23IBT3405M	BIOPROCESS VALIDATION AND CGMP	M	BVcGMP	3	0	0	0	3	3	
17	PEC	PE-4	23IBT3406M	FOOD TECHNOLOGY	M	FT	3	0	0	0	3	3	
18	PEC	PE-5	23IBT3507	MICROBIAL TECHNOLOGY	R	MBT	3	0	0	0	3	3	
19	PEC	PE-5	23IBT3508	METABOLIC ENGINEERING	R	MBE	3	0	0	0	3	3	

69. SPECIALIZATION: INTELLIGENT MULTIMEDIA PROCESSING

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDEC15A	NATURAL LANGUAGE PROCESSING USING TENSOR FLOW	A	NLPTF	0	0	6	4	4	10	DNA(1)
2	SDC	SDP-4	23SDEC15E	NATURAL LANGUAGE PROCESSING USING TENSOR FLOW	E	NLPTF	0	0	6	4	4	10	DNA(1)
3	SDC	SDP-4	23SDEC15R	NATURAL LANGUAGE PROCESSING USING TENSOR FLOW	R	NLPTF	0	0	2	4	2	6	DNA(1)
4	PEC	PE-1	23IMP3101A	NATURAL LANGUAGE PROCESSING & APPLICATIONS	A	NLPA	4	0	4	4	7	12	DNA(1)
5	PEC	PE-1	23IMP3101E	NATURAL LANGUAGE PROCESSING & APPLICATIONS	E	NLPA	4	0	4	4	7	12	DNA(1)
6	PEC	PE-1	23IMP3101R	NATURAL LANGUAGE PROCESSING & APPLICATIONS	R	NLPA	3	0	2	4	5	9	DNA(1)
7	PEC	PE-2	23IMP3202A	DATA ENGINEERING	A	DE	5	0	0	0	5	5	DNA(1)
8	PEC	PE-2	23IMP3202E	DATA ENGINEERING	E	DE	5	0	0	0	5	5	DNA(1)

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
9	PEC	PE-2	23IMP3202R	DATA ENGINEERING	R	DE	3	0	0	0	3	3	DNA(1)
10	PEC	PE-3	23IMP3303A	BIO MEDICAL SIGNAL AND IMAGE ANALYSIS	A	AMSIA	4	0	4	4	7	12	DNA(1)
11	PEC	PE-3	23IMP3303E	BIO MEDICAL SIGNAL AND IMAGE ANALYSIS	E	AMSIA	4	0	4	4	7	12	DNA(1)
12	PEC	PE-3	23IMP3303R	BIO MEDICAL SIGNAL AND IMAGE ANALYSIS	R	AMSIA	3	0	2	4	5	9	DNA(1)
13	PEC	PE-4	23IMP3404M	DATA VISUALIZATION	M	DV	3	0	0	0	3	3	DNA(1)
14	PEC	PE-5	23IMP3505	MULTI MEDIA PROCESSING	R	MMP	3	0	0	0	3	3	DNA(1)
15	PEC	PE-5	23IMP3506	INTRODUCTION TO QUANTUM COMPUTING	R	MMP	3	0	0	0	3	3	DNA(1)

70. SPECIALIZATION: IOT ANALYTICS

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDIN05A	IOT ANALYTICS FOR THE CLOUD	A	IOTAC	0	0	6	4	4	10	FITS(1)
2	SDC	SDP-4	23SDIN05E	IOT ANALYTICS FOR THE CLOUD	E	IOTAC	0	0	6	4	4	10	FITS(1)
3	SDC	SDP-4	23SDIN05R	IOT ANALYTICS FOR THE CLOUD	R	IOTAC	0	0	2	4	2	6	FITS(1)
4	PEC	PE-1	23IOT3101A	INDUSTRIAL INTERNET OF THINGS	A	IIOT	4	0	4	4	7	12	FITS(1)
5	PEC	PE-1	23IOT3101E	INDUSTRIAL INTERNET OF THINGS	E	IIOT	4	0	4	4	7	12	FITS(1)
6	PEC	PE-1	23IOT3101R	INDUSTRIAL INTERNET OF THINGS	R	IIOT	3	0	2	4	5	9	FITS(1)
7	PEC	PE-2	23IOT3202A	EDGE COMPUTING	A	EC	5	0	0	0	5	5	FITS(1)
8	PEC	PE-2	23IOT3202E	EDGE COMPUTING	E	EC	5	0	0	0	5	5	FITS(1)
9	PEC	PE-2	23IOT3202R	EDGE COMPUTING	R	EC	3	0	0	0	3	3	FITS(1)
10	PEC	PE-2	23IOT3203A	PRECISION AGRICULTURE	A	PA	5	0	0	0	5	5	FITS(1)
11	PEC	PE-2	23IOT3203E	PRECISION AGRICULTURE	E	PA	5	0	0	0	5	5	FITS(1)
12	PEC	PE-2	23IOT3203R	PRECISION AGRICULTURE	R	PA	3	0	0	0	3	3	FITS(1)
13	PEC	PE-2	23IOT3204A	SMART FARMING	A	SF	5	0	0	0	5	5	FITS(1)
14	PEC	PE-2	23IOT3204E	SMART FARMING	E	SF	5	0	0	0	5	5	FITS(1)
15	PEC	PE-2	23IOT3204R	SMART FARMING	R	SF	3	0	0	0	3	3	FITS(1)
16	PEC	PE-3	23IOT3305A	DEEP LEARNING	A	DL	4	0	4	4	7	12	FITS(1) Rule:1
17	PEC	PE-3	23IOT3305E	DEEP LEARNING	E	DL	4	0	4	4	7	12	FITS(1) Rule:1
18	PEC	PE-3	23IOT3305R	DEEP LEARNING	R	DL	3	0	2	4	5	9	FITS(1) Rule:1
19	PEC	PE-4	23IOT3406M	DATA VISUALIZATION TECHNIQUES	M	DVT	3	0	0	0	3	3	

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
20	PEC	PE-4	23IOT3407M	APPLIED MACHINE LEARNING FOR AGRICULTURE	M	AMLA	3	0	0	0	3	3	FITS(1)
21	PEC	PE-5	23IOT3508	BIG DATA ANALYTICS	R	BDA	3	0	0	0	3	3	FITS(1)

71. SPECIALIZATION: MEDICAL BIOTECHNOLOGY

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDBT09A	CELL CULTURE TECHNOLOGIES	A	CCT	0	0	6	4	4	10	
2	SDC	SDP-4	23SDBT09E	CELL CULTURE TECHNOLOGIES	E	CCT	0	0	6	4	4	10	
3	SDC	SDP-4	23SDBT09R	CELL CULTURE TECHNOLOGIES	R	CCT	0	0	2	4	2	6	
4	PEC	PE-1	23MBT3101A	STEM CELL TECHNOLOGY	A	SCT	4	0	4	4	7	12	
5	PEC	PE-1	23MBT3101E	STEM CELL TECHNOLOGY	E	SCT	4	0	4	4	7	12	
6	PEC	PE-1	23MBT3101R	STEM CELL TECHNOLOGY	R	SCT	3	0	2	4	5	9	
7	PEC	PE-2	23MBT3202A	VIROLOGY	A	VR	5	0	0	0	5	5	
8	PEC	PE-2	23MBT3202E	VIROLOGY	E	VR	5	0	0	0	5	5	
9	PEC	PE-2	23MBT3202R	VIROLOGY	R	VR	3	0	0	0	3	3	
10	PEC	PE-2	23MBT3203A	NANOBIOTECHNOLOGY	A	NBT	5	0	0	0	5	5	
11	PEC	PE-2	23MBT3203E	NANOBIOTECHNOLOGY	E	NBT	5	0	0	0	5	5	
12	PEC	PE-2	23MBT3203R	NANOBIOTECHNOLOGY	R	NBT	3	0	0	0	3	3	
13	PEC	PE-3	23MBT3304A	HEALTHCARE BIOTECHNOLOGY	A	HBT	4	0	4	4	7	12	AIML(1)
14	PEC	PE-3	23MBT3304E	HEALTHCARE BIOTECHNOLOGY	E	HBT	4	0	4	4	7	12	AIML(1)
15	PEC	PE-3	23MBT3304R	HEALTHCARE BIOTECHNOLOGY	R	HBT	3	0	2	4	5	9	AIML(1)
16	PEC	PE-4	23MBT3405M	CANCER BIOLOGY	M	CB	3	0	0	0	3	3	
17	PEC	PE-4	23MBT3406M	NEUROBIOLOGY	M	NB	3	0	0	0	3	3	
18	PEC	PE-5	23MBT3507	BIOELECTRONICS & BIOSENSORS	R	BB	3	0	0	0	3	3	
19	PEC	PE-5	23MBT3508	TISSUE ENGINEERING	R	NBT	3	0	0	0	3	3	

72. SPECIALIZATION: ROBOTICS AND AUTOMATION

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDEC17A	ROBOT DESIGN AND ANALYSIS	A	RDA	0	0	6	4	4	10	FOR(1)
2	SDC	SDP-4	23SDEC17E	ROBOT DESIGN AND ANALYSIS	E	RDA	0	0	6	4	4	10	FOR(1)
3	SDC	SDP-4	23SDEC17R	ROBOT DESIGN AND ANALYSIS	R	RDA	0	0	2	4	2	6	FOR(1)
4	PEC	PE-1	23RAN3101A	ROBOT MOTION PLANNING, DYNAMICS AND CONTROL	A	RMPDC	4	0	4	4	7	12	FOR(1)

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
5	PEC	PE-1	23RAN3101E	ROBOT MOTION PLANNING, DYNAMICS AND CONTROL	E	RMPDC	4	0	4	4	7	12	FOR(1)
6	PEC	PE-1	23RAN3101R	ROBOT MOTION PLANNING, DYNAMICS AND CONTROL	R	RMPDC	3	0	2	4	5	9	FOR(1)
7	PEC	PE-1	23RAN3102A	AUTONOMOUS MOBILE ROBOT SYSTEMS	A	AMRS	4	0	4	4	7	12	FOR(1)
8	PEC	PE-1	23RAN3102E	AUTONOMOUS MOBILE ROBOT SYSTEMS	E	AMRS	4	0	4	4	7	12	FOR(1)
9	PEC	PE-1	23RAN3102R	AUTONOMOUS MOBILE ROBOT SYSTEMS	R	AMRS	3	0	2	4	5	9	FOR(1)
10	PEC	PE-2	23RAN3202A	AUTONOMOUS VEHICLES & AUTOMOTIVE ELECTRONICS	A	AVAE	5	0	0	0	5	5	FOR(1)
11	PEC	PE-2	23RAN3202E	AUTONOMOUS VEHICLES & AUTOMOTIVE ELECTRONICS	E	AVAE	5	0	0	0	5	5	FOR(1)
12	PEC	PE-2	23RAN3202R	AUTONOMOUS VEHICLES & AUTOMOTIVE ELECTRONICS	R	AVAE	3	0	0	0	3	3	FOR(1)
13	PEC	PE-2	23RAN3203A	ROBOT MANIPULATION AND WHEELED MOBILE ROBOTS	A	RMWR	5	0	0	0	5	5	FOR(1)
14	PEC	PE-2	23RAN3203E	ROBOT MANIPULATION AND WHEELED MOBILE ROBOTS	E	RMWR	5	0	0	0	5	5	FOR(1)
15	PEC	PE-2	23RAN3203R	ROBOT MANIPULATION AND WHEELED MOBILE ROBOTS	R	RMWR	3	0	0	0	3	3	FOR(1)
16	PEC	PE-3	23RAN3304A	ADVANCED ROBOTICS	A	AR	4	0	4	4	7	12	FOR(1)
17	PEC	PE-3	23RAN3304E	ADVANCED ROBOTICS	E	AR	4	0	4	4	7	12	FOR(1)
18	PEC	PE-3	23RAN3304R	ADVANCED ROBOTICS	R	AR	3	0	2	4	5	9	FOR(1)
19	PEC	PE-4	23RAN3405M	ARTIFICIAL INTELLIGENCE FOR ROBOTICS	M	AIR	3	0	0	0	3	3	FOR(1)
20	PEC	PE-4	23RAN3406M	HUMAN MACHINE INTERFACE & BRAIN MACHINE INTERFACE	M	HMIBMI	3	0	0	0	3	3	FOR(1)
21	PEC	PE-5	23RAN3507	COMPUTER VISION & APPLICATIONS	R	CVA	3	0	0	0	3	3	FOR(1)

73. SPECIALIZATION: SMART GRID TECHNOLOGIES

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDEE04A	AI & ML FOR SMART GRIDS	A	AIMLSG	0	0	6	4	4	10	EPE(1)
2	SDC	SDP-4	23SDEE04E	AI & ML FOR SMART GRIDS	E	AIMLSG	0	0	6	4	4	10	EPE(1)
3	SDC	SDP-4	23SDEE04R	AI & ML FOR SMART GRIDS	R	AIMLSG	0	0	2	4	2	6	EPE(1)
4	PEC	PE-1	23SGT3101A	DISTRIBUTED ENERGY RESOURCES AND SMART GRIDS	A	DERSG	4	0	4	4	7	12	EPE(1)
5	PEC	PE-1	23SGT3101E	DISTRIBUTED ENERGY RESOURCES AND SMART GRIDS	E	DERSG	4	0	4	4	7	12	EPE(1)
6	PEC	PE-1	23SGT3101R	DISTRIBUTED ENERGY RESOURCES AND SMART GRIDS	R	DERSG	3	0	2	4	5	9	EPE(1)

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
7	PEC	PE-2	23SGT3202A	DISTRIBUTION SYSTEM PRACTICES	A	DSP	5	0	0	0	5	5	EPE(1)
8	PEC	PE-2	23SGT3202E	DISTRIBUTION SYSTEM PRACTICES	E	DSP	5	0	0	0	5	5	EPE(1)
9	PEC	PE-2	23SGT3202R	DISTRIBUTION SYSTEM PRACTICES	R	DSP	3	0	0	0	3	3	EPE(1)
10	PEC	PE-2	23SGT3203A	MICROGRID DYNAMICS AND CONTROL	A	MDC	5	0	0	0	5	5	EPE(1)
11	PEC	PE-2	23SGT3203E	MICROGRID DYNAMICS AND CONTROL	E	MDC	5	0	0	0	5	5	EPE(1)
12	PEC	PE-2	23SGT3203R	MICROGRID DYNAMICS AND CONTROL	R	MDC	3	0	0	0	3	3	EPE(1)
13	PEC	PE-3	23SGT3304A	ENERGY MANAGEMENT SYSTEMS AND SCADA	A	EMSAS	4	0	4	4	7	12	EPE(1)
14	PEC	PE-3	23SGT3304E	ENERGY MANAGEMENT SYSTEMS AND SCADA	E	EMSAS	4	0	4	4	7	12	EPE(1)
15	PEC	PE-3	23SGT3304R	ENERGY MANAGEMENT SYSTEMS AND SCADA	R	EMSAS	3	0	2	4	5	9	EPE(1)
16	PEC	PE-4	23SGT3405M	SMART GRID COMMUNICATION AND CYBERSECURITY	M	SGCCS	3	0	0	0	3	3	EPE(1)
17	PEC	PE-4	23SGT3406M	SMART METERS AND SMART CITIES	M	SMSC	3	0	0	0	3	3	EPE(1)
18	PEC	PE-5	23SGT3507	INTERNET OF THINGS AND SMART GRID ANALYTICS	R	IOTSGA	3	0	0	0	3	3	EPE(1)
19	PEC	PE-5	23SGT3508	SMART GRID PROTECTION	R	SGP	3	0	0	0	3	3	EPE(1)

74. SPECIALIZATION: SMART MANUFACTURING

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDME06A	GEOMETRIC DIMENSIONING AND TOLERANCING	A	GDAT	0	0	6	4	4	10	MSM(1)
2	SDC	SDP-4	23SDME06E	GEOMETRIC DIMENSIONING AND TOLERANCING	E	GDAT	0	0	6	4	4	10	MSM(1)
3	SDC	SDP-4	23SDME06R	GEOMETRIC DIMENSIONING AND TOLERANCING	R	GDAT	0	0	2	4	2	6	MSM(1)
4	PEC	PE-1	23SMF3101A	REVERSE ENGINEERING & RAPID PROTOTYPING	A	MMC	4	0	4	4	7	12	MSM(1)
5	PEC	PE-1	23SMF3101E	REVERSE ENGINEERING & RAPID PROTOTYPING	E	MMC	4	0	4	4	7	12	MSM(1)
6	PEC	PE-1	23SMF3101R	REVERSE ENGINEERING & RAPID PROTOTYPING	R	MMC	3	0	2	4	5	9	MSM(1)
7	PEC	PE-2	23SMF3202A	ADVANCED MATERIALS MANUFACTURING & TESTING	A	AMMT	5	0	0	0	5	5	MSM(1)
8	PEC	PE-2	23SMF3202E	ADVANCED MATERIALS MANUFACTURING & TESTING	E	AMMT	5	0	0	0	5	5	MSM(1)
9	PEC	PE-2	23SMF3202R	ADVANCED MATERIALS MANUFACTURING & TESTING	R	AMMT	3	0	0	0	3	3	MSM(1)

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
10	PEC	PE-2	23SMF3203A	MODERN MANUFACTURING PROCESSES	A	MMP	5	0	0	0	5	5	MSM(1)
11	PEC	PE-2	23SMF3203E	MODERN MANUFACTURING PROCESSES	E	MMP	5	0	0	0	5	5	MSM(1)
12	PEC	PE-2	23SMF3203R	MODERN MANUFACTURING PROCESSES	R	MMP	3	0	0	0	3	3	MSM(1)
13	PEC	PE-3	23SMF3304A	SUSTAINABLE DESIGN & SOCIAL INNOVATION IN SMART MANUFACTURING	A	SDSISM	4	0	4	4	7	12	MSM(1)
14	PEC	PE-3	23SMF3304E	SUSTAINABLE DESIGN & SOCIAL INNOVATION IN SMART MANUFACTURING	E	SDSISM	4	0	4	4	7	12	MSM(1)
15	PEC	PE-3	23SMF3304R	SUSTAINABLE DESIGN & SOCIAL INNOVATION IN SMART MANUFACTURING	R	SDSISM	3	0	2	4	5	9	MSM(1)
16	PEC	PE-4	23SMF3405M	ROBOTICS & INDUSTRIAL AUTOMATION	M	RIA	3	0	0	0	3	3	MSM(1)
17	PEC	PE-4	23SMF3406M	MECHANICAL MEASUREMENTS AND METROLOGY	M	MMM	3	0	0	0	3	3	MSM(1)
18	PEC	PE-5	23SMF3507	MACHINE TO MACHINE COMMUNICATION	R	MMC	3	0	0	0	3	3	MSM(1)
19	PEC	PE-5	23SMF3508	FLEXIBLE MANUFACTURING SYSTEMS	R	FMS	3	0	0	0	3	3	MSM(1)

75. SPECIALIZATION: SOCIAL AND DIGITAL MEDIA ANALYTICS

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDAD06A	ANALYSIS OF DIGITAL MARKETING	A	ADM	0	0	6	4	4	10	DBMS(1) Rule:1
2	SDC	SDP-4	23SDAD06E	ANALYSIS OF DIGITAL MARKETING	E	ADM	0	0	6	4	4	10	DBMS(1) Rule:1
3	SDC	SDP-4	23SDAD06R	ANALYSIS OF DIGITAL MARKETING	R	ADM	0	0	2	4	2	6	DBMS(1) Rule:1
4	PEC	PE-1	23SDM3101A	SENTIMENT ANALYSIS	A	SA	4	0	4	4	7	12	DBMS(1)
5	PEC	PE-1	23SDM3101E	SENTIMENT ANALYSIS	E	SA	4	0	4	4	7	12	DBMS(1)
6	PEC	PE-1	23SDM3101R	SENTIMENT ANALYSIS	R	SA	3	0	2	4	5	9	DBMS(1)
7	PEC	PE-2	23SDM3202A	OPINION MINING & RECOMMENDER SYSTEMS	A	OMRS	5	0	0	0	5	5	DBMS(1)
8	PEC	PE-2	23SDM3202E	OPINION MINING & RECOMMENDER SYSTEMS	E	OMRS	5	0	0	0	5	5	DBMS(1)
9	PEC	PE-2	23SDM3202R	OPINION MINING & RECOMMENDER SYSTEMS	R	OMRS	3	0	0	0	3	3	DBMS(1)
10	PEC	PE-2	23SDM3203A	META SOCIAL MEDIA ANALYTICS	A	MSMA	5	0	0	0	5	5	DBMS(1)
11	PEC	PE-2	23SDM3203E	META SOCIAL MEDIA ANALYTICS	E	MSMA	5	0	0	0	5	5	DBMS(1)

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
12	PEC	PE-2	23SDM3203R	META SOCIAL MEDIA ANALYTICS	R	MSMA	3	0	0	0	3	3	DBMS(1)
13	PEC	PE-3	23SDM3304A	SOCIAL MEDIA MARKETING ANALYTICS	A	SMMA	4	0	4	4	7	12	DBMS(1)
14	PEC	PE-3	23SDM3304E	SOCIAL MEDIA MARKETING ANALYTICS	E	SMMA	4	0	4	4	7	12	DBMS(1)
15	PEC	PE-3	23SDM3304R	SOCIAL MEDIA MARKETING ANALYTICS	R	SMMA	3	0	2	4	5	9	DBMS(1)
16	PEC	PE-4	23SDM3405M	DIGITAL MEDIA ANALYTICS	M	DMA	3	0	0	0	3	3	DBMS(1)
17	PEC	PE-4	23SDM3406M	ETHICAL SOCIAL MEDIA	M	ESM	3	0	0	0	3	3	DBMS(1)
18	PEC	PE-5	23SDM3508	SEARCH ENGINE OPTIMIZATION	R	SEO	3	0	0	0	3	3	DBMS(1) Rule:1
19	PEC	PE-5	23SDM3507	INTELLIGENT SOCIAL MEDIA MONITORING SYSTEMS	R	ISMMS	3	0	0	0	3	3	DBMS(1)

76. SPECIALIZATION: SOFTWARE MODELLING AND DEVOPS

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	FCC	FC-1	23CS2239F	SOFTWARE VERIFICATION & VALIDATION	F	SVV	2	0	2	0	3	4	
2	FCC	FC-1	23CS2228F	CROSS-PLATFORM USER INTERFACE DESIGN	F	CPUIF	2	0	2	0	3	4	
3	SDC	SDP-4	23SDCI05A	CLOUD DEVOPS	A	CDP	0	0	6	4	4	10	
4	SDC	SDP-4	23SDCI05E	CLOUD DEVOPS	E	CDP	0	0	6	4	4	10	
5	SDC	SDP-4	23SDCI05R	CLOUD DEVOPS	R	CDP	0	0	2	4	2	6	
6	PEC	PE-1	23SMD3101A	SOFTWARE VERIFICATION & VALIDATION	A	SVV	4	0	4	4	7	12	
7	PEC	PE-1	23SMD3101E	SOFTWARE VERIFICATION & VALIDATION	E	SVV	4	0	4	4	7	12	
8	PEC	PE-1	23SMD3101R	SOFTWARE VERIFICATION & VALIDATION	R	SVV	3	0	2	4	5	9	
9	PEC	PE-2	23SMD3202A	DESIGN PATTERNS & CLEAN CODING TECHNIQUES	A	DPCT	5	0	0	0	5	5	
10	PEC	PE-2	23SMD3202E	DESIGN PATTERNS & CLEAN CODING TECHNIQUES	E	DPCT	5	0	0	0	5	5	
11	PEC	PE-2	23SMD3202R	DESIGN PATTERNS & CLEAN CODING TECHNIQUES	R	DPCT	3	0	0	0	3	3	
12	PEC	PE-3	23SMD3303A	CONTINUOUS DELIVERY & DEVOPS	A	CDD	4	0	4	4	7	12	
13	PEC	PE-3	23SMD3303E	CONTINUOUS DELIVERY & DEVOPS	E	CDD	4	0	4	4	7	12	
14	PEC	PE-3	23SMD3303R	CONTINUOUS DELIVERY & DEVOPS	R	CDD	3	0	2	4	5	9	
15	PEC	PE-4	23SMD3404M	SOFTWARE PROJECT MANAGEMENT	M	SPM	3	0	0	0	3	3	
16	PEC	PE-4	23SMD3405M	SOFTWARE ARCHITECTURE & DESIGN	M	SAD	3	0	0	0	3	3	

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
17	PEC	PE-5	23SMD3506	SOFTWARE RELIABILITY	R	SR	3	0	0	0	3	3	
18	PEC	PE-5	23SMD3507	CROSS-PLATFORM USER INTERFACE DESIGN	R	CPUIF	3	0	0	0	3	3	

77. SPECIALIZATION: SPATIAL COMPUTING AND IMMERSIVE TECHNOLOGIES

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDCS06A	CERTIFIED GAME DEVELOPER	A	CGD	0	0	6	4	4	10	CTSD(1)
2	SDC	SDP-4	23SDCS06E	CERTIFIED GAME DEVELOPER	E	CGD	0	0	6	4	4	10	CTSD(1)
3	SDC	SDP-4	23SDCS06R	CERTIFIED GAME DEVELOPER	R	CGD	0	0	2	4	2	6	CTSD(1)
4	PEC	PE-1	23SCI3101A	SPATIAL DATA VISUALIZATION	A	SDV	4	0	4	4	7	12	
5	PEC	PE-1	23SCI3101E	SPATIAL DATA VISUALIZATION	E	SDV	4	0	4	4	7	12	
6	PEC	PE-1	23SCI3101R	SPATIAL DATA VISUALIZATION	R	SDV	3	0	2	4	5	9	
7	PEC	PE-2	23SCI3202A	SPATIAL COMPUTING FOR MULTIPLAYER GAMES	A	SCMG	5	0	0	0	5	5	
8	PEC	PE-2	23SCI3202E	SPATIAL COMPUTING FOR MULTIPLAYER GAMES	E	SCMG	5	0	0	0	5	5	
9	PEC	PE-2	23SCI3202R	SPATIAL COMPUTING FOR MULTIPLAYER GAMES	R	SCMG	3	0	0	0	3	3	
10	PEC	PE-3	23SCI3303A	SPATIAL COMPUTING PLATFORMS AND DEVELOPMENT	A	SCPD	4	0	4	4	7	12	
11	PEC	PE-3	23SCI3303E	SPATIAL COMPUTING PLATFORMS AND DEVELOPMENT	E	SCPD	4	0	4	4	7	12	
12	PEC	PE-3	23SCI3303R	SPATIAL COMPUTING PLATFORMS AND DEVELOPMENT	R	SCPD	3	0	2	4	5	9	
13	PEC	PE-4	23SCI3404M	SPATIAL AND SOCIAL IMPLICATIONS OF IMMERSIVE TECHNOLOGIES	M	SSIIT	3	0	0	0	3	3	
14	PEC	PE-4	23SCI3405M	SPATIAL AUDIO DESIGN FOR GAMES	M	SADG	3	0	0	0	3	3	
15	PEC	PE-5	23SCI3506	SPATIAL USER INTERFACE DESIGN	R	SUID	3	0	0	0	3	3	
16	PEC	PE-5	23SCI3507	INTRODUCTION TO SPATIAL COMPUTING	R	ITSC	3	0	0	0	3	3	

78. SPECIALIZATION: STRUCTURAL ENGINEERING

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDCE06A	DESIGN & ANALYSIS OF DIFFERENT STRUCTURAL COMPONENTS USING SAFE	A	SAFE	0	0	6	4	4	10	
2	SDC	SDP-4	23SDCE06E	DESIGN & ANALYSIS OF DIFFERENT STRUCTURAL COMPONENTS USING SAFE	E	SAFE	0	0	6	4	4	10	
3	SDC	SDP-4	23SDCE06R	DESIGN & ANALYSIS OF DIFFERENT STRUCTURAL COMPONENTS USING SAFE	R	SAFE	0	0	2	4	2	6	

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
4	PEC	PE-1	23STE3101A	ADVANCED CONCRETE TECHNOLOGY	A	ACT	4	0	4	4	7	12	
5	PEC	PE-1	23STE3101E	ADVANCED CONCRETE TECHNOLOGY	E	ACT	4	0	4	4	7	12	
6	PEC	PE-1	23STE3101R	ADVANCED CONCRETE TECHNOLOGY	R	ACT	3	0	2	4	5	9	
7	PEC	PE-2	23STE3202A	ADVANCED STRUCTURAL ANALYSIS	A	ASA	5	0	0	0	5	5	
8	PEC	PE-2	23STE3202E	ADVANCED STRUCTURAL ANALYSIS	E	ASA	5	0	0	0	5	5	
9	PEC	PE-2	23STE3202R	ADVANCED STRUCTURAL ANALYSIS	R	ASA	3	0	0	0	3	3	
10	PEC	PE-2	23STE3203A	ADVANCED DESIGN OF STEEL STRUCTURE	A	ADSS	5	0	0	0	5	5	
11	PEC	PE-2	23STE3203E	ADVANCED DESIGN OF STEEL STRUCTURE	E	ADSS	5	0	0	0	5	5	
12	PEC	PE-2	23STE3203R	ADVANCED DESIGN OF STEEL STRUCTURE	R	ADSS	3	0	0	0	3	3	
13	PEC	PE-3	23STE3304A	BRIDGE ENGINEERING	A	BE	4	0	4	4	7	12	
14	PEC	PE-3	23STE3304E	BRIDGE ENGINEERING	E	BE	4	0	4	4	7	12	
15	PEC	PE-3	23STE3304R	BRIDGE ENGINEERING	R	BE	3	0	2	4	5	9	
16	PEC	PE-4	23STE3405M	PRESTRESSED CONCRETE	M	PC	3	0	0	0	3	3	
17	PEC	PE-4	23STE3406M	PRECAST AND PREFABRICATED STRUCTURES	M	PPS	3	0	0	0	3	3	
18	PEC	PE-5	23STE3507	PRE ENGINEERING STRUCTURES	R	PES	3	0	0	0	3	3	
19	PEC	PE-5	23STE3508	ADVANCED DESIGN OF REINFORCED CONCRETE STRUCTURES	R	ADRCS	3	0	0	0	3	3	

79. SPECIALIZATION: VERY LARGE SCALE INTEGRATION

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDEC18A	VLSI SUBMICRON DESIGN AND VERIFICATION	A	VSDV	0	0	6	4	4	10	
2	SDC	SDP-4	23SDEC18E	VLSI SUBMICRON DESIGN AND VERIFICATION	E	VSDV	0	0	6	4	4	10	
3	SDC	SDP-4	23SDEC18R	VLSI SUBMICRON DESIGN AND VERIFICATION	R	VSDV	0	0	2	4	2	6	
4	PEC	PE-1	23VLS3101A	ANALOG VLSI DESIGN	A	AVD	4	0	4	4	7	12	
5	PEC	PE-1	23VLS3101E	ANALOG VLSI DESIGN	E	AVD	4	0	4	4	7	12	
6	PEC	PE-1	23VLS3101R	ANALOG VLSI DESIGN	R	AVD	3	0	2	4	5	9	
7	PEC	PE-2	23VLS3202A	TESTING AND VERIFICATION OF VLSI CIRCUITS	A	TVVC	5	0	0	0	5	5	
8	PEC	PE-2	23VLS3202E	TESTING AND VERIFICATION OF VLSI CIRCUITS	E	TVVC	5	0	0	0	5	5	

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
9	PEC	PE-2	23VLS3202R	TESTING AND VERIFICATION OF VLSI CIRCUITS	R	TVVC	3	0	0	0	3	3	
10	PEC	PE-3	23VLS3303A	VLSI PHYSICAL DESIGN AUTOMATION	A	VPDA	4	0	4	4	7	12	
11	PEC	PE-3	23VLS3303E	VLSI PHYSICAL DESIGN AUTOMATION	E	VPDA	4	0	4	4	7	12	
12	PEC	PE-3	23VLS3303R	VLSI PHYSICAL DESIGN AUTOMATION	R	VPDA	3	0	2	4	5	9	
13	PEC	PE-4	23VLS3404M	SYSTEM-ON-CHIP	M	SOC	3	0	0	0	3	3	
14	PEC	PE-5	23VLS3505	MIXED SIGNAL IC DESIGN	R	MSID	3	0	0	0	3	3	

80. SPECIALIZATION: WATER RESOURCE AND ENVIRONMENTAL ENGINEERING

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	SDC	SDP-4	23SDCE07A	ARC-GIS (PRO)	A	ARC	0	0	6	4	4	10	
2	SDC	SDP-4	23SDCE07E	ARC-GIS (PRO)	E	ARC	0	0	6	4	4	10	
3	SDC	SDP-4	23SDCE07R	ARC-GIS (PRO)	R	ARC	0	0	2	4	2	6	
4	PEC	PE-1	23WRE3101A	RIVER ENGINEERING	A	RE	4	0	4	4	7	12	
5	PEC	PE-1	23WRE3101E	RIVER ENGINEERING	E	RE	4	0	4	4	7	12	
6	PEC	PE-1	23WRE3101R	RIVER ENGINEERING	R	RE	3	0	2	4	5	9	
7	PEC	PE-2	23WRE3202A	SOLID WASTE MANAGEMENT AND LANDFILLS	A	SWML	5	0	0	0	5	5	
8	PEC	PE-2	23WRE3202E	SOLID WASTE MANAGEMENT AND LANDFILLS	E	SWML	5	0	0	0	5	5	
9	PEC	PE-2	23WRE3202R	SOLID WASTE MANAGEMENT AND LANDFILLS	R	SWML	3	0	0	0	3	3	
10	PEC	PE-2	23WRE3203A	DESIGN OF HYDRAULIC STRUCTURES	A	DHS	5	0	0	0	5	5	
11	PEC	PE-2	23WRE3203E	DESIGN OF HYDRAULIC STRUCTURES	E	DHS	5	0	0	0	5	5	
12	PEC	PE-2	23WRE3203R	DESIGN OF HYDRAULIC STRUCTURES	R	DHS	3	0	0	0	3	3	
13	PEC	PE-3	23WRE3304A	WATER RESOURCES FIELD METHODS	A	WRFM	4	0	4	4	7	12	
14	PEC	PE-3	23WRE3304E	WATER RESOURCES FIELD METHODS	E	WRFM	4	0	4	4	7	12	
15	PEC	PE-3	23WRE3304R	WATER RESOURCES FIELD METHODS	R	WRFM	3	0	2	4	5	9	
16	PEC	PE-4	23WRE3405M	URBAN WATER HYDROLOGY AND HYDRAULICS	M	UWHH	3	0	0	0	3	3	
17	PEC	PE-4	23WRE3406M	ENVIRONMENTAL IMPACT ASSESSMENT AND LIFE CYCLE ANALYSES	M	EIALCA	3	0	0	0	3	3	

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
18	PEC	PE-5	23WRE3507	PHYSICO-CHEMICAL PROCESSES FOR WATER AND WASTEWATER TREATMENT	R	PCPWWT	3	0	0	0	3	3	
19	PEC	PE-5	23WRE3508	SUSTAINABLE ENGINEERING AND TECHNOLOGY	R	SET	3	0	0	0	3	3	

81. PROJECT: RESEARCH PROJECT

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	PRI	PRI-CORE	23IE4057A	RESEARCH PROJECT - 1	A	RCP-1	0	0	10	20	10	30	
2	PRI	PRI-CORE	23IE4058A	RESEARCH PROJECT - 2	A	RCP-2	0	0	10	20	10	30	

82. PROJECT: INNOVATION PROJECT

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	PRI	PRI-CORE	23IE4055A	INNOVATION PROJECT - 1	A	ICP-1	0	0	10	20	10	30	
2	PRI	PRI-CORE	23IE4056A	INNOVATION PROJECT - 2	A	ICP-2	0	0	10	20	10	30	

83. PROJECT: CAPSTONE PROJECT

S#	Cat	Sub-Cat	CourseCode	Course Title	Mode	Acrym	L	T	P	S	CR	CH	Pre-req
1	PRI	PRI-CORE	23IE4053A	CAPSTONE PROJECT - 1	A	CP-1	0	0	10	20	10	30	
2	PRI	PRI-CORE	23IE4053R	CAPSTONE PROJECT - 1	R	CP-1	0	0	8	16	8	24	
3	PRI	PRI-CORE	23IE4054A	CAPSTONE PROJECT - 2	A	CP-2	0	0	10	20	10	30	
4	PRI	PRI-CORE	23IE4054R	CAPSTONE PROJECT - 2	R	CP-2	0	0	8	16	8	24	

Program Articulation Matrix

S#	Cat	Course	CO	CO Description	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
1	HAS	23FL3054 - FLG	CO1	Acquire a working knowledge of the basic elements of the French language viz. letters, vowels, accents, articles, useful expressions, etc.						2		2						
2	HAS	23FL3054 - FLG	CO2	Classify questions and respond in the affirmative or negative with ?tre and avoir and form plurals							2			3				
3	HAS	23FL3054 - FLG	CO3	Utilize and apply the adjectives and essential verbs.								2		2				
4	HAS	23FL3054 - FLG	CO4	Construct and use in speech, vocabulary, reading, questions and answers								2				3		
5	HAS	23FL3055 - GLG	CO1	classify their understanding of greeting wishes, alphabets and numbers learning. to understand the greetings in formal and informal way					2			2						
6	HAS	23FL3055 - GLG	CO2	Apply their knowledge of essential daily expressions, present, past and future tense. Conjugating the verbs in the Singular and Plural groups, Past participle tense and the futertense and relations with the verbs										3				
7	HAS	23FL3055 - GLG	CO3	Utilize their understanding with suitable prepositions, questions, and possessive pronouns, and the importance of four German cases. Prepositions in Akkusativ and Dativ										2				
8	HAS	23FL3055 - GLG	CO4	Develop their knowledge about how to move in public places, such as shopping centres, restaurants, tourist places, etc, and preparation of them for German A1 level examination.								2				3		

S#	Cat	Course	CO	CO Description	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
9	HAS	23FL3058 - JLG	CO1	Classify Hiragana, Katakana, and basic Kanji characters used in greetings and simple scripts					2			2						
10	HAS	23FL3058 - JLG	CO2	Apply their knowledge of essential daily expressions, numbers, months, dates, time, body parts, colors, and common vocabulary to effectively communicate in basic everyday situations										3				
11	HAS	23FL3058 - JLG	CO3	Utilize their understanding of present, past, and future tenses, along with the ability to construct interrogative sentences, to express themselves in various timeframes and ask questions effectively in different conversational contexts. pen_spark								2		2				
12	HAS	23FL3058 - JLG	CO4	Develop their knowledge of verbs, including negative conjugations, and prepositions to discuss hobbies, deliver self-introductions, and navigate basic interview scenarios in Japanese								2				3		
13	HAS	23MB0001 - BME	CO1	Understand the basic concepts of marketing management						2								
14	HAS	23MB0001 - BME	CO2	Understand the concepts of Marketing environment, consumer behaviour and Segmentation, Targeting and Positioning (STP)						2								
15	HAS	23MB0001 - BME	CO3	Apply the marketing mix strategies with special focus on technology products			2											
16	HAS	23MB0001 - BME	CO4	Apply promotion and distribution strategies for marketing of high tech products and services			2											
17	HAS	23MB0002 - PIMT	CO1	Understand the basic management concepts along with an insight into levels of management									1					
18	HAS	23MB0002 - PIMT	CO2	Understand the key contributions of classical approach to Management												1		

S#	Cat	Course	CO	CO Description	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
19	HAS	23MB0002 - PIMT	CO3	Understand and apply Quantitative methods to improve Management performance.									2					
20	HAS	23MB0002 - PIMT	CO4	Understand the key contributions of Behavioural and contemporary approaches to Management.									3			2		
21	HAS	23MB0003 - FMFE	CO1	Understanding the comprehension of finance functions and diverse business models to facilitate informed decision-making processes within financial management.		2												
22	HAS	23MB0003 - FMFE	CO2	Analyzing the investment decisions through an understanding of capital budgeting techniques, both traditional and modern, emphasizing long-term implications.				2										
23	HAS	23MB0003 - FMFE	CO3	Analyze and make informed decisions regarding working capital management, considering the short-term financial health of organizations through practical case studies.				2										
24	HAS	23MB0003 - FMFE	CO4	Developing proficiency in comprehending and utilizing various sources of finance while discerning the implications of different dividend policies in practical financial scenarios.				2										
25	HAS	23MB0004 - OMG	CO1	Remember and understand the various management theories and management approaches.									2					
26	HAS	23MB0004 - OMG	CO2	Remember and understand organization theories, structures and organization principles												2		

S#	Cat	Course	CO	CO Description	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
27	HAS	23MB0004 - OMG	CO3	Have basic knowledge and understanding of motivation, motivational theories, leadership theories, moral and behavioral sciences and also understand the management concept, administration and management objectives.									2					
28	HAS	23MB0004 - OMG	CO4	Remember and understand the various issues in industrial relations, trade unions and college bargaining and industrial safety									2					
29	HAS	23MB0005 - MPF	CO1	Understanding Personal Finance	2													
30	HAS	23MB0005 - MPF	CO2	Applying and Building a Budget and Tracking Your Money		3												
31	HAS	23MB0005 - MPF	CO3	Applying and Managing Debt and Credit in present context			3											
32	HAS	23MB0005 - MPF	CO4	Applying Saving and Investment to present scenario			3											
33	HAS	23MB4062 - CPM	CO1	Understand construction principles and techniques to effectively plan and execute construction projects.	1										2		1	
34	HAS	23MB4062 - CPM	CO2	Understand about construction project scheduling, cost estimation, and resource allocation for a particular project	1										2		1	
35	HAS	23MB4062 - CPM	CO3	Understand about the Construction Methods and Techniques	1										2		1	
36	HAS	23MB4062 - CPM	CO4	Understand about the Construction Project Cost Estimation and Control.	1										2		1	
37	HAS	23MB4062 - CPM	CO5	Understand about project cost control and monitoring, earned value management in construction.	1										2		1	

S#	Cat	Course	CO	CO Description	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
38	HAS	23MB4063 - RSQM	CO1	Demonstrate an understanding of resource management principles and the importance of resource optimization in construction projects by applying best practices to enhance efficiency.	2				3				2					3
39	HAS	23MB4063 - RSQM	CO2	Interpret and ensure compliance with construction safety regulations and standards set by regulatory bodies, ensuring legal and regulatory adherence in project execution.	2				3				2					3
40	HAS	23MB4063 - RSQM	CO3	Apply hazard identification techniques and conduct comprehensive risk assessments to prioritize and mitigate risks in construction environments, enhancing safety management.	2				3				2					3
41	HAS	23MB4063 - RSQM	CO4	Develop and implement safety management systems and quality assurance strategies, including safety policies, training, and communication, to ensure high safety standards and quality control in construction projects.	2				3				2					3
42	HAS	23MB4064 - SDM	CO1	Understand the concepts of digital marketing and search engine optimization											2			
43	HAS	23MB4064 - SDM	CO2	Apply the search engine optimization technique to website											2			
44	HAS	23MB4064 - SDM	CO3	Apply the key PPC concepts to draw visitors to a business websites											2			
45	HAS	23MB4064 - SDM	CO4	Analyse the search engine optimization technique for marketing strtegy											2			
46	HAS	23MB4065 - DET	CO1	Introduction to Digital Economics	2													
47	HAS	23MB4065 - DET	CO2	Digital Business Models		2												
48	HAS	23MB4065 - DET	CO3	Tokenomics Fundamentals			3											

S#	Cat	Course	CO	CO Description	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
49	HAS	23MB4065 - DET	CO4	Economic Implications of Cryptocurrencies			3											
50	HAS	23MB4066 - DFDFS	CO1	Define and explain the key concepts of DeFi, including decentralization, blockchain technology, smart contracts, and decentralized applications (DApps). Differentiate between traditional financial systems and DeFi ecosystems, identifying the advantages and challenges of each.	2												2	
51	HAS	23MB4066 - DFDFS	CO2	Describe the fundamentals of blockchain technology, including consensus mechanisms, cryptographic principles, and transaction processing. Evaluate different blockchain platforms and their suitability for supporting DeFi applications.			2											2
52	HAS	23MB4066 - DFDFS	CO3	Analyze and evaluate various decentralized financial protocols, such as decentralized exchanges (DEXs), lending platforms, liquidity provision mechanisms, and asset management protocols. Demonstrate the ability to interact with DeFi protocols through hands-on exercises and practical applications.						3							3	
53	HAS	23MB4066 - DFDFS	CO4	Explain the concept of stablecoins and synthetic assets, including their design principles, use cases, and implications for the broader financial ecosystem. Assess the role of stablecoins and synthetic assets in mitigating volatility and enabling new financial services in DeFi.						3								3
54	HAS	23MB4067 - IMPP	CO1	Understand the basic management concepts along with an insight into production and control											2			2

S#	Cat	Course	CO	CO Description	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
55	HAS	23MB4067 - IMPP	CO2	Select best forecasting models to predict future demand	2													2
56	HAS	23MB4067 - IMPP	CO3	Solve various production scheduling problems to optimize productivity				2										2
57	HAS	23MB4067 - IMPP	CO4	Understand concept of Inventory control, Method study and time study											2			2
58	HAS	23MB4068 - IMG	CO1	Identify and analyze market needs and gaps, using tools such as SWOT and PESTLE analysis, to pinpoint real-world problems suitable for entrepreneurial solutions.		1									2		1	
59	HAS	23MB4068 - IMG	CO2	Develop innovative solutions through creative thinking and apply lean startup principles to build and iterate a minimum viable product (MVP).		1									2		2	
60	HAS	23MB4068 - IMG	CO3	Create a comprehensive lean business model canvas that outlines customer segments, value propositions, channels, revenue streams, and key metrics.		1									2		2	
61	HAS	23MB4068 - IMG	CO4	Create a compelling business plan and investor pitch, including financial projections and funding strategies, demonstrating effective communication and persuasion skills.		1									2		1	
62	HAS	23UC0026 - HGP	CO1	Understanding the basic concepts of value education								2	1					
63	HAS	23UC0026 - HGP	CO2	Gain basic understanding of the principles in harmony among the human beings								2	1					
64	HAS	23UC0026 - HGP	CO3	Gain knowledge in the concept of Harmony in the family and society								3	3					
65	HAS	23UC0026 - HGP	CO4	Acquire knowledge in the concepts of harmony in the nature								3	3					

S#	Cat	Course	CO	CO Description	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
66	HAS	23UC0027 - LAMS	CO1	Understand basic leadership, skills and perspectives and leadership styles									2					
67	HAS	23UC0027 - LAMS	CO2	Understand different managerial skills and apply them to develop high performance teams											3			
68	HAS	23UC0027 - LAMS	CO3	Analyse effective communicative strategies and apply them in team tasks										3				
69	HAS	23UC0027 - LAMS	CO4	Apply strategic planning fundamentals and decision-making techniques, through exercises and case studies											3			
70	HAS	23UC1101 - IPE	CO1	Language Mechanics and Basic Grammar										2				
71	HAS	23UC1101 - IPE	CO2	Integrated Reading and Techniques of Writing										2				
72	HAS	23UC1202 - EP	CO1	Express the ability in interactive skills of listening, speaking, reading, and writing that are better suited for the corporate environment.										3				
73	HAS	23UC1202 - EP	CO2	Apply various strategies for LSRW skills and apply them in interpreting the text										3				
74	HAS	23UC1203 - DTI	CO1	Understand the importance of Design thinking mindset for identifying contextualized problems		1												
75	HAS	23UC1203 - DTI	CO2	Analyze the problem statement by empathizing with user							2							
76	HAS	23UC1203 - DTI	CO3	Develop ideation and test the prototypes made			2											
77	HAS	23UC1203 - DTI	CO4	Explore the fundamentals of entrepreneurship skills for transforming the challenge into an opportunity											1			

S#	Cat	Course	CO	CO Description	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
78	BSC	23BT2103 - BCM	CO1	Relate the importance of biological buffers and its preparation,explain cellular thermodynamics, structure, function andmetabolism of carbohydrates; the association between structureand function of carbohydrate at a chemical level with a biologicalperspective with a hands-on approach on laboratory techniques	3	3		3									3	
79	BSC	23BT2103 - BCM	CO2	Make use of the information about the folding, conformation, dynamics and intrinsicproperties of proteins and apply the knowledge to learnt aboutabnormalities related with protein dysfunction and their clinicalimportance	3	3		3									3	
80	BSC	23BT2103 - BCM	CO3	Apply the structural and biological roles that nucleic acids andlipids play in cellular physiology, and application of thatknowledge in biotechniques to identify these biomolecules anddiscuss about disorders related with nucleotide metabolism	3	3		3									3	
81	BSC	23BT2103 - BCM	CO4	Categorize enzymes, catalysis, kinetics, applications; immobilizedenzymes and their catalytic mechanisms	3	3		3									3	
82	BSC	23BT2103 - BCM	CO5	Analyze the biomolecules from various sources by employing experimental techniques and laboratory studies	3	3		3									3	
83	BSC	23CS3202 - NSC	CO1	Understand a diverse array of optimization algorithms, each distinguished by its specific approach and techniques aimed at achieving optimal solutions to complex problems.	3	2	2										2	

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84	BSC	23CS3202 - NSC	CO2	Apply a diverse set of metaheuristic optimization algorithms, showcasing their efficacy in tackling complex problems through comprehensive simulations and practical case studies.	3	2	2		2								2	
85	BSC	23CS3202 - NSC	CO3	Analyze a range of advanced neural network architectures and computational intelligence techniques to solve complex problems in machine learning and pattern recognition, demonstrating practical proficiency through implementation and evaluation.	2	3	2										2	
86	BSC	23CS3202 - NSC	CO4	contrast in between Apply a variety of advanced computational intelligence techniques and hybrid algorithms to address complex optimization and problem-solving challenges, showcasing practical proficiency through implementation and evaluation.	2	3		2									2	
87	BSC	23CS3202 - NSC	CO5	Design and Build bio inspired algorithms for Function Optimization to solve numerical problems.	2		3	3									2	
88	BSC	23CS3203 - QCP	CO1	Demonstrate the basics of quantum computing, including quantum states, qubits, single and multiple qubit gates, entanglement, quantum circuits, and various applications.	3				3									3
89	BSC	23CS3203 - QCP	CO2	Apply quantum computing algorithms like Deutsch-Jozsa, Bernstein-Vazirani, Simon\\'s, Shor\\'s, and Grover\\'s algorithms, as well as the Quantum Fourier Transform and superdense coding.				3										3

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90	BSC	23CS3203 - QCP	CO3	Implement quantum circuits and the intricacies of quantum measurement, providing hands-on experience in these areas.		3												3
91	BSC	23CS3203 - QCP	CO4	Demonstrate Quantum machine learning techniques like unsupervised and supervised learning, pattern recognition, neural networks, support vector machines, regression analysis, and boosting.	3													3
92	BSC	23CY1002 - PCT	CO1	Understand the basic concepts of physical chemistry and thermodynamics	2													
93	BSC	23CY1002 - PCT	CO2	Apply the laws of thermodynamics to various systems to evaluate their efficiency		2												
94	BSC	23CY1002 - PCT	CO3	Apply the principles of chemical reactions to understand the phase transitions and rate of reaction		2												
95	BSC	23CY1002 - PCT	CO4	Apply the concepts of kinetic theory of gases to understand the behaviour of mixture of gases		2												
96	BSC	23CY1002 - PCT	CO5	Verify the gas laws, reaction rates and perfromance of different thermal systems using simulation softwares			2											
97	BSC	23CY1003 - CPC	CO1	Interpret the importance and applications of computational techniques in solving chemical problems.	3		2											
98	BSC	23CY1003 - CPC	CO2	Interpret simulation results to predict molecular behavior, reaction mechanisms, and material properties.	3		2											
99	BSC	23CY1003 - CPC	CO3	Apply frequency calculations, and molecular dynamics, and leveraging cheminformatics tools for data mining in chemistry.	3		2		2									

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100	BSC	23CY1003 - CPC	CO4	Utilize computational resources to address and solve contemporary chemical challenges.	3		2											
101	BSC	23CY1003 - CPC	CO5	Analyze results to gain insights into molecular behaviors, reaction mechanisms, and properties.	3				2									
102	BSC	23MT1001 - LACE	CO1	Apply matrix algebra to the real-world applications in engineering, physical and biological sciences, computer science, finance, economics and solving the system of equations.	3													
103	BSC	23MT1001 - LACE	CO2	Apply multivariate differential calculus to find maxima & minima of functions and understand the concepts of second order differential equations and its applications.	3													
104	BSC	23MT1001 - LACE	CO3	Apply beta and gamma functions to evaluate improper integrals. Evaluate double and triple integrals techniques over a region in two dimensional and three-dimensional geometry.	3													
105	BSC	23MT1001 - LACE	CO4	interpret the physical meaning of different operators such as gradient, curl and compute the line integrals of vector functions and learn their applications.	3													
106	BSC	23MT1002 - DIS	CO1	Apply the knolodge of sets and function to the real world problems and computer problems to analize and draw ven diagrams	3													
107	BSC	23MT1002 - DIS	CO2	Apply basic and computational techniques on discrete structures like relations, orders, functions & FSM, Lattices, and propositional &predicate logic	3													

S#	Cat	Course	CO	CO Description	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
108	BSC	23MT1002 - DIS	CO3	Apply the knowledge of counting techniques, Recurrence Relations, Generating functions to solve the real world problems	3													
109	BSC	23MT1002 - DIS	CO4	Apply graph theory to solving real world structures and their related applications	3													
110	BSC	23MT2004 - MPG	CO1	Apply various methods for finding the optimal solution of Linear Programming Problem.	2	2											2	
111	BSC	23MT2004 - MPG	CO2	Apply Integer and Fractional programming approaches for solving optimization problems.	2	3											2	
112	BSC	23MT2004 - MPG	CO3	Apply quadratic programming and geometric programing techniques for solving non linear programming problems	2		3										3	
113	BSC	23MT2004 - MPG	CO4	Apply evolutionary algorithms, nature-inspired metaheuristics, to solve complex optimization problems in infinite-dimensional spaces.		3	3										3	
114	BSC	23MT2005 - PSQT	CO1	Apply the importance of probabilistic concepts in a wide spectrum of problems arising in engineering applied science.	1		2	3									1	
115	BSC	23MT2005 - PSQT	CO2	Assess the relationship between variables using corelation and regression techinques	1		2	3										1
116	BSC	23MT2005 - PSQT	CO3	Apply the role of Statistical tests of significance in solving real world engineering problems	1		2	3										1
117	BSC	23MT2005 - PSQT	CO4	Formulate Stochastic process in terms of Markov chains and solve problems in queueing systems, and network	1		2	3									1	
118	BSC	23MT2014 - TOC	CO1	Identify how finite automata (DFA & NFA) recognize regular languages, their equivalence to regular expressions (Arden\'s Theorem), and DFA minimization techniques.	1	2	3											

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119	BSC	23MT2014 - TOC	CO2	Analyze context-free languages using closure properties to prove languages are non-regular and derive normal forms	1	2	3											
120	BSC	23MT2014 - TOC	CO3	Examine context-free languages and their recognition by pushdown automata (PDA) and differentiate between decidable and undecidable langauges using Turing machines.	1	2	3											
121	BSC	23MT2014 - TOC	CO4	Explore interactibility by exploring complexity classes, and polynomial-time reductions.	1	2	3											
122	BSC	23PH1009 - CMP	CO1	Describe computational principles, historical context, and practical applications in computer science and scientific computation	2												2	
123	BSC	23PH1009 - CMP	CO2	Understanding of integration using stochastic methods, including quadrature techniques such as direct fit polynomials		2											2	
124	BSC	23PH1009 - CMP	CO3	Explain solid grasp of sampling and integration techniques, including the Metropolis algorithm, its applications in statistical physics		2											2	
125	BSC	23PH1009 - CMP	CO4	Summarize general behavior of classical systems, basic methods for many-body systems, including the Verlet algorithm		2											2	
126	BSC	23PH1009 - CMP	CO5	Analysing the data by apply the knowledge of physics to executethe related experiments and develop some inter disciplinaryprojects		2												2
127	BSC	23PH1010 - EMT	CO1	Understand the concept of Electromagnetic theory: similarities and differences, free-space propagation, and the classification of photons and electrons.		2												
128	BSC	23PH1010 - EMT	CO2	Understand the concept of nonlinear optics and it applications.	3			3										

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129	BSC	23PH1010 - EMT	CO3	Understand the concept of Photonic Crystals.														
130	BSC	23PH1010 - EMT	CO4	Understand the concepts of Allications of Nanophotonics					3									
131	BSC	23PH1010 - EMT	CO5	Undersgtand the experimental techniques														
132	BSC	23PH1011 - CMR	CO1	Understand fundamental concepts in computational mechanics and their relevance to robotics	2	2											2	
133	BSC	23PH1011 - CMR	CO2	Apply the proficiency in modeling for robotic systems development	2	2											2	
134	BSC	23PH1011 - CMR	CO3	Apply the various methods of trajectory planning for motion control in robotics.		2	2										2	
135	BSC	23PH1011 - CMR	CO4	Apply computational techniques to solve practical problems in robotics		2	2										2	
136	BSC	23PH1011 - CMR	CO5	Analyze the Robot navigation and path following using simulation software					2								2	
137	BSC	23SC3201 - DSS	CO1	Understand the fundamental principles of data science and statistics	2	2												2
138	BSC	23SC3201 - DSS	CO2	Gain basic understanding the role of data science in various scenarios in the real-world problems.		2		3										2
139	BSC	23SC3201 - DSS	CO3	Apply probabilistic predictions and decisions based on data analysis.		2		3										2
140	BSC	23SC3201 - DSS	CO4	Implement and apply proficiency in statistical inference techniques			2											2
141	ESC	23AD2001O - AIML	CO1	Apply a variety of artificial intelligence algorithms and techniques to effectively solve complex problems in diverse real-world environments		3	3	3									3	

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142	ESC	23AD2001O - AIML	CO2	Solve constraint satisfaction problems, employ knowledge engineering principles to perform inferencing, reasoning and probability theory.		3	3	3									3	
143	ESC	23AD2001O - AIML	CO3	Apply various machine learning techniques to analyze and solve real-world problems		3	3	3									3	
144	ESC	23AD2001O - AIML	CO4	solve complex real-world problems using advanced supervised and unsupervised learning techniques.		3	3	3									3	
145	ESC	23AD2001O - AIML	CO5	Evaluate solutions for various AI & ML related problems.			3	3	3								3	
146	ESC	23EC1101 - FITS	CO1	Apply the basic concepts of IoT and its implementation using the Development Hardware	2	1											3	
147	ESC	23EC1101 - FITS	CO2	Apply the different sensors interfacing with Development Hardware		1	3										3	
148	ESC	23EC1101 - FITS	CO3	Apply the different actuators interfacing with Development Hardware		1	3										3	
149	ESC	23EC1101 - FITS	CO4	Analyze the IoT concept to solve real time insights		1	3	2									3	
150	ESC	23EC1101 - FITS	CO5	Analyze the concept of IoT by interfacing with sensors and Development Hardware		1	3										3	
151	ESC	23EC1202 - DDCA	CO1	Build the combinational and programmable digital logic circuits using logic gates and optimization methods	2	2											3	
152	ESC	23EC1202 - DDCA	CO2	Construct the sequential and memory circuits using flip-flops	2	2											3	
153	ESC	23EC1202 - DDCA	CO3	Organize computer architecture and instructions sequence	2	2											3	

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154	ESC	23EC1202 - DDCA	CO4	Model the Memory Architecture and I/O Organization modules	2	2											3	
155	ESC	23EC1202 - DDCA	CO5	Develop and analyze computer architecture modules using basic combinational, sequential and memory logics	2		2		2								3	
156	ESC	23EC1203 - BEEC	CO1	Understand the basic concepts of circuits and its fundamentals	2												3	
157	ESC	23EC1203 - BEEC	CO2	Grasp the principles of AC circuits, including sinusoidal waveforms, impedance, and power factor.	2												3	
158	ESC	23EC1203 - BEEC	CO3	Comprehend the behavior of basic electronic components, such as diodes, and transistors.	2												3	
159	ESC	23EC1203 - BEEC	CO4	Understand the basic functional Principles of analog and digital ICs.	2												3	
160	ESC	23ME1103 - DTW	CO1	Demonstrate proficiency in typing sentence , paragraph , report , presentations along spread sheets using office tools, LaTeX tools and PowerBI					2									2
161	ESC	23ME1103 - DTW	CO2	Build a static website and blog with using html along with Special features of HTML5, CSS and Javascript					3									
162	ESC	23ME1103 - DTW	CO3	Develop a virtual environment with cospace and construct a marker based Augumented Reality			3							3			3	
163	ESC	23ME1103 - DTW	CO4	Utilising the softwares of Autodesk Fusion 360 and the same can be printed in 3D printer as physical prototype			3										3	
164	ESC	23SC1101 - CTSD	CO1	Develop and apply logical building blocks to solve real world problems	3	3	3											

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165	ESC	23SC1101 - CTSD	CO2	Apply computational thinking for designing solutions	3	3	3											
166	ESC	23SC1101 - CTSD	CO3	Develop and apply the CRUD operations on arrays	3	3												
167	ESC	23SC1101 - CTSD	CO4	Apply CRUD operations on Linear Data Structures			3	3										
168	ESC	23SC1101 - CTSD	CO5	Apply the structured programming paradigm with logic building skills onBasic and Linear Data Structures for solving real world problems	3	3			3									
169	ESC	23SC1101 - CTSD	CO6	Skill the students in such a way that students will be able to develop logicthat help them to create programs as well as applications in C	3	3			3									
170	ESC	23SC1202 - DS	CO1	Understand various sorting algorithms and analyse the efficiency of the algorithms.	3	3												
171	ESC	23SC1202 - DS	CO2	Implement and evaluate Linear Data Structures and Demonstrate their applications.	3	3	3											
172	ESC	23SC1202 - DS	CO3	Implement and evaluate tree data structures and understand hashing techniques	3	3	3						3					
173	ESC	23SC1202 - DS	CO4	Understand graph data structures and apply graphs to solve problems	3	3							3					
174	ESC	23SC1202 - DS	CO5	"Design, Develop and evaluate common practical applications for linear and nonlinear data structures."			3						3					
175	ESC	23SC1202 - DS	CO6	"Skill the students in such a way that students will be able to develop logic that help them to create programs on both linear and non-linear datastructuresand its applications."			3						3					
176	ESC	23SC1203 - CTOD	CO1	Apply Object oriented paradigm for code reusability			3		3									3

S#	Cat	Course	CO	CO Description	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
177	ESC	23SC1203 - CTOD	CO2	Design object-oriented solutions to the real-world problems through SOLID design principles			3		3									3
178	ESC	23SC1203 - CTOD	CO3	Build Abstract Data Types by applying generic classes and java API			3		3									3
179	ESC	23SC1203 - CTOD	CO4	Demonstrate Exception handling and String manipulation techniques			3		3									3
180	ESC	23SC1203 - CTOD	CO5	Develop solutions for real time problems by using object-oriented programming concepts.	3	3		3										3
181	ESC	23SC1203 - CTOD	CO6	Develop a real time project by using object-oriented programming concepts.	3	3		3	3									3
182	PCC	23AD2102R - DBMS	CO1	Apply fundamental database design principles, including normalization, to model complex data relationships.	2	2			2									
183	PCC	23AD2102R - DBMS	CO2	Develop efficient and secure database applications using Structured Query Language (SQL) and PL/SQL for data manipulation and retrieval.	2	2			2								1	
184	PCC	23AD2102R - DBMS	CO3	Implement and manage database systems with appropriate indexing techniques, query optimization strategies, and concurrency control mechanisms for efficient and reliable data access.	2	2			2								2	
185	PCC	23AD2102R - DBMS	CO4	Analyze and evaluate the suitability of emerging database technologies like NoSQL and Big Data for modern applications, considering security implications.	2	2			2								2	
186	PCC	23AD2102R - DBMS	CO5	Design and implement practical database solutions to solve real-world problems using appropriate database management system technologies and best practices.	2	2			3								3	

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187	PCC	23CI2001 - ASE	CO1	Demonstrate understanding of software development life cycle, associated process models, and reverse engineering techniques.	2	2	2											2
188	PCC	23CI2001 - ASE	CO2	Apply requirement modeling, Agile, and Extreme Programming methodologies.		2	2	2										2
189	PCC	23CI2001 - ASE	CO3	Analyze Agile models including Scrum, Kanban, and SAFe methodology, analyze the use of JIRA for project monitoring, and identify appropriate design patterns for project development.		3	3											3
190	PCC	23CI2001 - ASE	CO4	Analyze testing strategies and evaluate risk management factors	3	3	3											3
191	PCC	23CS2103R - AOOP	CO1	Apply software development practices for designing reusable solutions (design patterns), writing clean and maintainable code (clean coding techniques), and ensuring code quality through tests (Test Driven Development).	2				2									2
192	PCC	23CS2103R - AOOP	CO2	Analyze the advanced Java programming concepts for working with typed collections (generics & collections framework) and exploring advanced data structures like trees and graphs.	2				3									2
193	PCC	23CS2103R - AOOP	CO3	Apply the techniques for running multiple tasks concurrently in Java, including thread creation, synchronization, and advanced concepts like parallel programming.		2			3									2
194	PCC	23CS2103R - AOOP	CO4	Develop the applications that connect databases with JDBC and building web applications using servlets and JSP for dynamic content.		3			3									2

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195	PCC	23CS2103R - AOOP	CO5	Experiment with different design patterns (creational, structural, and behavioral) to structure your code effectively, while solidifying your understanding of key OOP principles like encapsulation, inheritance, and polymorphism to create reusable, maintainable, and adaptable object-oriented applications.	2	2			3									2
196	PCC	23CS2103R - AOOP	CO6	Dissect real-world challenges by applying object-oriented design patterns and principles in open-source projects to craft reusable solutions.		3			3				3					2
197	PCC	23CS2104R - OS	CO1	Understand subsystem components of the Kernel and apply the CPU Scheduling algorithms.	2	3	3	3									2	
198	PCC	23CS2104R - OS	CO2	Apply the memory and process virtualization and Paging, apply Page Replacement Algorithms		3	3	3									3	
199	PCC	23CS2104R - OS	CO3	Apply the Concurrency and apply different Lock Algorithms & Solve Synchronisation Problems	2	2	3	3									2	
200	PCC	23CS2104R - OS	CO4	Apply the knowledge of persistence concepts, Redundant disk arrays, File System Implementation, Inter-process Communication and Distributed Systems. Apply Disk Scheduling Algorithms.	1	3	3	3										
201	PCC	23CS2104R - OS	CO5	Evaluate Unix System Calls. Use C Programming Language to implement Operating System Concepts (Practicals)	2	2	3	3									3	
202	PCC	23CS2205R - DAA	CO1	Apply concepts of mathematics to find space complexity, time complexities of various algorithms	2												1	

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203	PCC	23CS2205R - DAA	CO2	Analyze the problems that can be solved by using Divide and Conquer and Greedy Method		2	2		3								2	
204	PCC	23CS2205R - DAA	CO3	Analyze the problems that can be solved by using Dynamic Programming and Backtracking		2	2		3								2	
205	PCC	23CS2205R - DAA	CO4	Analyze the problems that can be solved by using Branch and Bound, NP-Hard Graph problems and Approximation Algorithms		2	2		3								2	
206	PCC	23CS2205R - DAA	CO5	Solve and Analyze the complex and realtime problems by various design strategies	2	2	2		3									2
207	PCC	23CS2205R - DAA	CO6	Analyze the various design techniques to solve any real-world problems	2	2	2		3									2
208	PCC	23CS4106 - DC	CO1	Understanding of Distributed Systems Principles	2	2	2											
209	PCC	23CS4106 - DC	CO2	Proficiency in Distributed Computing Technologies	2	2	3											
210	PCC	23CS4106 - DC	CO3	Ability to Design and Implement Distributed Algorithms	2	2	2											
211	PCC	23CS4106 - DC	CO4	Practical Experience with Distributed Systems Development	2	2	2											
212	PCC	23EC2210R - NPS	CO1	Apply the knowledge of communication to understand the concepts of physical layer and datalink layer.	2	2	2											2
213	PCC	23EC2210R - NPS	CO2	Analyze various MAC protocols and apply IP addressing concepts to subnet a network.		2	3											2
214	PCC	23EC2210R - NPS	CO3	Analyze static and dynamic routing algorithms and transport layer protocols.		2	3											2
215	PCC	23EC2210R - NPS	CO4	Analyze application layer protocols and various cryptographic algorithms.		2	3											2

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216	PCC	23EC2210R - NPS	CO5	Analyze the functionality of the network using different protocols and working of various cryptographic algorithms.		2			2									2
217	PRI	23IE2040 - SIP	CO1	Remember the fundamentals of the science of water cycle along with powerful tools that students can use to diagnose the health of the local water cycle as well as develop targeted action plans to restore the local natural water cycle and bring water prosperity			3										3	
218	PRI	23IE2040 - SIP	CO2	Remember the water sustainability and water resilience of village, city, residential facilities and households using multi-level water scorecards				3									3	
219	PRI	23IE2040 - SIP	CO3	Apply the design thinking positive action plan for a village, campus, residential facility and community neighbourhood.					3								3	
220	PRI	23IE2040 - SIP	CO4	Apply the water positive solutions within an urban watershed, a rural watershed, residential institutional and corporate community						3							3	
221	PRI	23IE3041 - TIP	CO1	Ability to demonstrate the impact of academic skills and logical thinking			3										3	
222	PRI	23IE3041 - TIP	CO2	Ability to integrate existing and new technical knowledge for industrial application				3									3	
223	PRI	23IE3041 - TIP	CO3	Ability to demonstrate the impact of the internship on their learning and professional development					3								3	
224	PRI	23IE3041 - TIP	CO4	Demonstrate the ability to harness resources by analyzing challenges						3							3	

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225	PRI	23IE4042 - IIP	CO1	Apply fundamental engineering/technology knowledge to solve real-world industrial problems.	2					3							2	
226	PRI	23IE4042 - IIP	CO2	Demonstrate effective communication and Technical skills in a professional setting.		2		3										2
227	PRI	23IE4042 - IIP	CO3	Collaborate effectively in a team environment.	3				2									3
228	PRI	23IE4048 - EPJ	CO1	Develop skills to conduct comprehensive searches, critically assess the credibility of sources, and apply research findings to support their academic or professional work.	2													
229	PRI	23IE4048 - EPJ	CO2	Analyze situations critically, identify underlying causes, and articulate clear problem statements that guide effective problem-solving strategies.		2												
230	PRI	23IE4048 - EPJ	CO3	Develop the ability to dissect research findings, identify strengths and weaknesses, and draw informed conclusions that contribute to their own academic or professional endeavors.						3							2	
231	PRI	23IE4048 - EPJ	CO4	Gain the skills to judge the validity, reliability, and impact of research, enabling them to make informed decisions and provide constructive feedback in academic or professional contexts.									2					2
232	PRI	23IE4051 - IIP-1	CO1	Apply theoretical knowledge to practical tasks in an industrial setting, ensuring that academic concepts are effectively translated into real-world applications.		3	3											
233	PRI	23IE4051 - IIP-1	CO2	Build problem-solving skills in real-world industrial projects by identifying challenges, brainstorming potential solutions, and implementing strategies to overcome obstacles.			3	3										

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234	PRI	23IE4051 - IIP-1	CO3	Analyze the industry-standard tools and techniques to complete assigned tasks with precision, staying updated with the latest advancements and best practices in the field.	3		3										3	
235	PRI	23IE4051 - IIP-1	CO4	Examine and interpret data collected during the internship to enhance process efficiency and effectiveness, utilizing statistical methods and data analytics to derive actionable insights and recommendations.			3		3									
236	PRI	23IE4052 - IIP-2	CO1	Apply theoretical knowledge to practical tasks in an industrial setting, ensuring that academic concepts are effectively translated into real-world applications.		3	3											
237	PRI	23IE4052 - IIP-2	CO2	Build problem-solving skills in real-world industrial projects by identifying challenges, brainstorming potential solutions, and implementing strategies to overcome obstacles.			3	3										
238	PRI	23IE4052 - IIP-2	CO3	Analyze the industry-standard tools and techniques to complete assigned tasks with precision, staying updated with the latest advancements and best practices in the field.	3		3										3	
239	PRI	23IE4052 - IIP-2	CO4	Examine and interpret data collected during the internship to enhance process efficiency and effectiveness, utilizing statistical methods and data analytics to derive actionable insights and recommendations.					3									
240	PRI	23IE4053R - CP-1	CO1	Exercise to acquire knowledge within the chosen area of technology for project development.	2													
241	PRI	23IE4053R - CP-1	CO2	Identify, discuss and justify the technical aspects of the chosen area for problem analysis		2												

S#	Cat	Course	CO	CO Description	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
242	PRI	23IE4053R - CP-1	CO3	Reproduce, improve and refine technical aspects for chosen problem												3		
243	PRI	23IE4053R - CP-1	CO4	Communicate and report effectively project related activities and findings.					3				2					
244	PRI	23IE4054R - CP-2	CO1	Develop skills to conduct comprehensive searches, critically assess the credibility of sources, and apply research findings to support their academic or professional work.	2													
245	PRI	23IE4054R - CP-2	CO2	Analyze situations critically, identify underlying causes, and articulate clear problem statements that guide effective problem-solving strategies.		3												
246	PRI	23IE4054R - CP-2	CO3	Develop the ability to dissect research findings, identify strengths and weaknesses, and draw informed conclusions that contribute to their own academic or professional endeavors.					2									
247	PRI	23IE4054R - CP-2	CO4	Gain the skills to judge the validity, reliability, and impact of research, enabling them to make informed decisions and provide constructive feedback in academic or professional contexts.									2					
248	SIL	22UC0021 - SIL-1	CO1	Apply effective communication and collaboration skills to work with diverse populations in addressing social issues within the community.									3			3		
249	SIL	22UC0021 - SIL-1	CO2	Build technological solutions to real-world problems or challenges with peers to achieve common goals.			3			3								
250	SIL	22UC0021 - SIL-1	CO3	Plan effectively to communicate ideas and collaborate with others to achieve artistic or recreational goals.									3			3		

S#	Cat	Course	CO	CO Description	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
251	SIL	22UC0021 - SIL-1	CO4	Develop innovative solutions by thinking critically and creatively within a collaborative social immersive learning environment.						3			3					
252	SIL	22UC0021 - SIL-1	CO5	Identify the strategies to promote personal well-being for healthy living through social interaction and shared experiences.			3			3								
253	SIL	22UC0022 - SIL-2	CO1	Apply effective communication and collaboration skills to work with diverse populations in addressing social issues within the community.								3	3					
254	SIL	22UC0022 - SIL-2	CO2	Build technological solutions to real-world problems or challenges with peers to achieve common goals.								3	3					
255	SIL	22UC0022 - SIL-2	CO3	Plan effectively to communicate ideas and collaborate with others to achieve artistic or recreational goals.				3				3	3					3
256	SIL	22UC0022 - SIL-2	CO4	Develop innovative solutions by thinking critically and creatively within a collaborative social immersive learning environment.				3				3	3					3
257	SIL	22UC0022 - SIL-2	CO5	Identify the strategies to promote personal well-being for healthy living through social interaction and shared experiences.						3		3	3					
258	SIL	22UC0023 - SIL-3	CO1	Apply effective communication and collaboration skills to work with diverse populations in addressing social issues within the community.								3	3					
259	SIL	22UC0023 - SIL-3	CO2	Build technological solutions to real-world problems or challenges with peers to achieve common goals.								3	3					

S#	Cat	Course	CO	CO Description	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
260	SIL	22UC0023 - SIL-3	CO3	Plan effectively to communicate ideas and collaborate with others to achieve artistic or recreational goals.			3						3				3	
261	SIL	22UC0023 - SIL-3	CO4	Develop innovative solutions by thinking critically and creatively within a collaborative social immersive learning environment.			3						3				3	
262	SIL	22UC0023 - SIL-3	CO5	Identify the strategies to promote personal well-being for healthy living through social interaction and shared experiences.			3			3			3					
					2.2	2.3	2.7	2.8	2.7	2.8	2	2.5	2.5	2.6	2.1	2.6	2.4	2.4

Stakeholder's Feedback

Q#	Question	No. of Stakeholder's						Rating (%)				
		STU	ALU	IE	AP	FAC	TOT	[5]	[4]	[3]	[2]	[1]
Q1	How would you rate the relevance of the current syllabus content in addressing industry needs and trends?	613	267	183	180	716	1959	56.3	34	8.9	0.8	0.1
Q2	How well do the course outcomes align with the skills required in the industry?	613	267	183	180	716	1959	52.8	37.1	9.3	0.7	0.1
Q3	How would you rate the inclusion of emerging technologies or methodologies in the syllabus?	613	267	183	180	716	1959	53.9	35.1	9.9	1.1	0.1
Q4	How effectively are the latest tools integrated into the curriculum?	613	267	183	0	0	1063	65	24.7	8.6	1.6	0.1
Q5	How beneficial are the global certifications included in the curriculum for industry readiness?	613	267	183	0	716	1779	58.6	30.3	9.9	1.1	0.2
Q6	How effectively does the curriculum incorporate practical lab experiments relevant to industry practices?	0	267	183	0	0	450	54.2	34.2	10.9	0.7	0
Q7	How does this curriculum compare with similar curricula at other institutions in terms of content and quality?	0	0	0	180	0	180	32.8	51.7	15.6	0	0
Q8	How effective is the integration of research opportunities into the curriculum?	0	0	0	180	716	896	40.2	44.8	13.3	1.8	0
Q9	How beneficial were the MOOCs recommended as part of the curriculum?	613	267	0	180	716	1776	53.8	34.9	10.3	0.9	0.1
Q10	How well does the course content map to skill council recommendations?	0	0	183	180	716	1079	42.3	45.7	11.5	0.5	0.1